The **gap-closing estimand**

Slides at
ianlundberg.org

Software at
ilundberg.github.io/gapclosing

A causal approach
to study **interventions**
that **close disparities**
across social categories

Ian Lundberg

UCLA Sociology and
California Center for Population Research
ianlundberg@ucla.edu

30 Jan 2025
Chinese University of Hong Kong

Paper in *Sociological Methods and Research*. Replication code on Dataverse. R package `gapclosing` on CRAN. Research reported in this publication was supported by The Eunice Kennedy Shriver National Institute of Child Health & Human Development of the National Institutes of Health under Award Number P2CHD047879 and by the National Science Foundation under Award Number 2104607.

The Gap-Closing Estimand:
A Causal Approach to Study Interventions That Close Disparities Across Social Categories

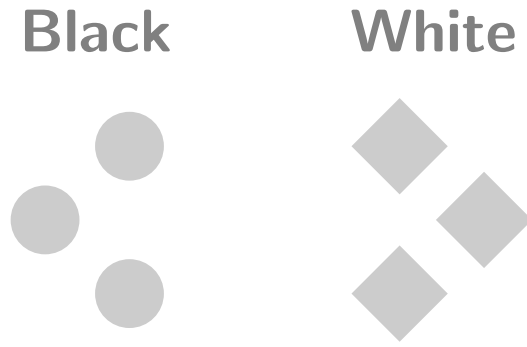
Ian Lundberg (UCLA)

The Gap-Closing Estimand:
A Causal Approach to Study Interventions That Close Disparities Across Social Categories

Ian Lundberg (UCLA)

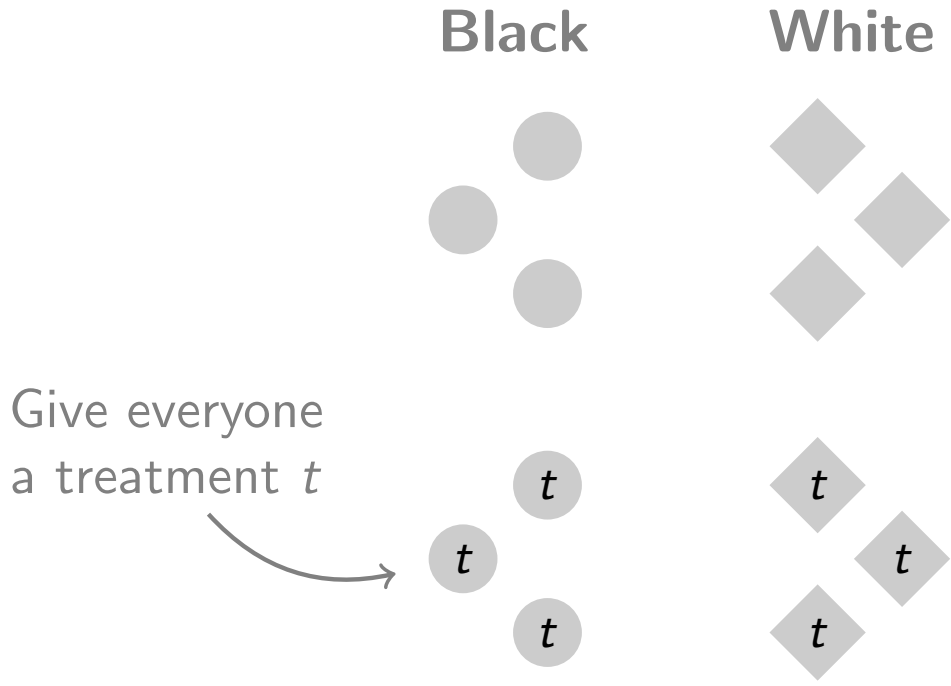
The Gap-Closing Estimand:
A Causal Approach to Study Interventions That Close Disparities Across Social Categories

Ian Lundberg (UCLA)



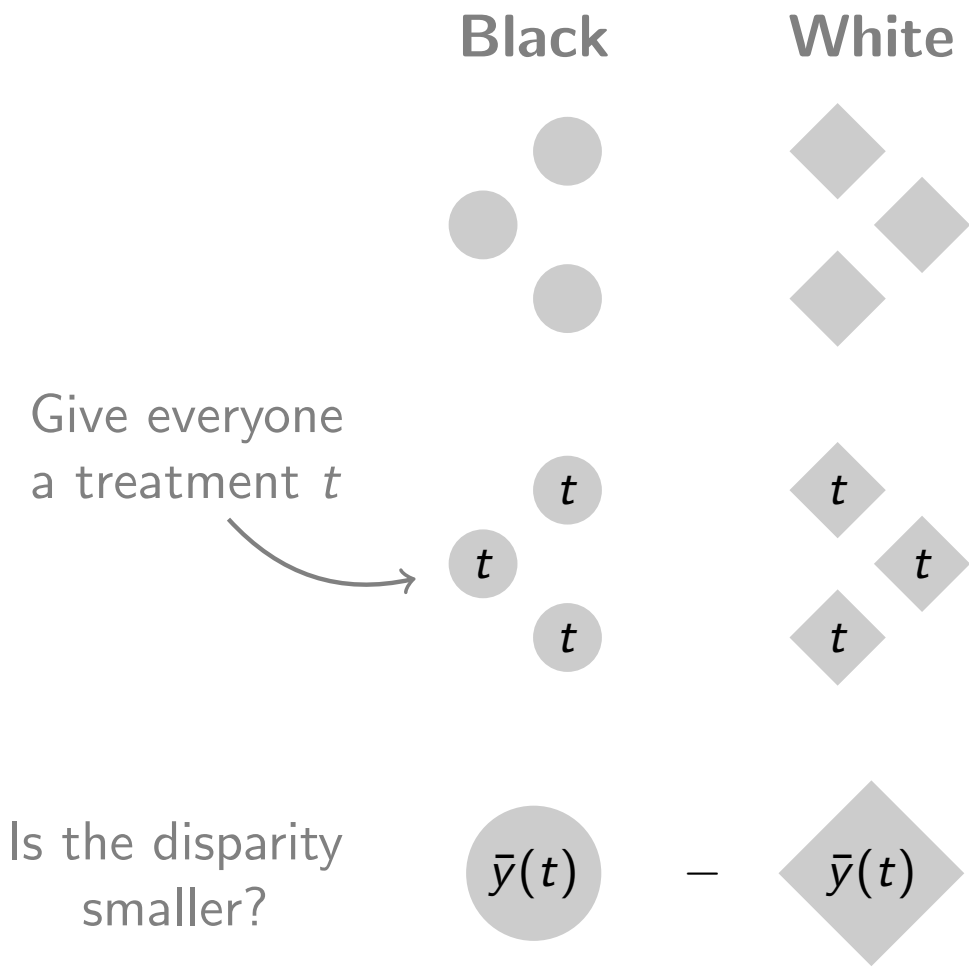
The Gap-Closing Estimand:
A Causal Approach to Study Interventions That Close Disparities Across Social Categories

Ian Lundberg (UCLA)



The Gap-Closing Estimand:
A Causal Approach to Study Interventions That Close Disparities Across Social Categories

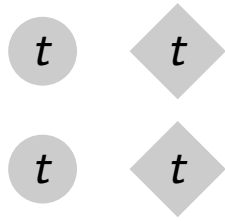
Ian Lundberg (UCLA)



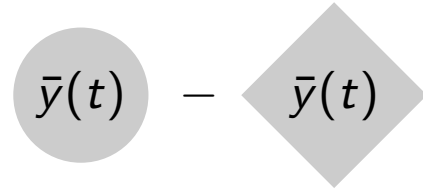
The Gap-Closing Estimand:
A Causal Approach to Study Interventions
That Close Disparities Across Social Categories

Ian Lundberg (UCLA)

Categories

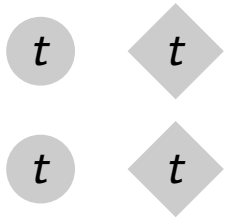


Treatment

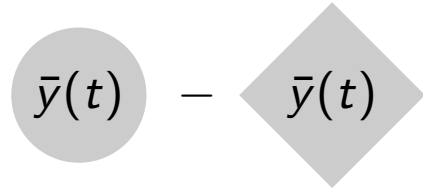


Counterfactual Disparity

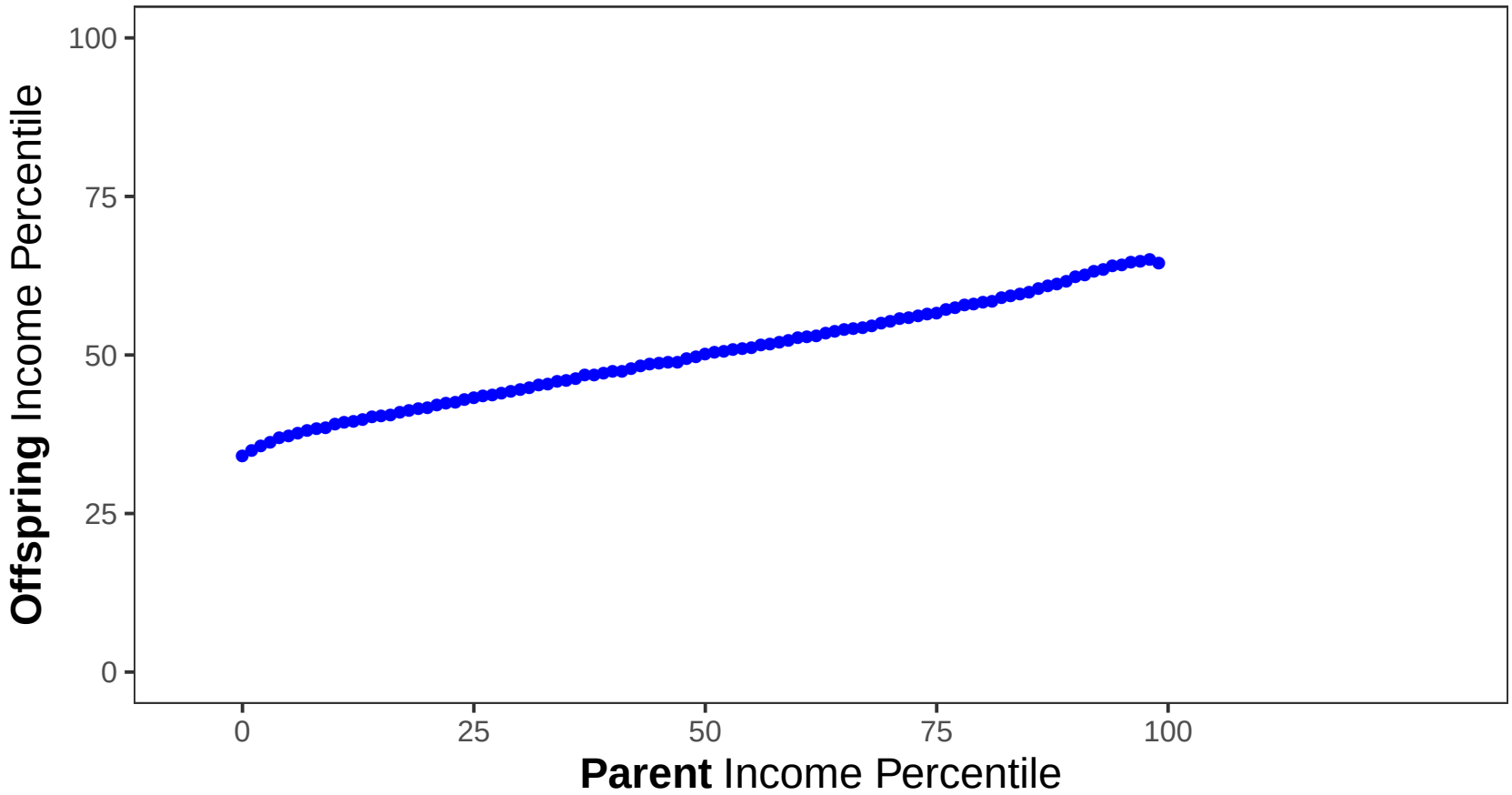
Categories



Treatment

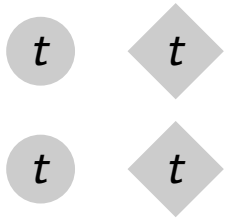


Counterfactual Disparity

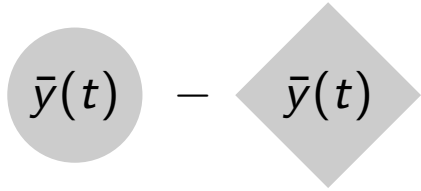


Chetty et al. 2017

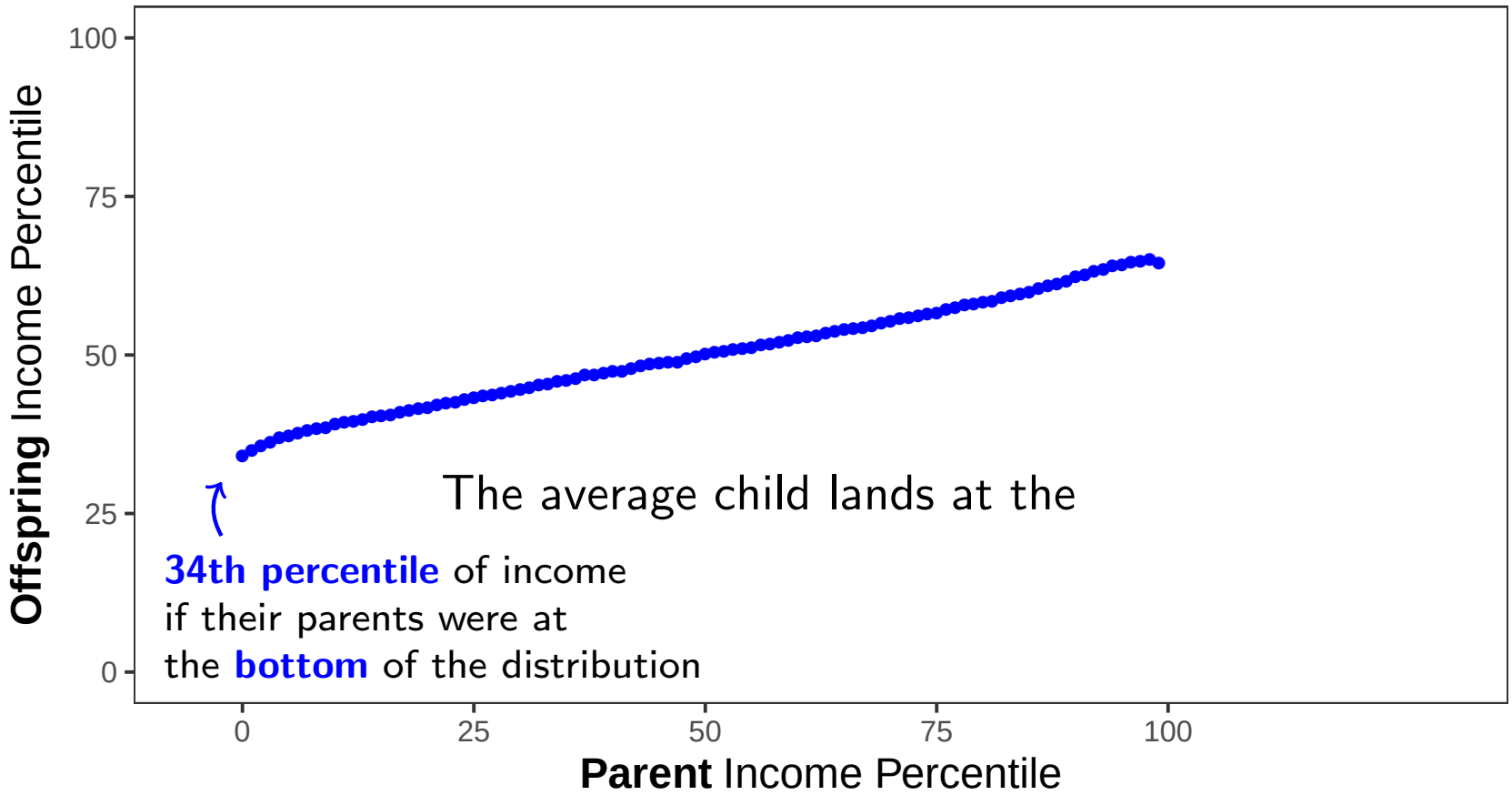
Categories



Treatment

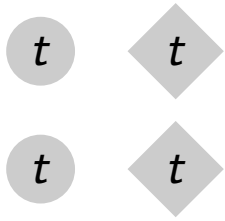


Counterfactual Disparity

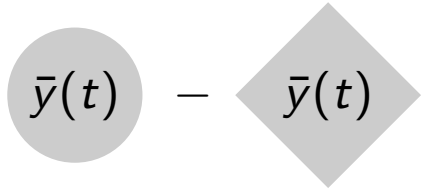


Chetty et al. 2017

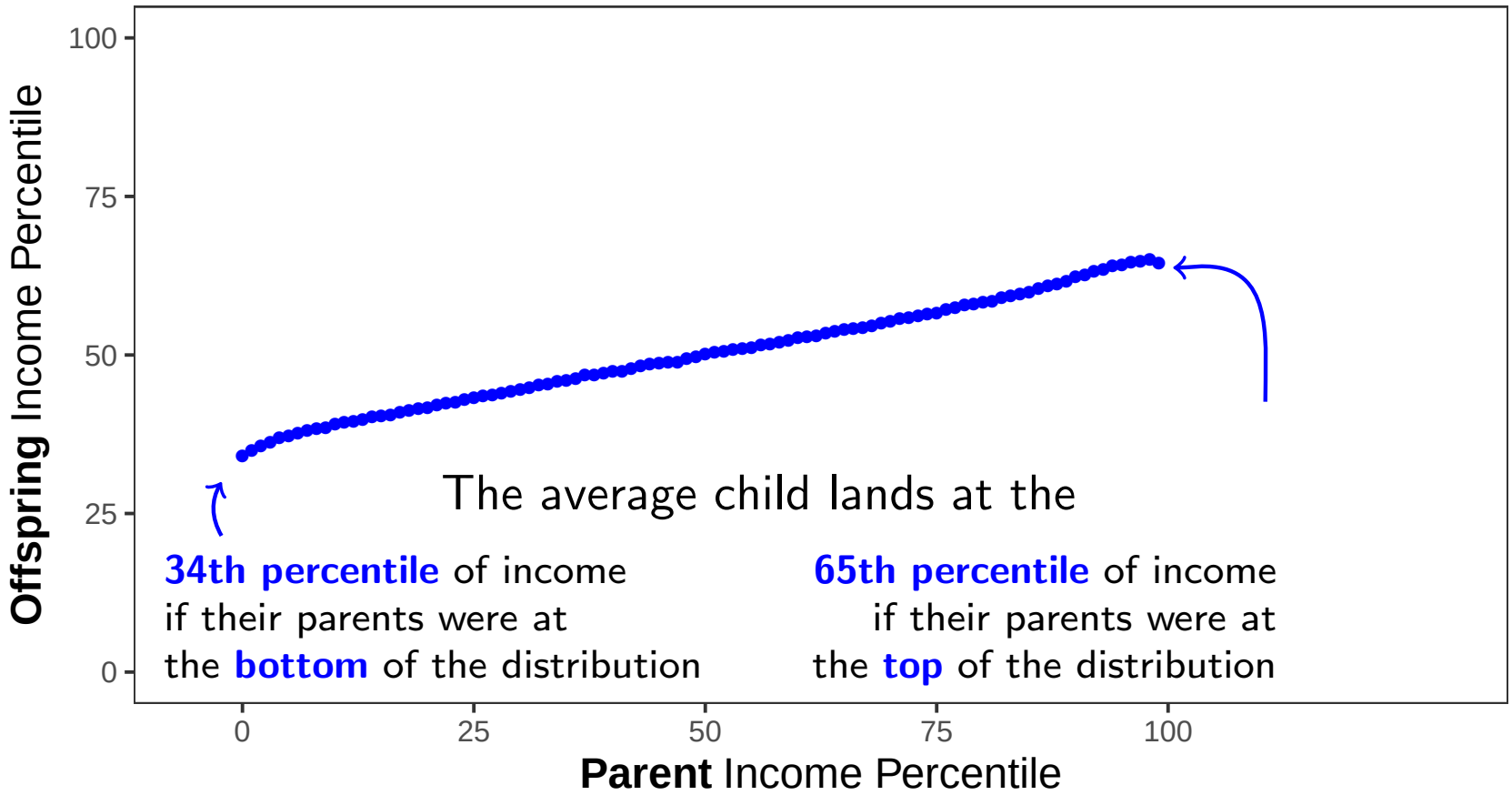
Categories



Treatment

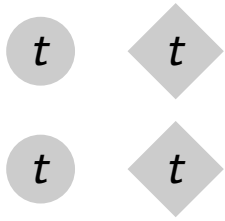


Counterfactual Disparity

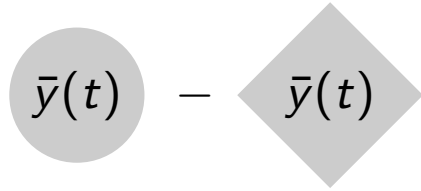


Chetty et al. 2017

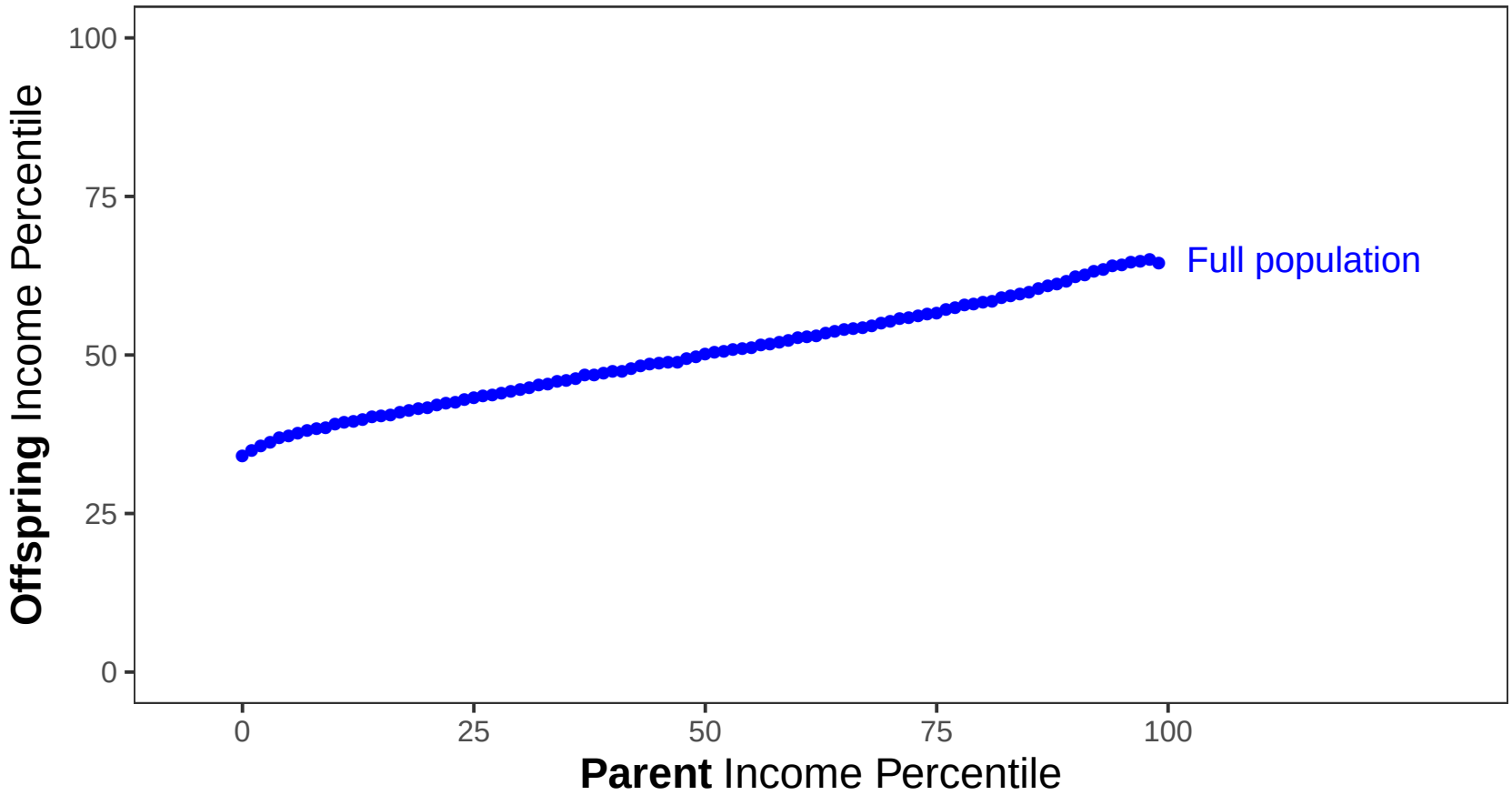
Categories



Treatment

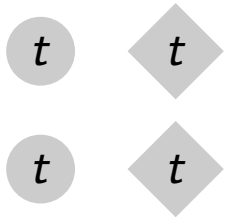


Counterfactual Disparity

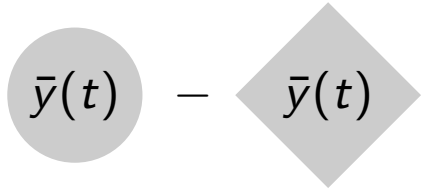


Chetty et al. 2017

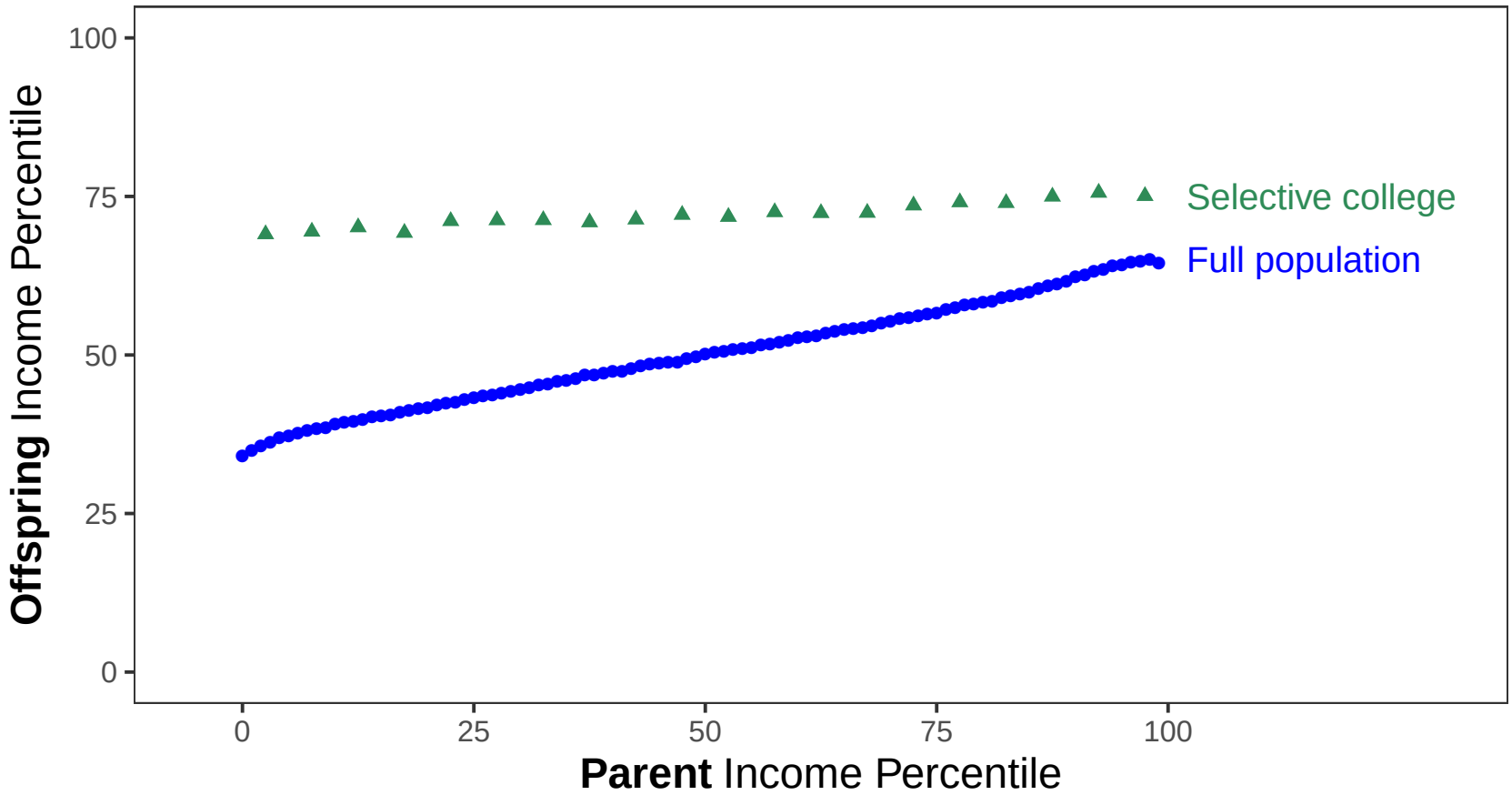
Categories



Treatment

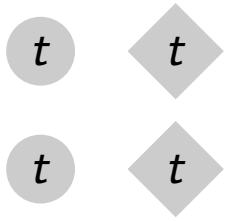


Counterfactual Disparity

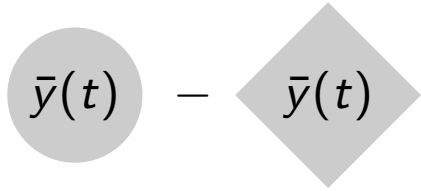


Chetty et al. 2017

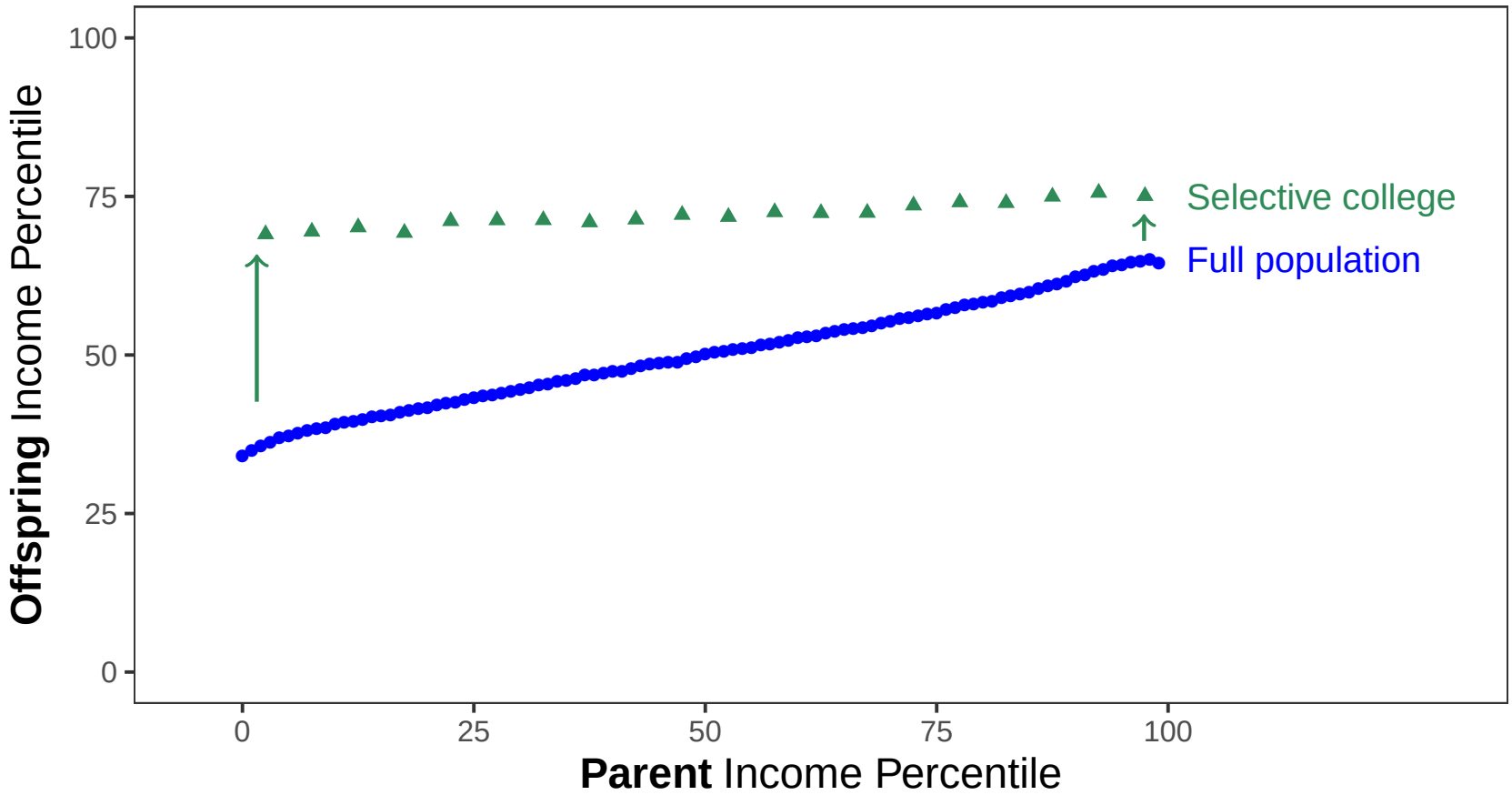
Categories



Treatment

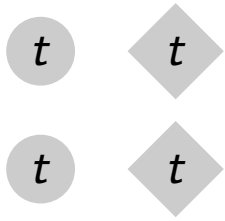


Counterfactual Disparity

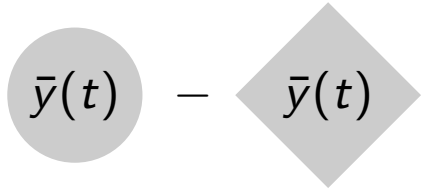


Chetty et al. 2017

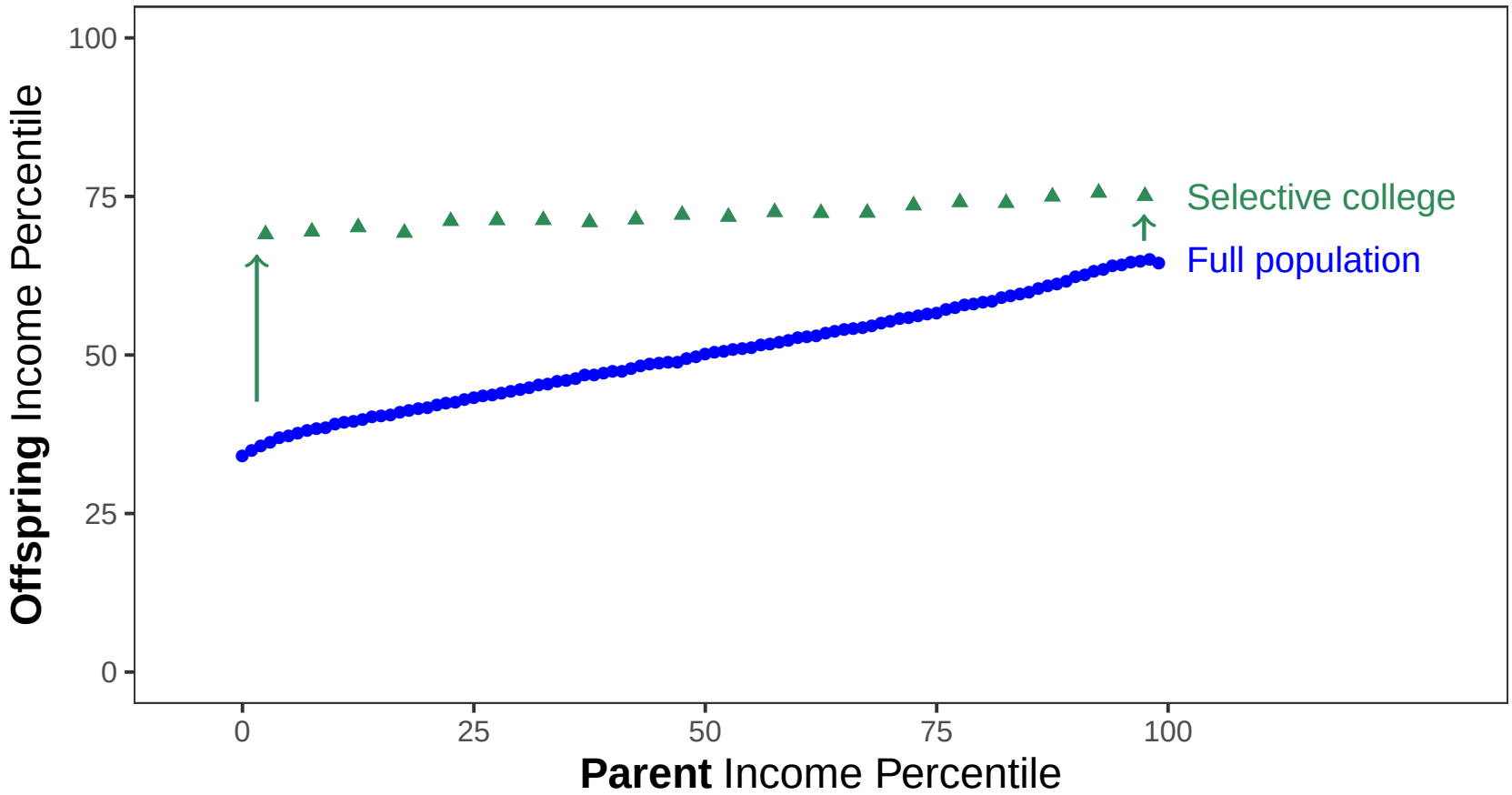
Categories
Parent Income



Treatment
Selective College



Counterfactual Disparity
Offspring Income



Chetty et al. 2017

The Gap-Closing Estimand

A causal approach to study interventions that close disparities across social categories

- Past work: Causal decomposition analysis
- Define the estimand
  - what is the treatment?
  - what is the scope of the intervention?
- Identify the estimand
  - choose a sufficient adjustment set
- Estimate
  - by outcome modeling
  - by weighting
  - by doubly robust estimation
- Produce software
- Impact: Others using these ideas

The Gap-Closing Estimand

A causal approach to study interventions that close disparities across social categories

- Past work: Causal decomposition analysis
- Define the estimand
  - what is the treatment?
  - what is the scope of the intervention? (contribution)
- Identify the estimand
  - choose a sufficient adjustment set
- Estimate
  - by outcome modeling
  - by weighting
  - by doubly robust estimation (contribution)
- Produce software (contribution)
- Impact: Others using these ideas

Past work: Causal decomposition analysis

Past work: Causal decomposition analysis

Unadjusted

$$Y = \beta(\texttt{Black}) + \epsilon$$

Adjusted

$$Y = \gamma(\texttt{Black}) + \vec{X}'\vec{\eta} + \delta$$

Past work: Causal decomposition analysis

Unadjusted

$$Y = \beta(\texttt{Black}) + \epsilon$$

Adjusted

$$Y = \gamma(\texttt{Black}) + \vec{X}'\vec{\eta} + \delta$$

Effect of race

Past work: Causal decomposition analysis

Unadjusted
$$Y = \beta(\texttt{Black}) + \epsilon$$

Adjusted
$$Y = \gamma(\texttt{Black}) + \vec{X}'\vec{\eta} + \delta$$

× Effect of race

Vanderweele & Robinson 2014

Past work: Causal decomposition analysis

Unadjusted

$$Y = \beta(\texttt{Black}) + \epsilon$$

Adjusted

$$Y = \gamma(\texttt{Black}) + \vec{X}'\vec{\eta} + \delta$$

- × Effect of race
- ✓ Disparity after intervention on $\vec{X}$

Vanderweele & Robinson 2014

Past work: Causal decomposition analysis

Unadjusted
$$Y = \beta(\texttt{Black}) + \epsilon$$

Adjusted
$$Y = \gamma(\texttt{Black}) + \vec{X}'\vec{\eta} + \delta$$

- × Effect of race — Vanderweele & Robinson 2014
- ✓ Disparity after intervention on $\vec{X}$

Choice of intervention targets — Jackson & Vanderweele 2018

Past work: Causal decomposition analysis

Unadjusted

$$Y = \beta(\texttt{Black}) + \epsilon$$

Adjusted

$$Y = \gamma(\texttt{Black}) + \vec{X}'\vec{\eta} + \delta$$

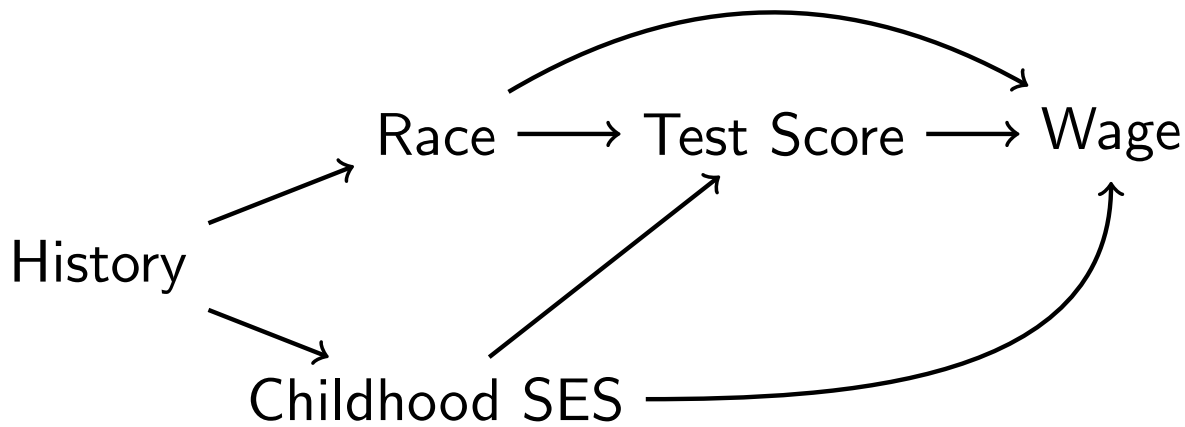
× Effect of race

✓ Disparity after intervention on $\vec{X}$

Vanderweele & Robinson 2014

Choice of intervention targets

Jackson & Vanderweele 2018



Past work: Causal decomposition analysis

Unadjusted
$$Y = \beta(\texttt{Black}) + \epsilon$$

Adjusted
$$Y = \gamma(\texttt{Black}) + \vec{X}'\vec{\eta} + \delta$$

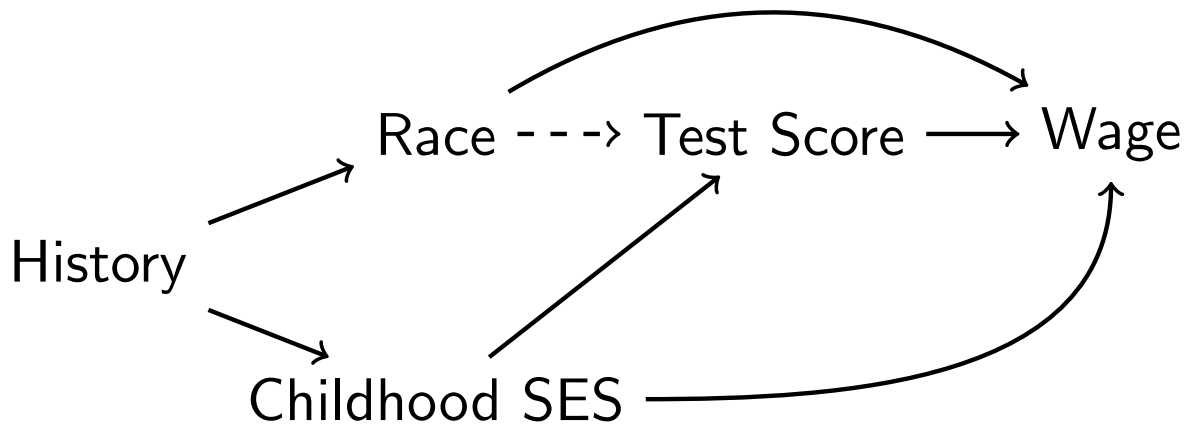
× Effect of race

✓ Disparity after intervention on $\vec{X}$

Vanderweele & Robinson 2014

Choice of intervention targets

Jackson & Vanderweele 2018



Past work: Causal decomposition analysis

Unadjusted

$$Y = \beta(\texttt{Black}) + \epsilon$$

Adjusted

$$Y = \gamma(\texttt{Black}) + \vec{X}'\vec{\eta} + \delta$$

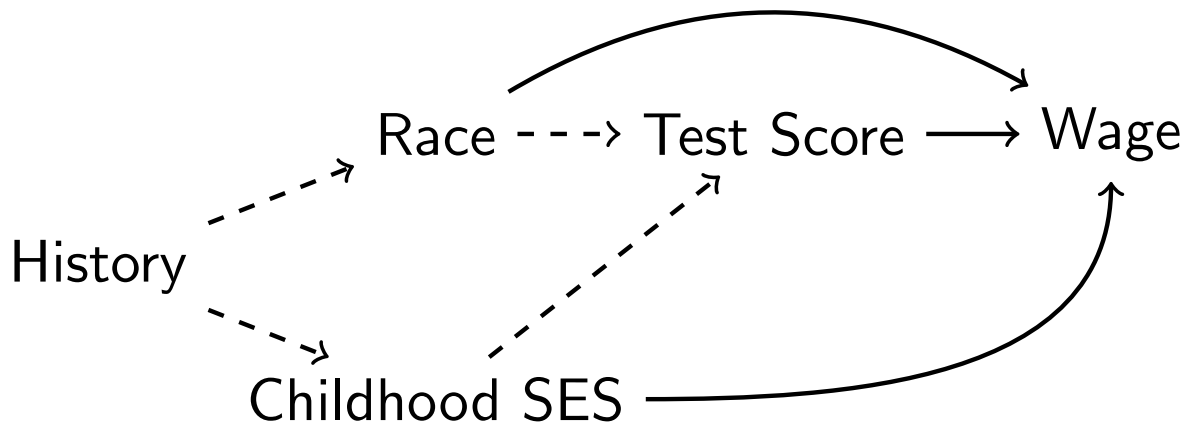
× Effect of race

✓ Disparity after intervention on $\vec{X}$

Vanderweele & Robinson 2014

Choice of intervention targets

Jackson & Vanderweele 2018



Past work: Causal decomposition analysis

Unadjusted
$$Y = \beta(\texttt{Black}) + \epsilon$$

Adjusted
$$Y = \gamma(\texttt{Black}) + \vec{X}'\vec{\eta} + \delta$$

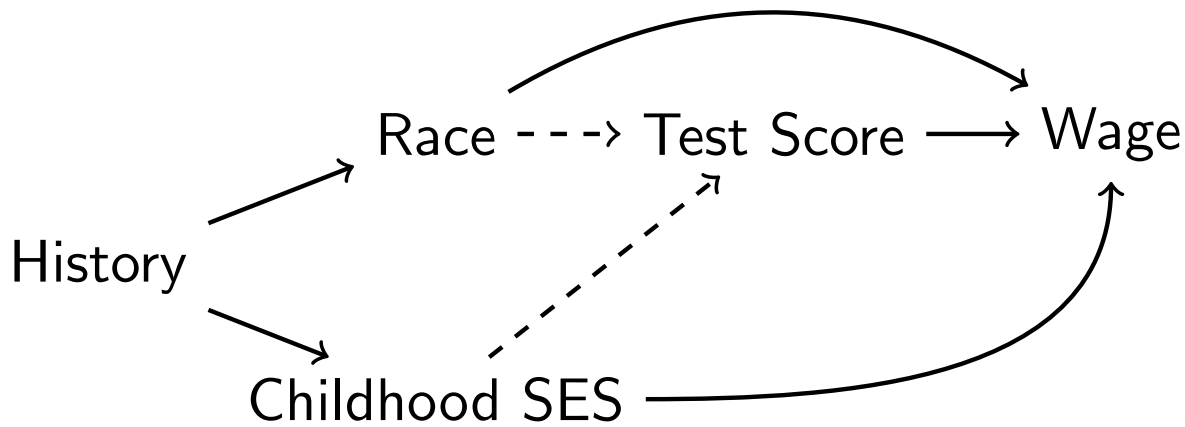
× Effect of race

✓ Disparity after intervention on $\vec{X}$

Vanderweele & Robinson 2014

Choice of intervention targets

Jackson & Vanderweele 2018



Past work: Causal decomposition analysis

Unadjusted	Adjusted
$Y = \beta(\texttt{Black}) + \epsilon$	$Y = \gamma(\texttt{Black}) + \vec{X}'\vec{\eta} + \delta$

- × Effect of race — Vanderweele & Robinson 2014
- ✓ Disparity after intervention on $\vec{X}$ — Vanderweele & Robinson 2014

Choice of intervention targets — Jackson & Vanderweele 2018

Equity: What should we equalize? — Jackson 2021

Past work: Causal decomposition analysis

Unadjusted	Adjusted
$Y = \beta(\texttt{Black}) + \epsilon$	$Y = \gamma(\texttt{Black}) + \vec{X}'\vec{\eta} + \delta$

×	Effect of race	Vanderweele & Robinson 2014
✓	Disparity after intervention on $\vec{X}$	
	Choice of intervention targets	Jackson & Vanderweele 2018
	Equity: What should we equalize?	Jackson 2021
	Systems may adapt to maintain inequity	Jackson & Arah 2020

The Gap-Closing Estimand

A causal approach to study interventions that close disparities across social categories

- Past work: Causal decomposition analysis
- **Define the estimand**
  - what is the treatment?
  - what is the scope of the intervention?
- Identify the estimand
  - choose a sufficient adjustment set
- Estimate
  - by outcome modeling
  - by weighting
  - by doubly robust estimation
- Produce software
- Impact: Others using these ideas

Define an intervention

Using the Chetty et al. 2017 example,

What gap in respondent incomes would remain
across categories of parent income
if we intervened to send people to selective colleges?

Interpret with respect to a **target trial** (Hernán & Robins 2016)

Interpret with respect to a **target trial** (Hernán & Robins 2016)

1. Sample $\mathcal{S}$ from the population

Interpret with respect to a **target trial** (Hernán & Robins 2016)

1. Sample $\mathcal{S}$ from the population
2. Assign treatment $T = 1$ to $\mathcal{S}$

Interpret with respect to a **target trial** (Hernán & Robins 2016)

1. Sample $\mathcal{S}$ from the population
2. Assign treatment $T = 1$ to $\mathcal{S}$
3. Observe the disparity
   across categories X

Interpret with respect to a **target trial** (Hernán & Robins 2016)

1. Sample $\mathcal{S}$ from the population
2. Assign treatment $T = 1$ to $\mathcal{S}$
3. Observe the disparity
   across categories X

Goal: Expected result over
hypothetical samples $\mathcal{S}$

Interpret with respect to a **target trial** (Hernán & Robins 2016)

Local intervention

1. Sample $\mathcal{S}$ from the population
2. Assign treatment $T = 1$ to $\mathcal{S}$
3. Observe the disparity
   across categories X

Goal: Expected result over
hypothetical samples $\mathcal{S}$

Interpret with respect to a **target trial** (Hernán & Robins 2016)

Local intervention

1. Sample $\mathcal{S}$ from the population
2. Assign treatment $T = 1$ to $\mathcal{S}$
3. Observe the disparity across categories X

Goal: Expected result over hypothetical samples $\mathcal{S}$

Global intervention

Interpret with respect to a **target trial** (Hernán & Robins 2016)

Local intervention

1. Sample $\mathcal{S}$ from the population
2. Assign treatment $T = 1$ to $\mathcal{S}$
3. Observe the disparity across categories X

Goal: Expected result over hypothetical samples $\mathcal{S}$

Global intervention

1. Take the entire population $\mathcal{P}$

Interpret with respect to a **target trial** (Hernán & Robins 2016)

Local intervention

1. Sample $\mathcal{S}$ from the population
2. Assign treatment $T = 1$ to $\mathcal{S}$
3. Observe the disparity across categories X

Goal: Expected result over hypothetical samples $\mathcal{S}$

Global intervention

1. Take the entire population $\mathcal{P}$
2. Assign treatment $T = 1$ to $\mathcal{P}$

Interpret with respect to a **target trial** (Hernán & Robins 2016)

Local intervention

1. Sample $\mathcal{S}$ from the population
2. Assign treatment $T = 1$ to $\mathcal{S}$
3. Observe the disparity across categories X

Goal: Expected result over hypothetical samples $\mathcal{S}$

Global intervention

1. Take the entire population $\mathcal{P}$
2. Assign treatment $T = 1$ to $\mathcal{P}$
3. Observe the disparity across categories X

Interpret with respect to a **target trial** (Hernán & Robins 2016)

Local intervention

1. Sample $\mathcal{S}$ from the population
2. Assign treatment $T = 1$ to $\mathcal{S}$
3. Observe the disparity across categories X

Goal: Expected result over hypothetical samples $\mathcal{S}$

Global intervention

1. Take the entire population $\mathcal{P}$
2. Assign treatment $T = 1$ to $\mathcal{P}$
3. Observe the disparity across categories X

Goal: Result of this procedure

Interpret with respect to a **target trial** (Hernán & Robins 2016)

Local intervention

1. Sample $\mathcal{S}$ from the population
2. Assign treatment $T = 1$ to $\mathcal{S}$
3. Observe the disparity across categories X

Goal: Expected result over hypothetical samples $\mathcal{S}$

Difficulty: Causal inference

Global intervention

1. Take the entire population $\mathcal{P}$
2. Assign treatment $T = 1$ to $\mathcal{P}$
3. Observe the disparity across categories X

Goal: Result of this procedure

Difficulty: Causal inference

Interpret with respect to a **target trial** (Hernán & Robins 2016)

Local intervention

1. Sample $\mathcal{S}$ from the population
2. Assign treatment $T = 1$ to $\mathcal{S}$
3. Observe the disparity across categories X

Goal: Expected result over hypothetical samples $\mathcal{S}$

Difficulty: Causal inference

Global intervention

1. Take the entire population $\mathcal{P}$
2. Assign treatment $T = 1$ to $\mathcal{P}$
3. Observe the disparity across categories X

Goal: Result of this procedure

Difficulty: Causal inference
Equilibrium dynamics

Interpret with respect to a **target trial** (Hernán & Robins 2016)



Local intervention

1. Sample $\mathcal{S}$ from the population
2. Assign treatment $T = 1$ to $\mathcal{S}$
3. Observe the disparity across categories X

Goal: Expected result over hypothetical samples $\mathcal{S}$

Difficulty: Causal inference

Global intervention

1. Take the entire population $\mathcal{P}$
2. Assign treatment $T = 1$ to $\mathcal{P}$
3. Observe the disparity across categories X

Goal: Result of this procedure

Difficulty: Causal inference
Equilibrium dynamics

Interpret with respect to a **target trial** (Hernán & Robins 2016)



Local intervention

1. Sample $\mathcal{S}$ from the population
2. Assign treatment $T = 1$ to $\mathcal{S}$
3. Observe the disparity across categories X

Goal: Expected result over hypothetical samples $\mathcal{S}$

Difficulty: Causal inference

Global intervention

1. Take the entire population $\mathcal{P}$
2. Assign treatment $T = 1$ to $\mathcal{P}$
3. Observe the disparity across categories X

Goal: Result of this procedure

Difficulty: Causal inference
Equilibrium dynamics

Interpret with respect to a **target trial** (Hernán & Robins 2016)



Local intervention

1. Sample $\mathcal{S}$ from the population
2. Assign treatment $T = 1$ to $\mathcal{S}$
3. Observe the disparity across categories X

Goal: Expected result over hypothetical samples $\mathcal{S}$

Difficulty: Causal inference

Global intervention

1. Take the entire population $\mathcal{P}$
2. Assign treatment $T = 1$ to $\mathcal{P}$
3. Observe the disparity across categories X

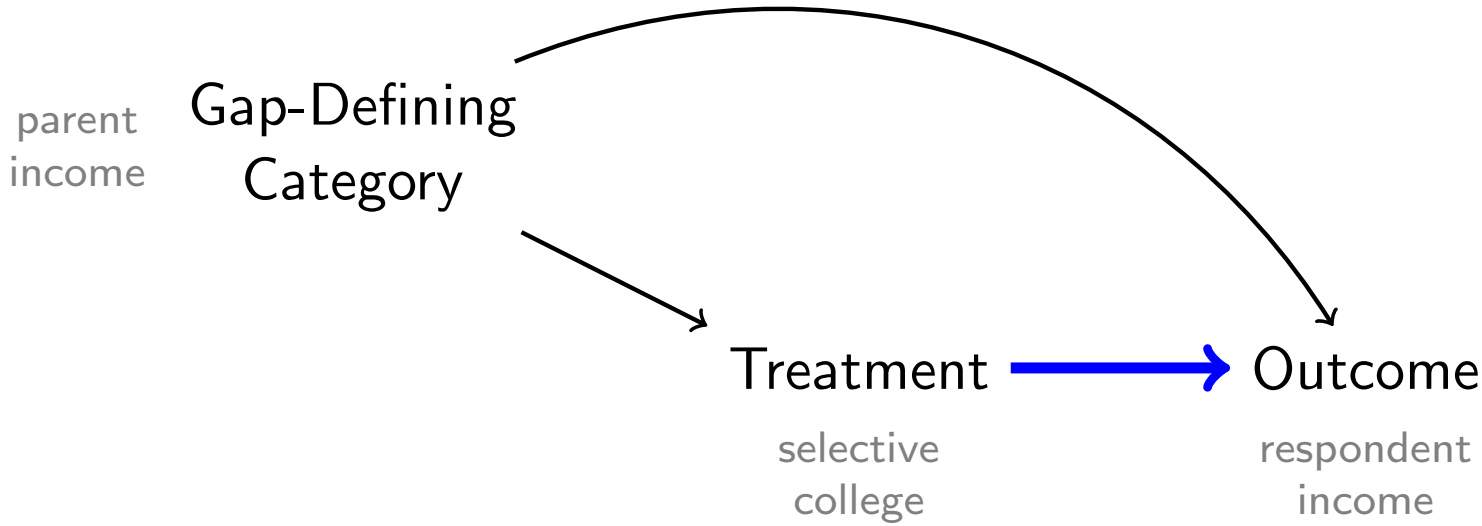
Goal: Result of this procedure

Difficulty: Causal inference
Equilibrium dynamics

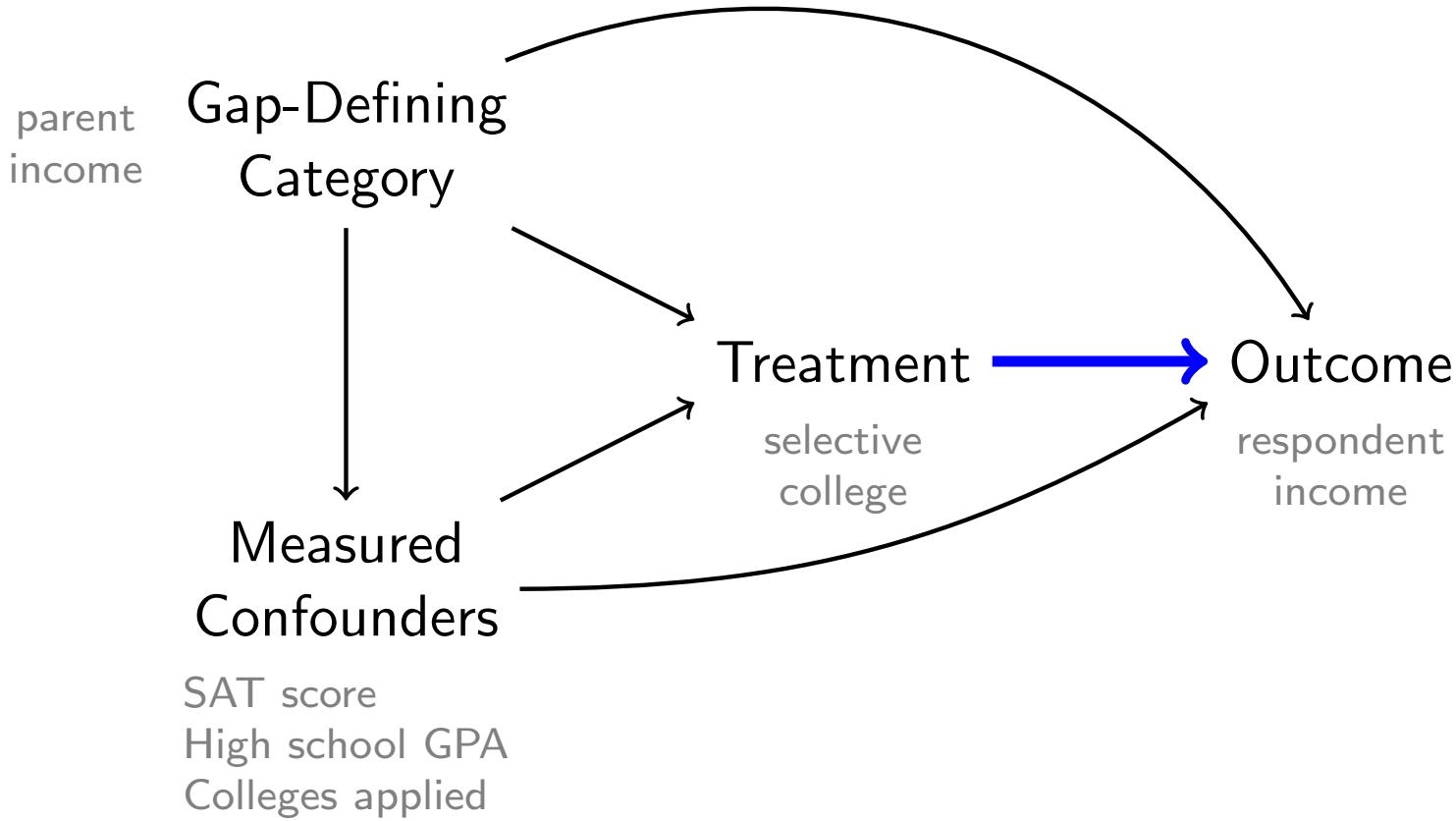
The Gap-Closing Estimand

A causal approach to study interventions that close disparities across social categories

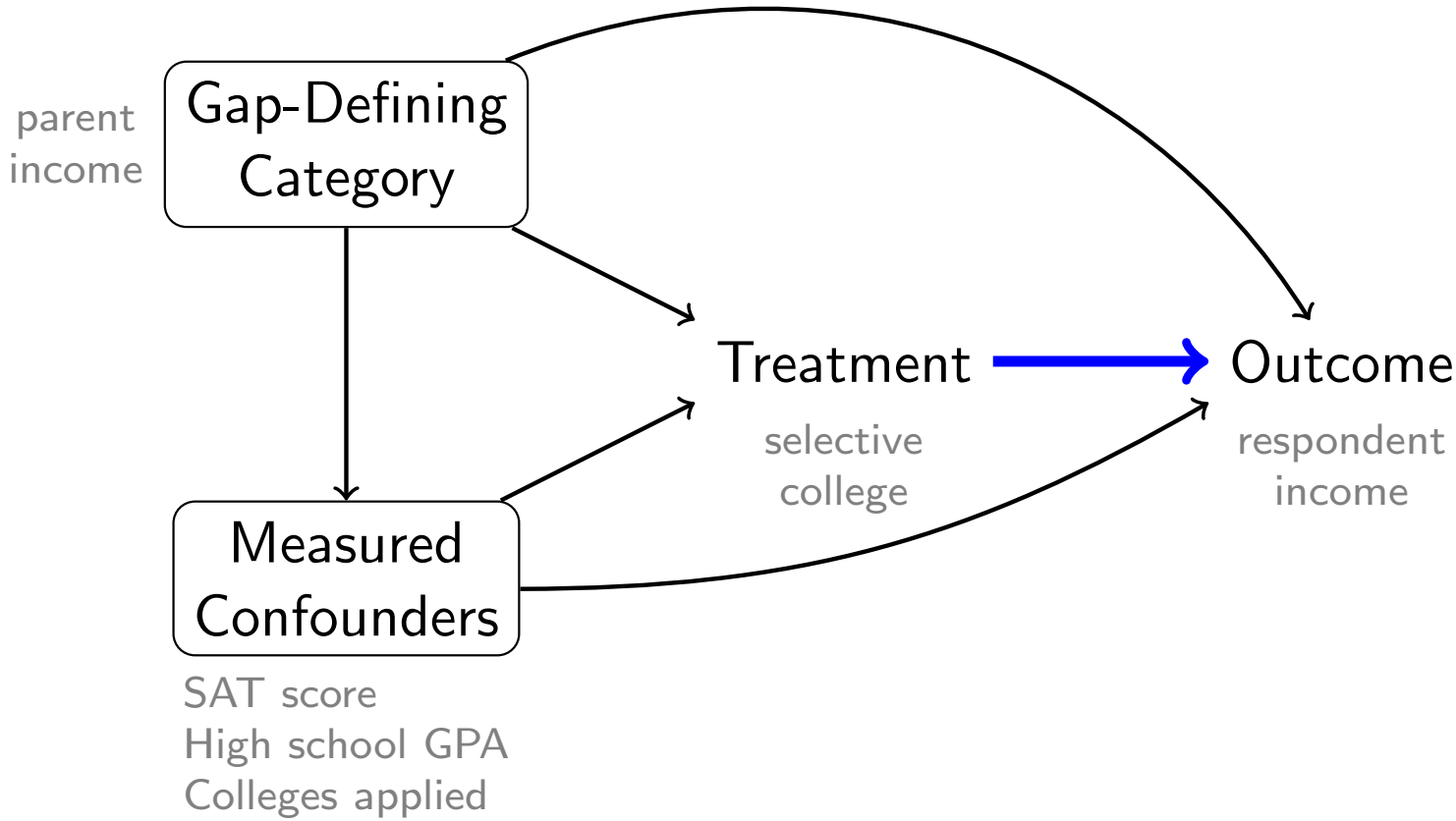
- Past work: Causal decomposition analysis
- Define the estimand
  - what is the treatment?
  - what is the scope of the intervention?
- **Identify the estimand**
  - choose a sufficient adjustment set
- Estimate
  - by outcome modeling
  - by weighting
  - by doubly robust estimation
- Produce software
- Impact: Others using these ideas


parent
income
Gap-Defining
Category
Treatment
selective
college
Outcome
respondent
income

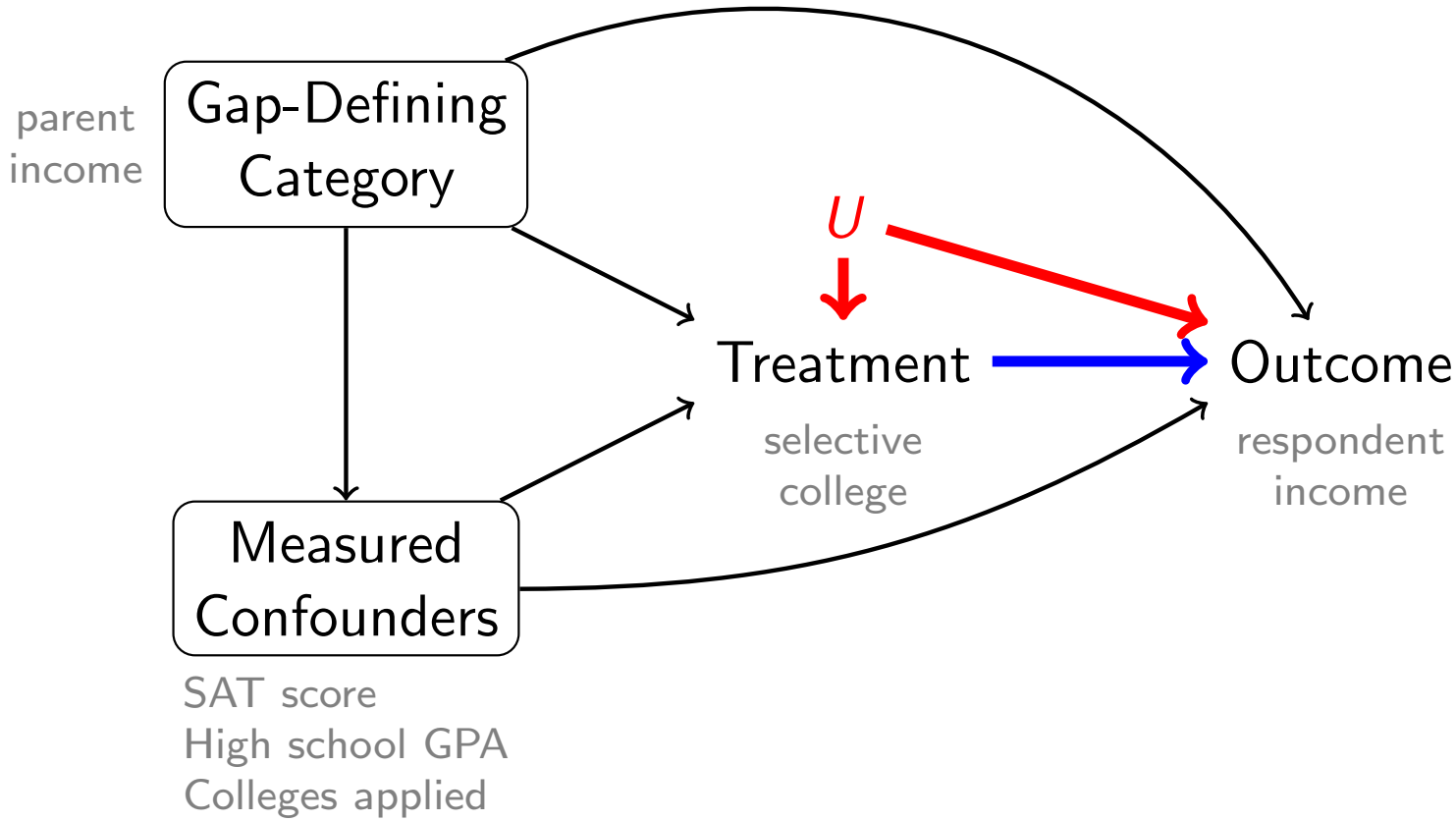
Pearl 2009


parent income
Gap-Defining Category
Treatment
selective college
Outcome
respondent income
Measured Confounders
SAT score
High school GPA
Colleges applied

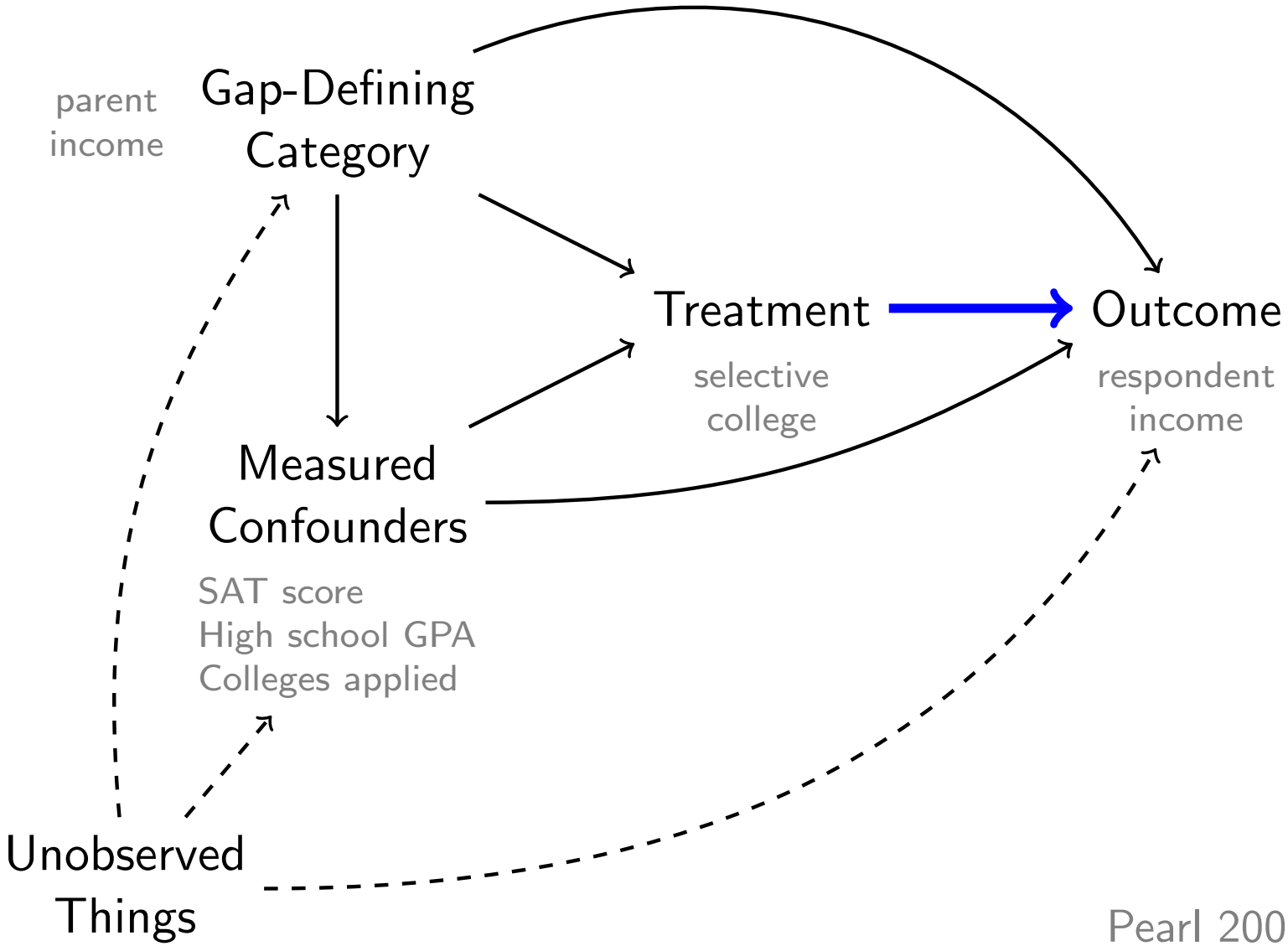
Pearl 2009


parent income
Gap-Defining Category
Treatment
selective college
Outcome
respondent income
Measured Confounders
SAT score
High school GPA
Colleges applied

Pearl 2009


parent
income
Gap-Defining
Category
U
Treatment
selective
college
Outcome
respondent
income
Measured
Confounders
SAT score
High school GPA
Colleges applied

Pearl 2009


parent income
Gap-Defining Category
Treatment
selective college
Outcome
respondent income
Measured Confounders
SAT score
High school GPA
Colleges applied
Unobserved Things
Pearl 2009

The Gap-Closing Estimand

A causal approach to study interventions that close disparities across social categories

- Past work: Causal decomposition analysis
- Define the estimand
  - what is the treatment?
  - what is the scope of the intervention?
- Identify the estimand
  - choose a sufficient adjustment set
- **Estimate**
  - by outcome modeling
  - by weighting
  - by doubly robust estimation
- Produce software
- Impact: Others using these ideas

		Outcome under treatment	Outcome under control
People in category 1	Person 1	?	Y_1
	Person 2	Y_2	?
	Person 3	Y_3	?
People in category 2	Person 4	?	Y_4
	Person 5	Y_5	?
	Person 6	?	Y_6

		Outcome under treatment	Outcome under control
People in category 1	Person 1	?	Y_1
	Person 2	Y_2	?
	Person 3	Y_3	?
People in category 2	Person 4	?	Y_4
	Person 5	Y_5	?
	Person 6	?	Y_6

Learn a prediction function

		Outcome under treatment	Outcome under control
People in category 1	Person 1	?	Y_1
	Person 2	Y_2	?
	Person 3	Y_3	?
People in category 2	Person 4	?	Y_4
	Person 5	Y_5	?
	Person 6	?	Y_6

Learn a prediction function

		Outcome under treatment	Outcome under control
People in category 1	Person 1	?	Y_1
	Person 2	Y_2	?
	Person 3	Y_3	?
People in category 2	Person 4	?	Y_4
	Person 5	Y_5	?
	Person 6	?	Y_6

Predict the whole table

		Outcome under treatment	Outcome under control
People in category 1	Person 1	$\hat{Y}_1(1)$	$\hat{Y}_1(0)$
	Person 2	$\hat{Y}_2(1)$	$\hat{Y}_2(0)$
	Person 3	$\hat{Y}_3(1)$	$\hat{Y}_3(0)$
People in category 2	Person 4	$\hat{Y}_4(1)$	$\hat{Y}_4(0)$
	Person 5	$\hat{Y}_5(1)$	$\hat{Y}_5(0)$
	Person 6	$\hat{Y}_6(1)$	$\hat{Y}_6(0)$

Robins 1986
Hahn 1998

Learn a prediction function

		Outcome under treatment	Outcome under control
People in category 1	Person 1	?	Y_1
	Person 2	Y_2	?
	Person 3	Y_3	?
People in category 2	Person 4	?	Y_4
	Person 5	Y_5	?
	Person 6	?	Y_6

Predict the whole table

		Outcome under treatment	Outcome under control
People in category 1	Person 1	$\hat{Y}_1(1)$	$\hat{Y}_1(0)$
	Person 2	$\hat{Y}_2(1)$	$\hat{Y}_2(0)$
	Person 3	$\hat{Y}_3(1)$	$\hat{Y}_3(0)$
People in category 2	Person 4	$\hat{Y}_4(1)$	$\hat{Y}_4(0)$
	Person 5	$\hat{Y}_5(1)$	$\hat{Y}_5(0)$
	Person 6	$\hat{Y}_6(1)$	$\hat{Y}_6(0)$

Robins 1986
Hahn 1998

Learn a prediction function

		Outcome under treatment	Outcome under control
People in category 1	Person 1	?	Y_1
	Person 2	Y_2	?
	Person 3	Y_3	?
People in category 2	Person 4	?	Y_4
	Person 5	Y_5	?
	Person 6	?	Y_6

Predict the whole table

		Outcome under treatment	Outcome under control
People in category 1	Person 1	$\hat{Y}_1(1)$	$\hat{Y}_1(0)$
	Person 2	$\hat{Y}_2(1)$	$\hat{Y}_2(0)$
	Person 3	$\hat{Y}_3(1)$	$\hat{Y}_3(0)$
People in category 2	Person 4	$\hat{Y}_4(1)$	$\hat{Y}_4(0)$
	Person 5	$\hat{Y}_5(1)$	$\hat{Y}_5(0)$
	Person 6	$\hat{Y}_6(1)$	$\hat{Y}_6(0)$

Problem: Optimization for the wrong task

Learn a prediction function

		Outcome under treatment	Outcome under control
People in category 1	Person 1	?	Y_1
	Person 2	Y_2	?
	Person 3	Y_3	?
People in category 2	Person 4	?	Y_4
	Person 5	Y_5	?
	Person 6	?	Y_6

Predict the whole table

		Outcome under treatment	Outcome under control
People in category 1	Person 1	$\hat{Y}_1(1)$	$\hat{Y}_1(0)$
	Person 2	$\hat{Y}_2(1)$	$\hat{Y}_2(0)$
	Person 3	$\hat{Y}_3(1)$	$\hat{Y}_3(0)$
People in category 2	Person 4	$\hat{Y}_4(1)$	$\hat{Y}_4(0)$
	Person 5	$\hat{Y}_5(1)$	$\hat{Y}_5(0)$
	Person 6	$\hat{Y}_6(1)$	$\hat{Y}_6(0)$

Problem: Optimization for the wrong task

Prediction error over
observed
cases

Learn a prediction function

		Outcome under treatment	Outcome under control
People in category 1	Person 1	?	Y_1
	Person 2	Y_2	?
	Person 3	Y_3	?
People in category 2	Person 4	?	Y_4
	Person 5	Y_5	?
	Person 6	?	Y_6

Predict the whole table

		Outcome under treatment	Outcome under control
People in category 1	Person 1	$\hat{Y}_1(1)$	$\hat{Y}_1(0)$
	Person 2	$\hat{Y}_2(1)$	$\hat{Y}_2(0)$
	Person 3	$\hat{Y}_3(1)$	$\hat{Y}_3(0)$
People in category 2	Person 4	$\hat{Y}_4(1)$	$\hat{Y}_4(0)$
	Person 5	$\hat{Y}_5(1)$	$\hat{Y}_5(0)$
	Person 6	$\hat{Y}_6(1)$	$\hat{Y}_6(0)$

Problem: Optimization for the wrong task

Prediction error over **observed** cases

vs

Prediction error over **all** cases

Solution: Reweight errors to approximate the correct task

Solution: Reweight errors to approximate the correct task

		Prediction under treatment
People in category 1	Person 1	$\hat{Y}_1(1)$
	Person 2	$\hat{Y}_2(1)$
	Person 3	$\hat{Y}_3(1)$
People in category 2	Person 4	$\hat{Y}_4(1)$
	Person 5	$\hat{Y}_5(1)$
	Person 6	$\hat{Y}_6(1)$

Solution: Reweight errors to approximate the correct task

		Prediction under treatment	Outcome under treatment
People in category 1	Person 1	$\hat{Y}_1(1)$	?
	Person 2	$\hat{Y}_2(1)$	Y_2
	Person 3	$\hat{Y}_3(1)$	Y_3
People in category 2	Person 4	$\hat{Y}_4(1)$	?
	Person 5	$\hat{Y}_5(1)$	Y_5
	Person 6	$\hat{Y}_6(1)$	?

Solution: Reweight errors to approximate the correct task

		Prediction under treatment	Outcome under treatment	Error
People in category 1	Person 1	$\hat{Y}_1(1)$	?	?
	Person 2	$\hat{Y}_2(1)$	Y_2	$\hat{Y}_2(1) - Y_2$
	Person 3	$\hat{Y}_3(1)$	Y_3	$\hat{Y}_3(1) - Y_3$
People in category 2	Person 4	$\hat{Y}_4(1)$	?	?
	Person 5	$\hat{Y}_5(1)$	Y_5	$\hat{Y}_5(1) - Y_5$
	Person 6	$\hat{Y}_6(1)$	?	?

Solution: Reweight errors to approximate the correct task

		Prediction under treatment	Outcome under treatment	Error	Weight on error
People in category 1	Person 1	$\hat{Y}_1(1)$	?	?	
	Person 2	$\hat{Y}_2(1)$	Y_2	$\hat{Y}_2(1) - Y_2$	3 / 2
	Person 3	$\hat{Y}_3(1)$	Y_3	$\hat{Y}_3(1) - Y_3$	3 / 2
People in category 2	Person 4	$\hat{Y}_4(1)$	?	?	
	Person 5	$\hat{Y}_5(1)$	Y_5	$\hat{Y}_5(1) - Y_5$	3
	Person 6	$\hat{Y}_6(1)$	?	?	

Solution: Reweight errors to approximate the correct task

Estimated bias: Mean($\hat{Y}_i - Y_i$) with
inverse probability of treatment weights

		Prediction under treatment	Outcome under treatment	Error	Weight on error
People in category 1	Person 1	$\hat{Y}_1(1)$	?	?	
	Person 2	$\hat{Y}_2(1)$	Y_2	$\hat{Y}_2(1) - Y_2$	3 / 2
	Person 3	$\hat{Y}_3(1)$	Y_3	$\hat{Y}_3(1) - Y_3$	3 / 2
People in category 2	Person 4	$\hat{Y}_4(1)$	?	?	
	Person 5	$\hat{Y}_5(1)$	Y_5	$\hat{Y}_5(1) - Y_5$	3
	Person 6	$\hat{Y}_6(1)$	?	?	

Solution: Reweight errors to approximate the correct task

Estimated bias:	Mean($\hat{Y}_i - Y_i$) with inverse probability of treatment weights
New Estimate:	(Original Estimate) − (Estimated Bias)

Solution: Reweight errors to approximate the correct task

Estimated bias: Mean($\hat{Y}_i - Y_i$) with
inverse probability of treatment weights

New Estimate: (Original Estimate) − (Estimated Bias)

Doubly Robust Estimation

Robins, Rotnitzky, & Zhao 1994
Bang & Robins 2005

Solution: Reweight errors to approximate the correct task

Estimated bias: Mean($\hat{Y}_i - Y_i$) with
inverse probability of treatment weights

New Estimate: (Original Estimate) − (Estimated Bias)

Doubly Robust Estimation



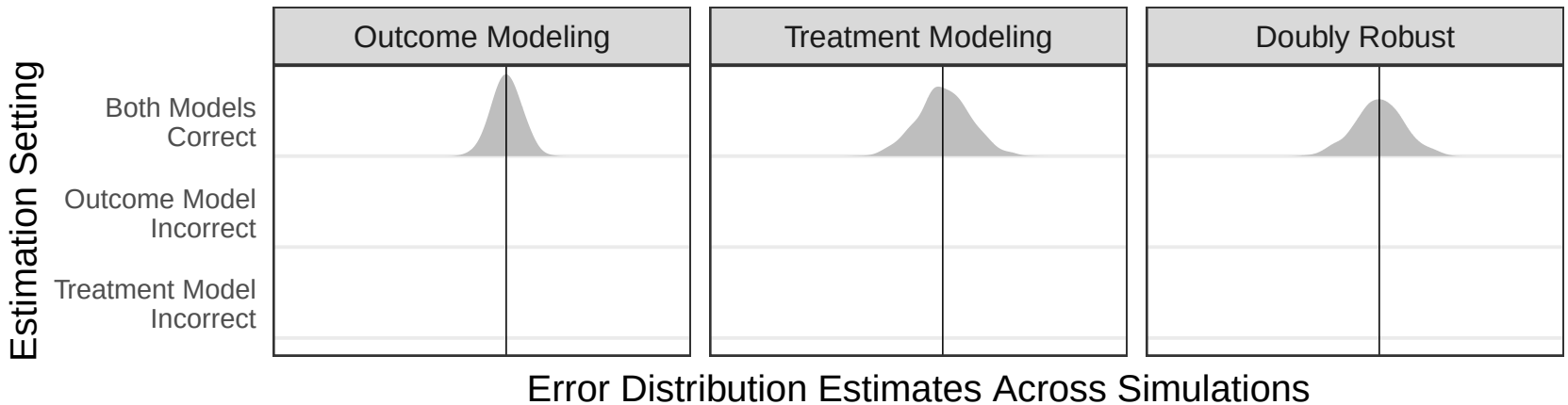
Robins, Rotnitzky, & Zhao 1994
Bang & Robins 2005

Solution: Reweight errors to approximate the correct task

Estimated bias: Mean($\hat{Y}_i - Y_i$) with
inverse probability of treatment weights

New Estimate: (Original Estimate) – (Estimated Bias)

Doubly Robust Estimation



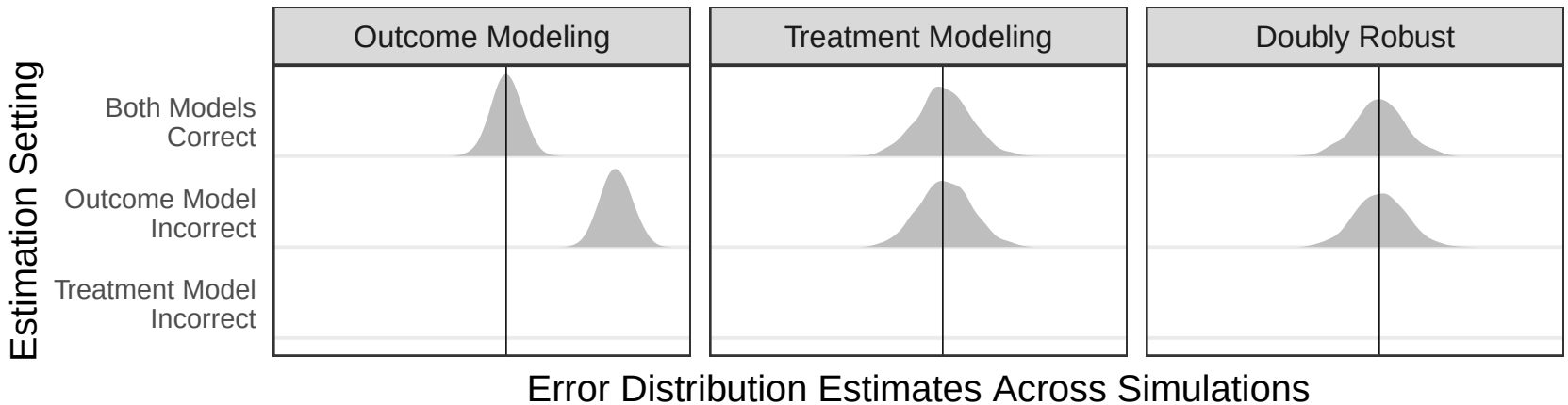
Robins, Rotnitzky, & Zhao 1994
Bang & Robins 2005

Solution: Reweight errors to approximate the correct task

Estimated bias: Mean($\hat{Y}_i - Y_i$) with
inverse probability of treatment weights

New Estimate: (Original Estimate) – (Estimated Bias)

Doubly Robust Estimation



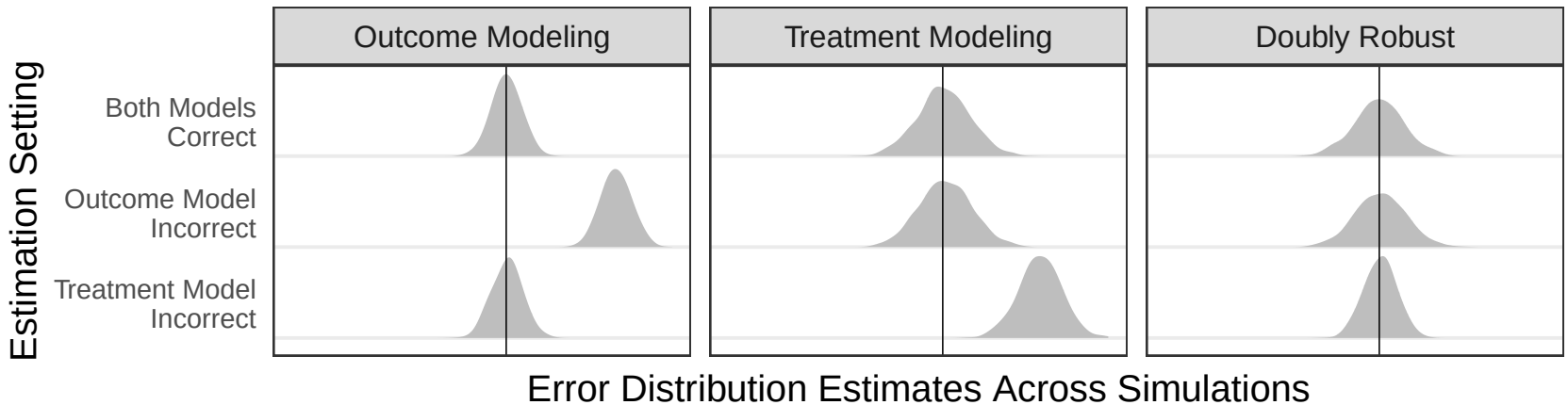
Robins, Rotnitzky, & Zhao 1994
Bang & Robins 2005

Solution: Reweight errors to approximate the correct task

Estimated bias: Mean($\hat{Y}_i - Y_i$) with
inverse probability of treatment weights

New Estimate: (Original Estimate) – (Estimated Bias)

Doubly Robust Estimation



Robins, Rotnitzky, & Zhao 1994
Bang & Robins 2005

Solution: Reweight errors to approximate the correct task

Estimated bias: Mean($\hat{Y}_i - Y_i$) with
inverse probability of treatment weights

New Estimate: (Original Estimate) − (Estimated Bias)

Doubly Robust Estimation

Even better:

Chernozhukov et al. 2018
Bickel 1982

Solution: Reweight errors to approximate the correct task

Estimated bias: Mean($\hat{Y}_i - Y_i$) with
inverse probability of treatment weights

New Estimate: (Original Estimate) − (Estimated Bias)

Doubly Robust Estimation

Even better: — Learn $\hat{Y}_i$ in sample A

— Estimate bias in sample B

Chernozhukov et al. 2018
Bickel 1982

Solution: Reweight errors to approximate the correct task

Estimated bias: Mean($\hat{Y}_i - Y_i$) with
inverse probability of treatment weights

New Estimate: (Original Estimate) − (Estimated Bias)

Doubly Robust Estimation

Even better:
— Learn $\hat{Y}_i$ in sample A
— Estimate bias in sample B
— Cross fit: Swap roles and average

Chernozhukov et al. 2018
Bickel 1982

Solution: Reweight errors to approximate the correct task

Estimated bias: Mean($\hat{Y}_i - Y_i$) with
inverse probability of treatment weights

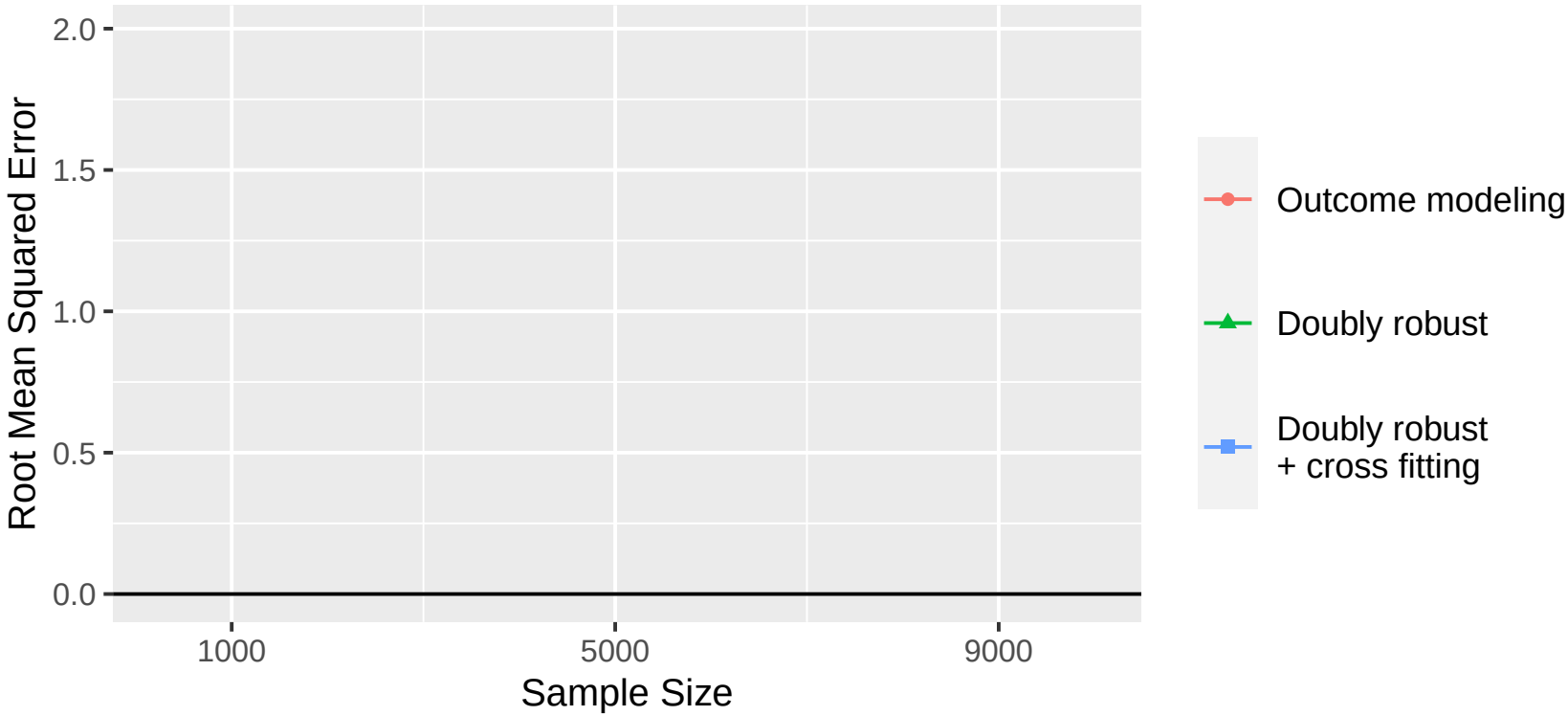
New Estimate: (Original Estimate) − (Estimated Bias)

Doubly Robust Estimation

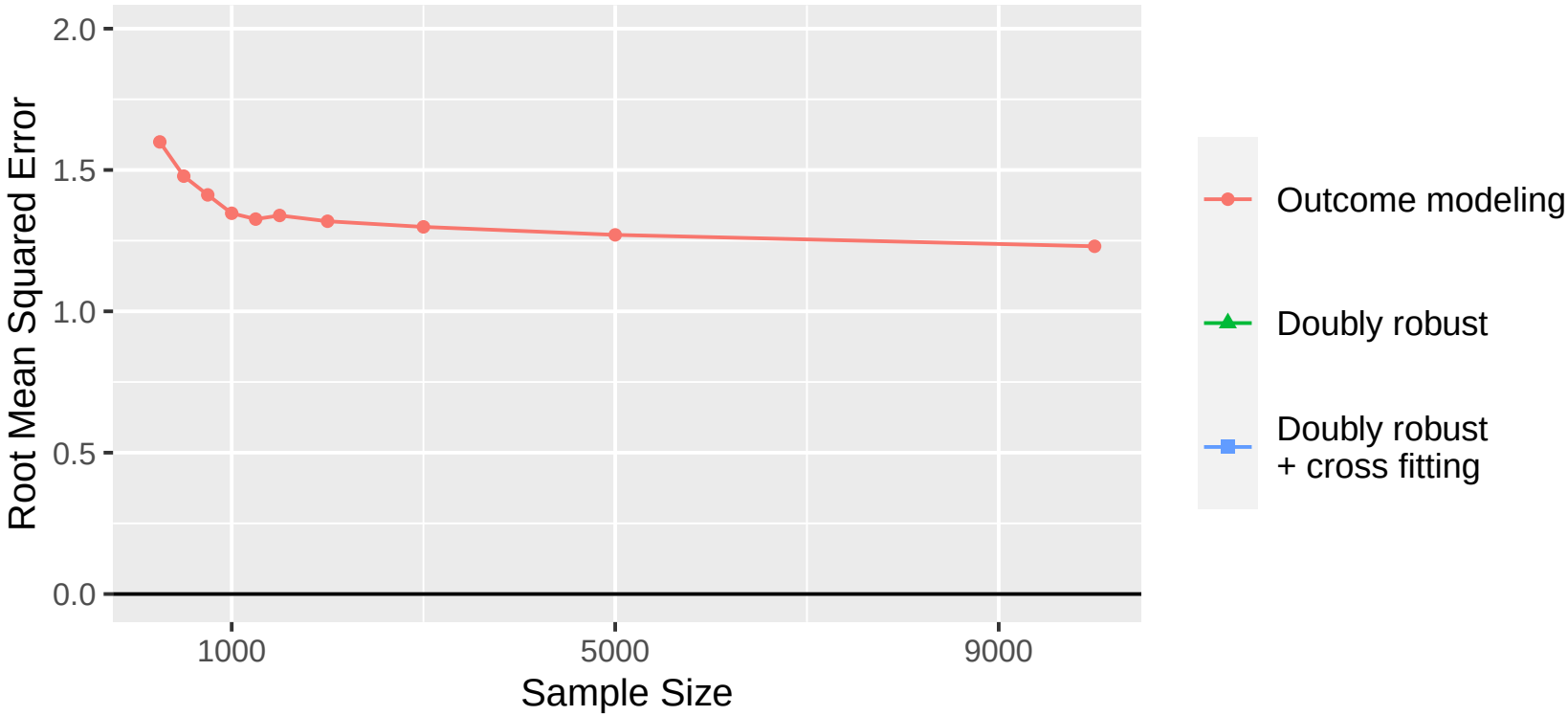
Even better:
— Learn $\hat{Y}_i$ in sample A
— Estimate bias in sample B
— Cross fit: Swap roles and average

Chernozhukov et al. 2018
Bickel 1982

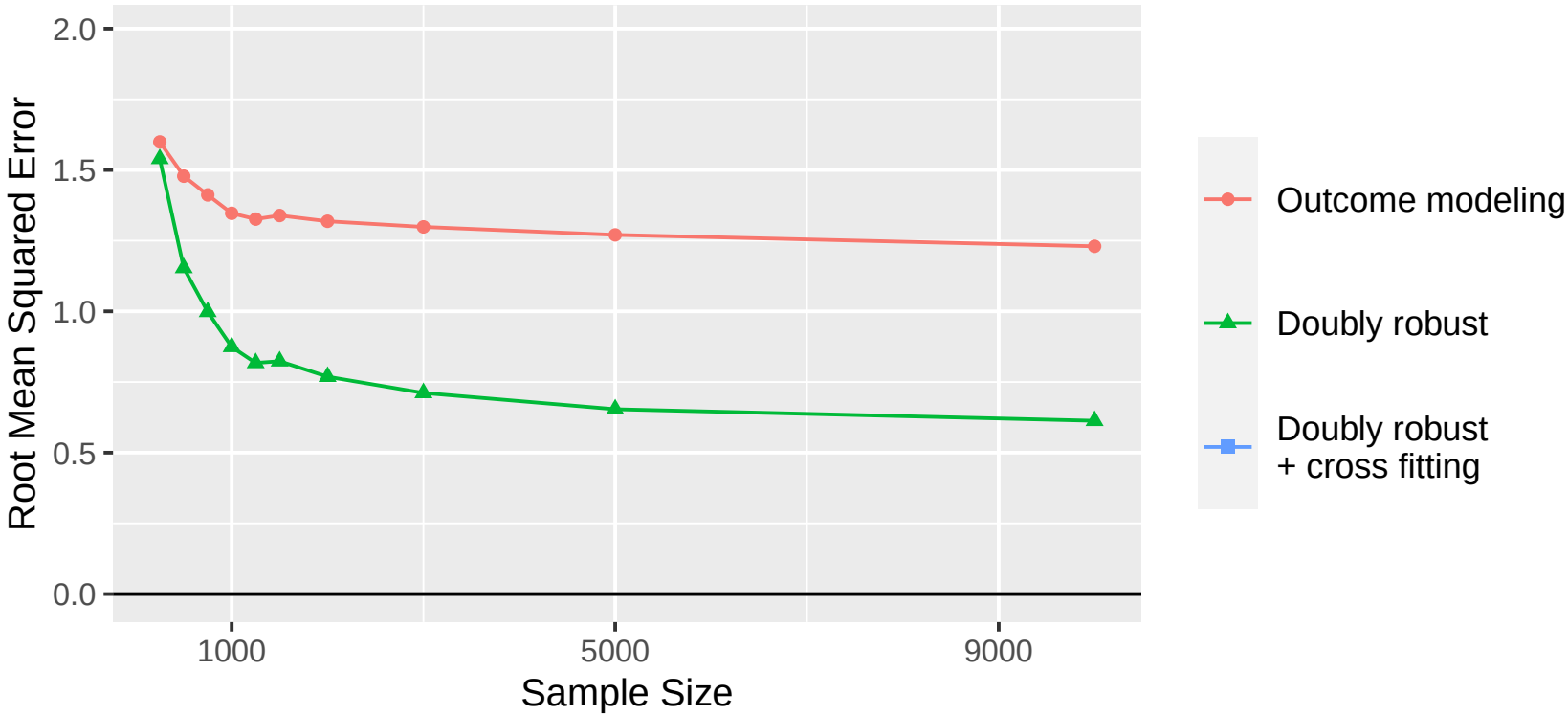
Solution: Reweight errors to approximate the correct task



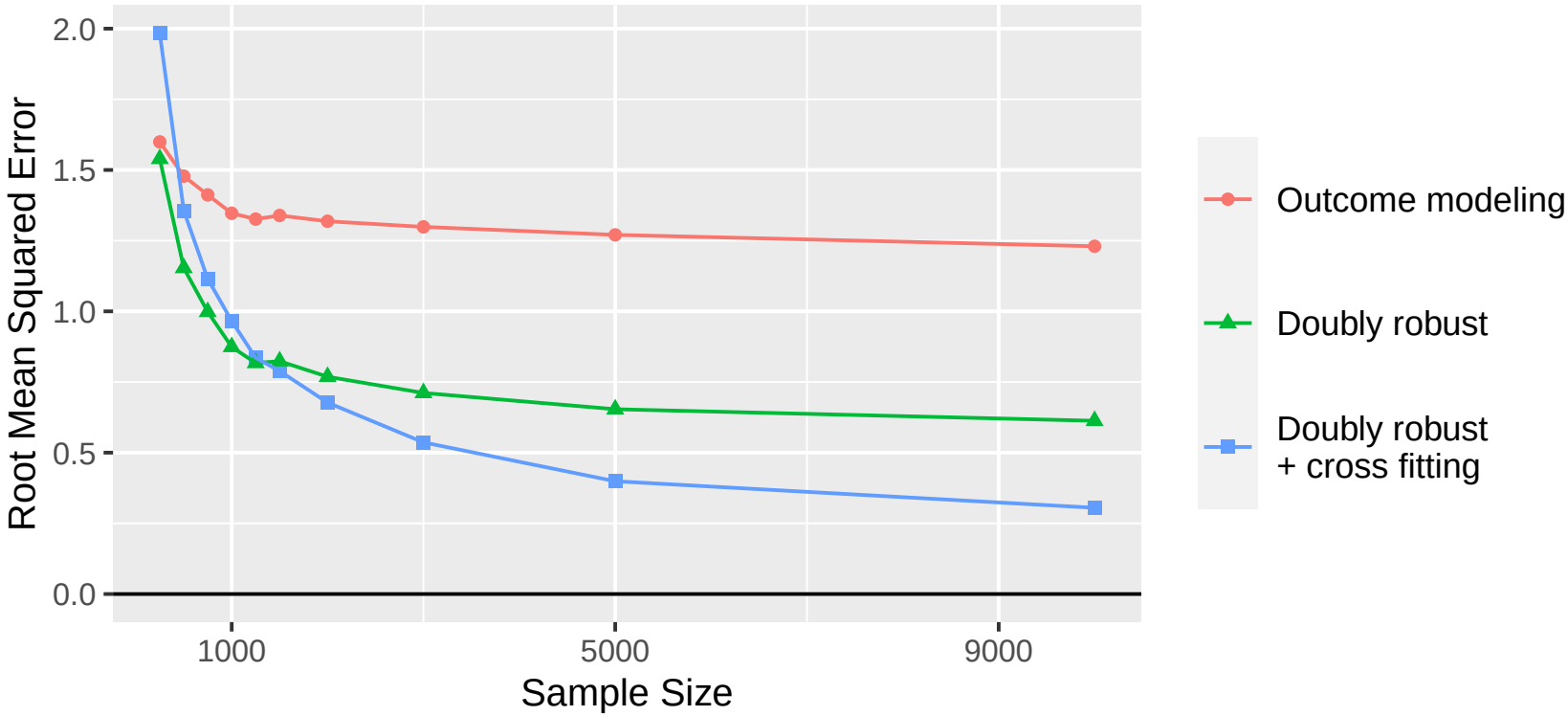
Solution: Reweight errors to approximate the correct task



Solution: Reweight errors to approximate the correct task



Solution: Reweight errors to approximate the correct task



Solution: Reweight errors to approximate the correct task

Estimated bias: Mean($\hat{Y}_i - Y_i$) with
inverse probability of treatment weights

New Estimate: (Original Estimate) − (Estimated Bias)

Doubly Robust Estimation

Even better:
— Learn $\hat{Y}_i$ in sample A
— Estimate bias in sample B
— Cross fit: Swap roles and average

Double Machine Learning

Chernozhukov et al. 2018
Bickel 1982

Solution: Reweight errors to approximate the correct task

Estimated bias: Mean($\hat{Y}_i - Y_i$) with
inverse probability of treatment weights

New Estimate: (Original Estimate) − (Estimated Bias)

Doubly Robust Estimation

Even better:
— Learn $\hat{Y}_i$ in sample A
— Estimate bias in sample B
— Cross fit: Swap roles and average

Double Machine Learning



The Gap-Closing Estimand

A causal approach to study interventions that close disparities across social categories

- Past work: Causal decomposition analysis
- Define the estimand
  - what is the treatment?
  - what is the scope of the intervention?
- Identify the estimand
  - choose a sufficient adjustment set
- Estimate
  - by outcome modeling
  - by weighting
  - by doubly robust estimation
- **Produce software**
- Impact: Others using these ideas

gapclosing

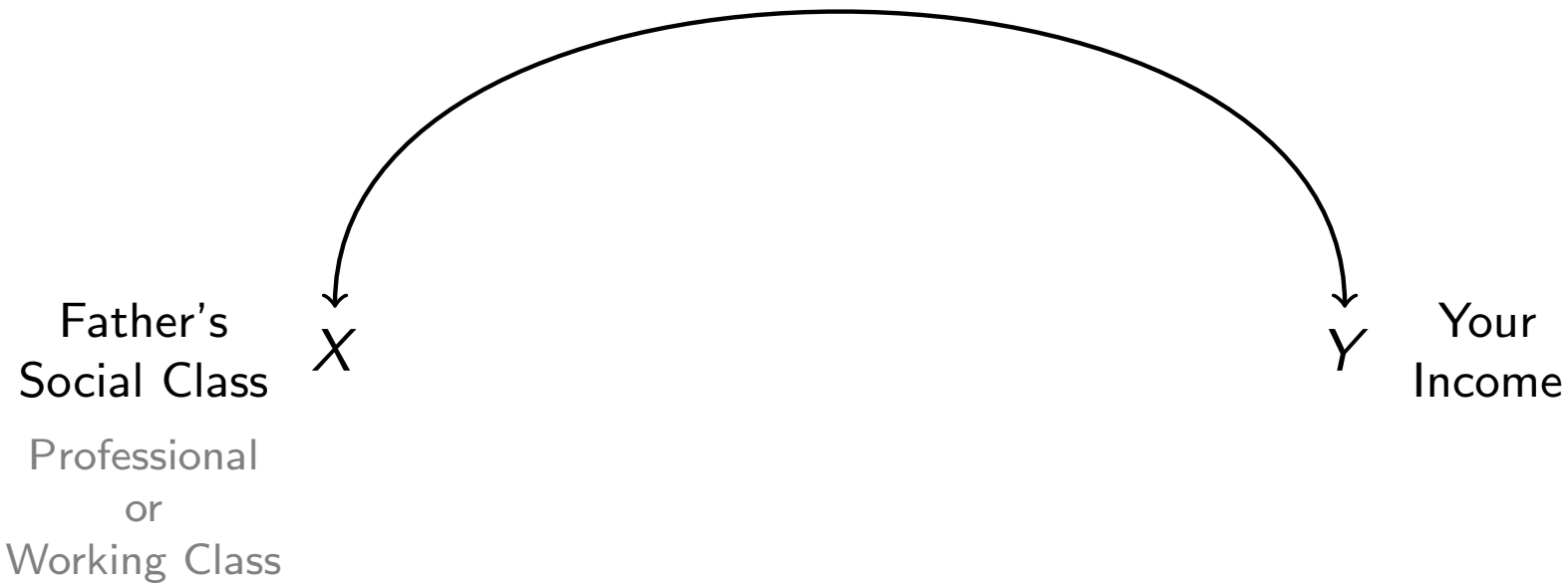
An R package to estimate gap closing estimands. Install this package with the command `devtools::install_github("ilundberg/gapclosing")`.

```
estimate <- gapclosing(
  data = simulated_data,
  outcome_formula = formula(outcome ~ category + confounder),
  treatment_formula = formula(treatment ~ category + confounder),
  category_name = "category",
  counterfactual_assignments = 1,
  outcome_algorithm = "ranger",
  treatment_algorithm = "ranger",
  sample_split = "cross_fit",
  se = T
)
```

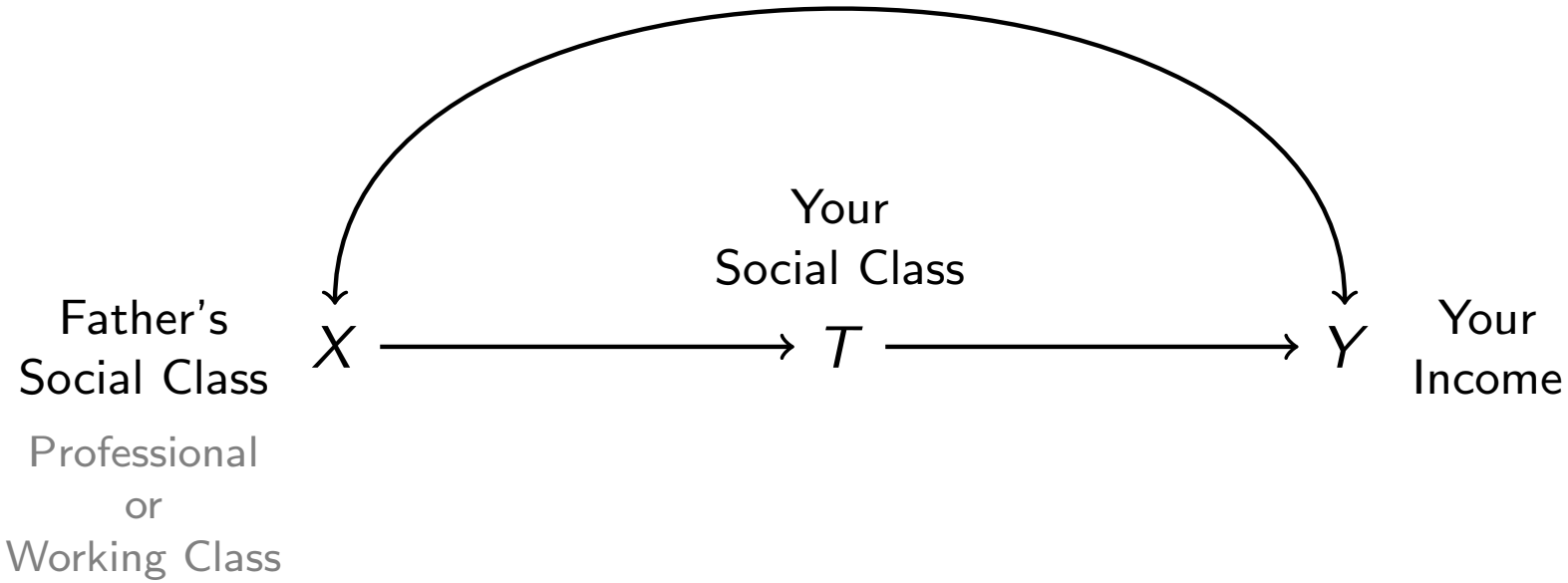
description	estimate	se	ci.min	ci.max
Factual gap	2.14	0.40	1.36	2.9
Counterfactual gap	0.67	0.44	-0.19	1.5

Empirical Examples

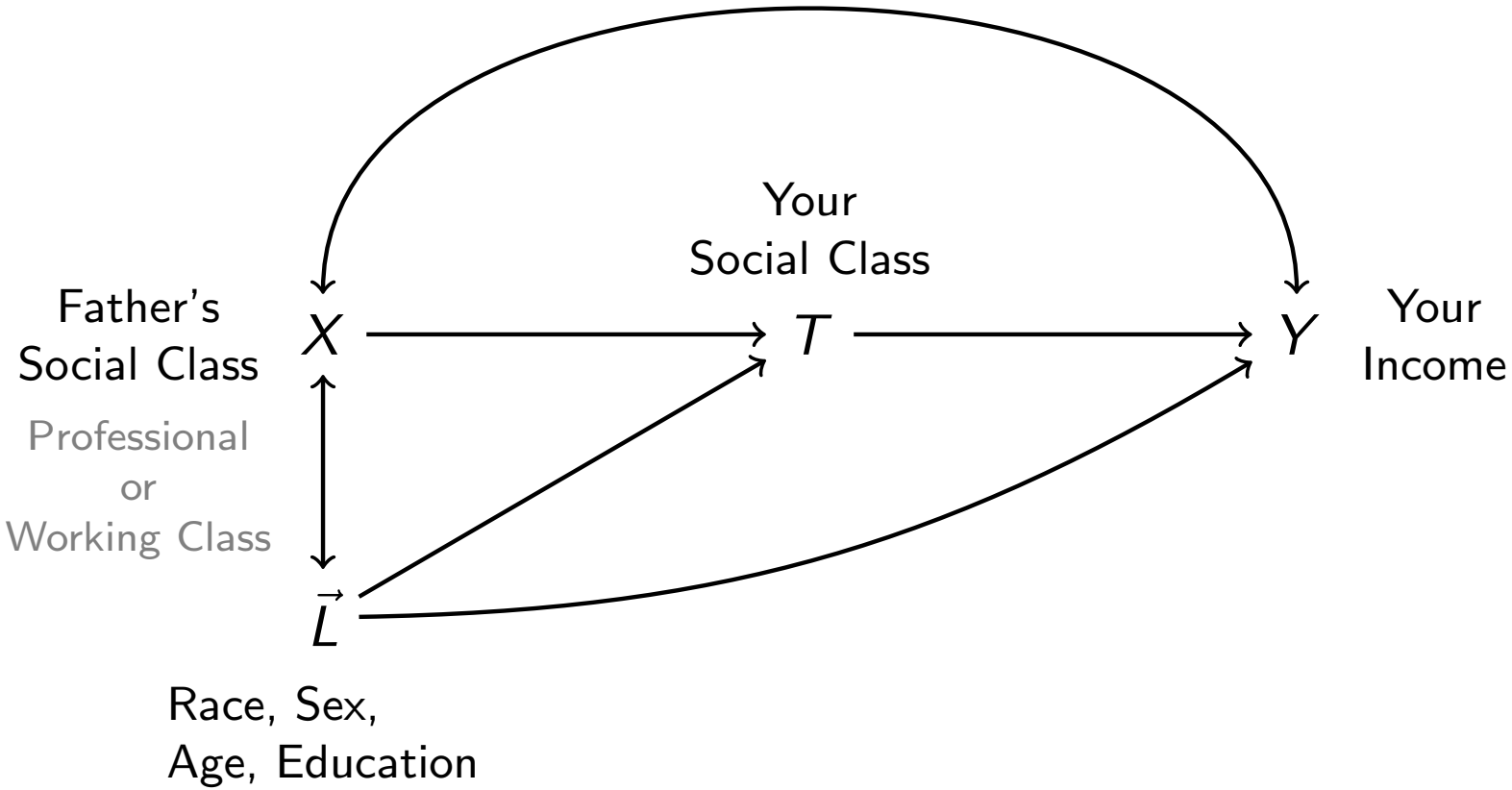
Empirical Example 1: Economic Mobility



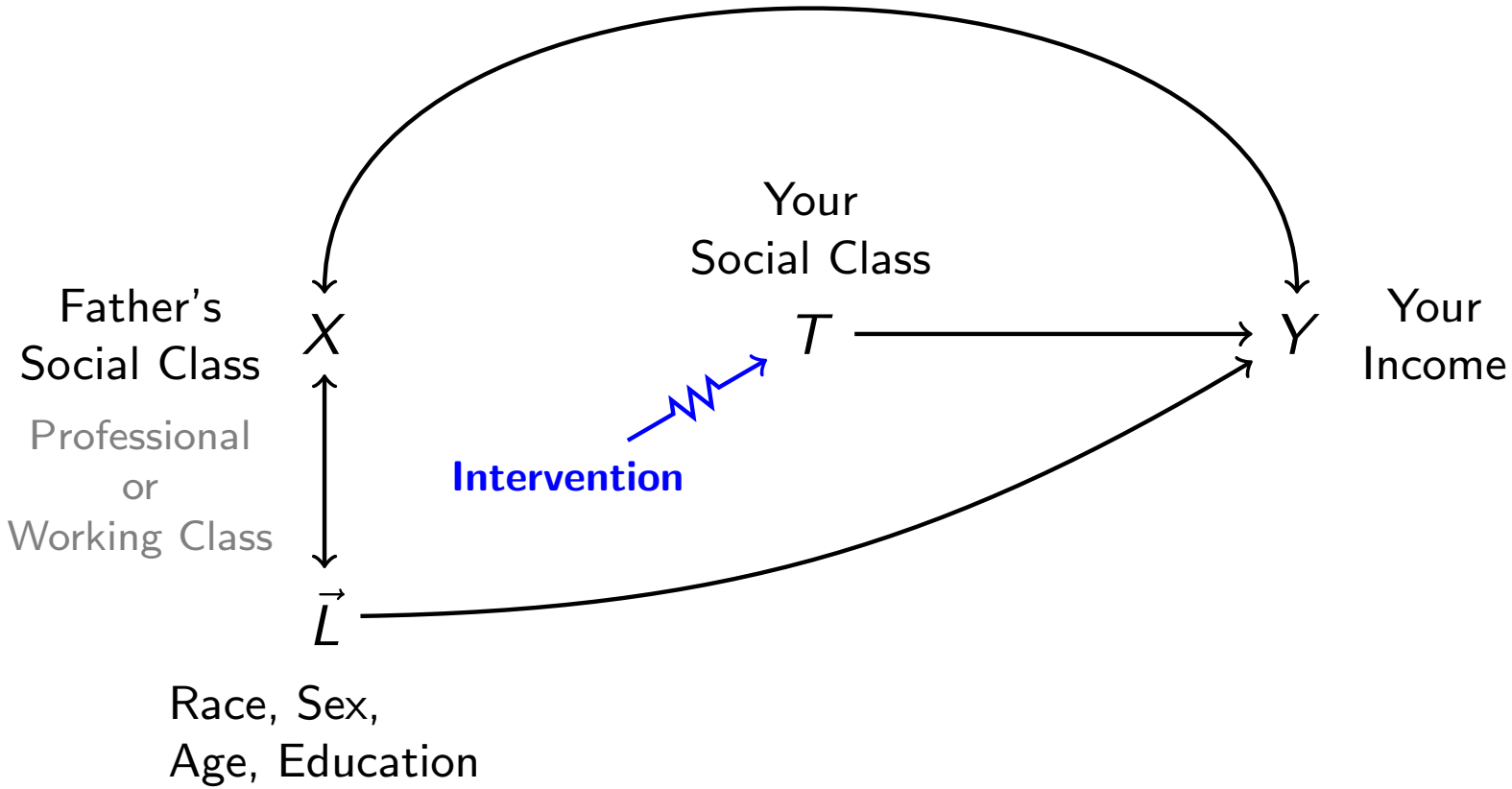
Empirical Example 1: Economic Mobility



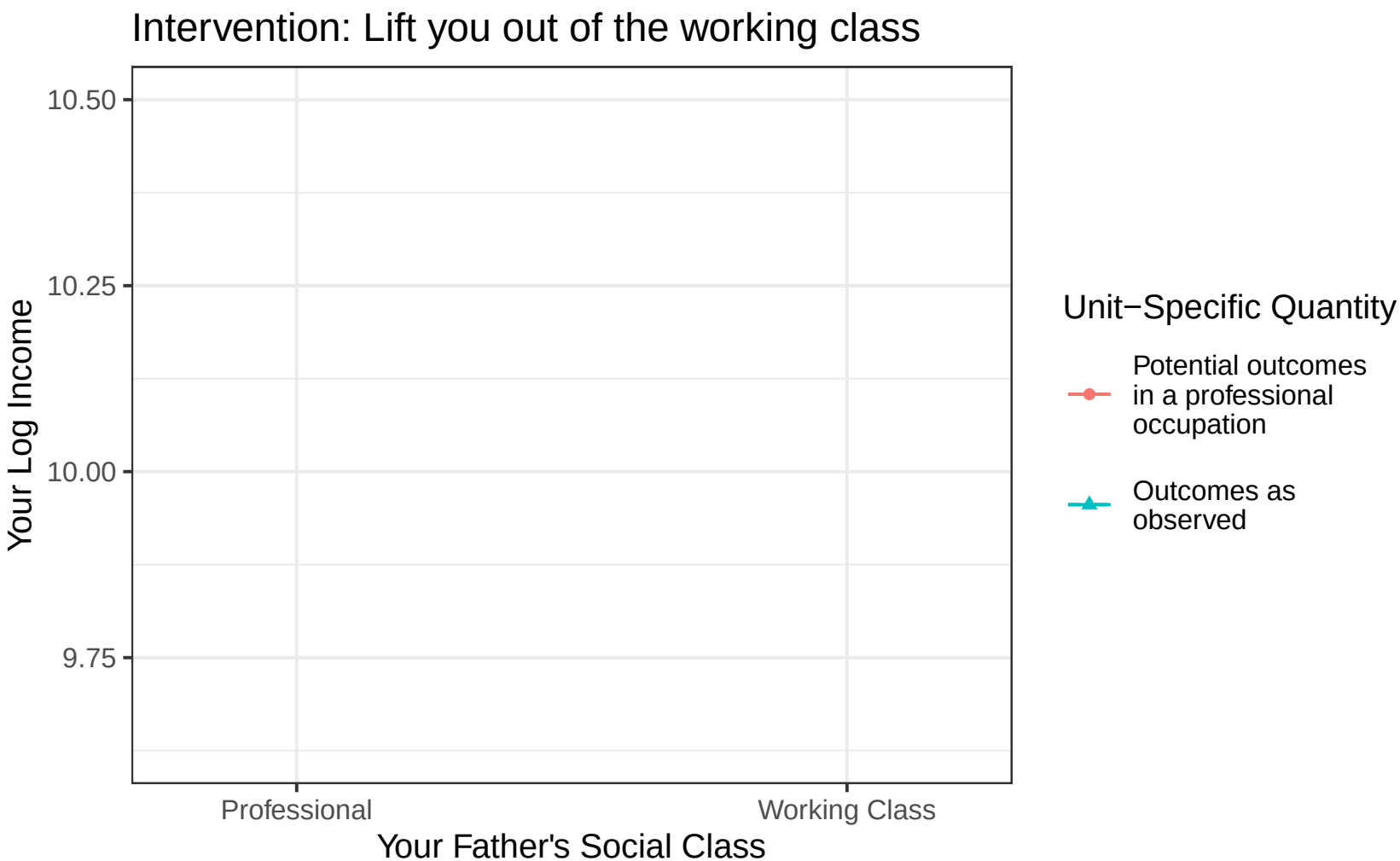
Empirical Example 1: Economic Mobility



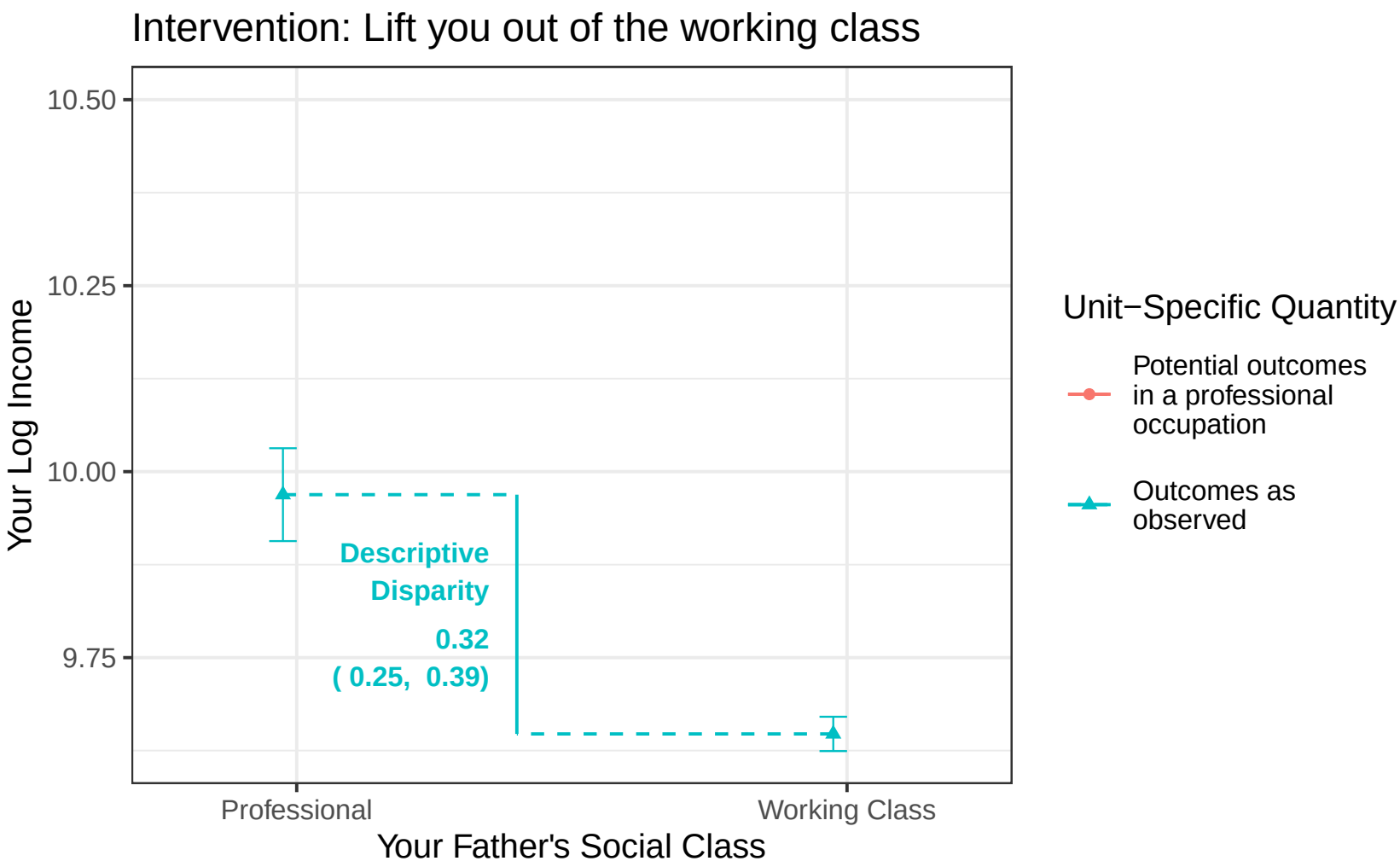
Empirical Example 1: Economic Mobility



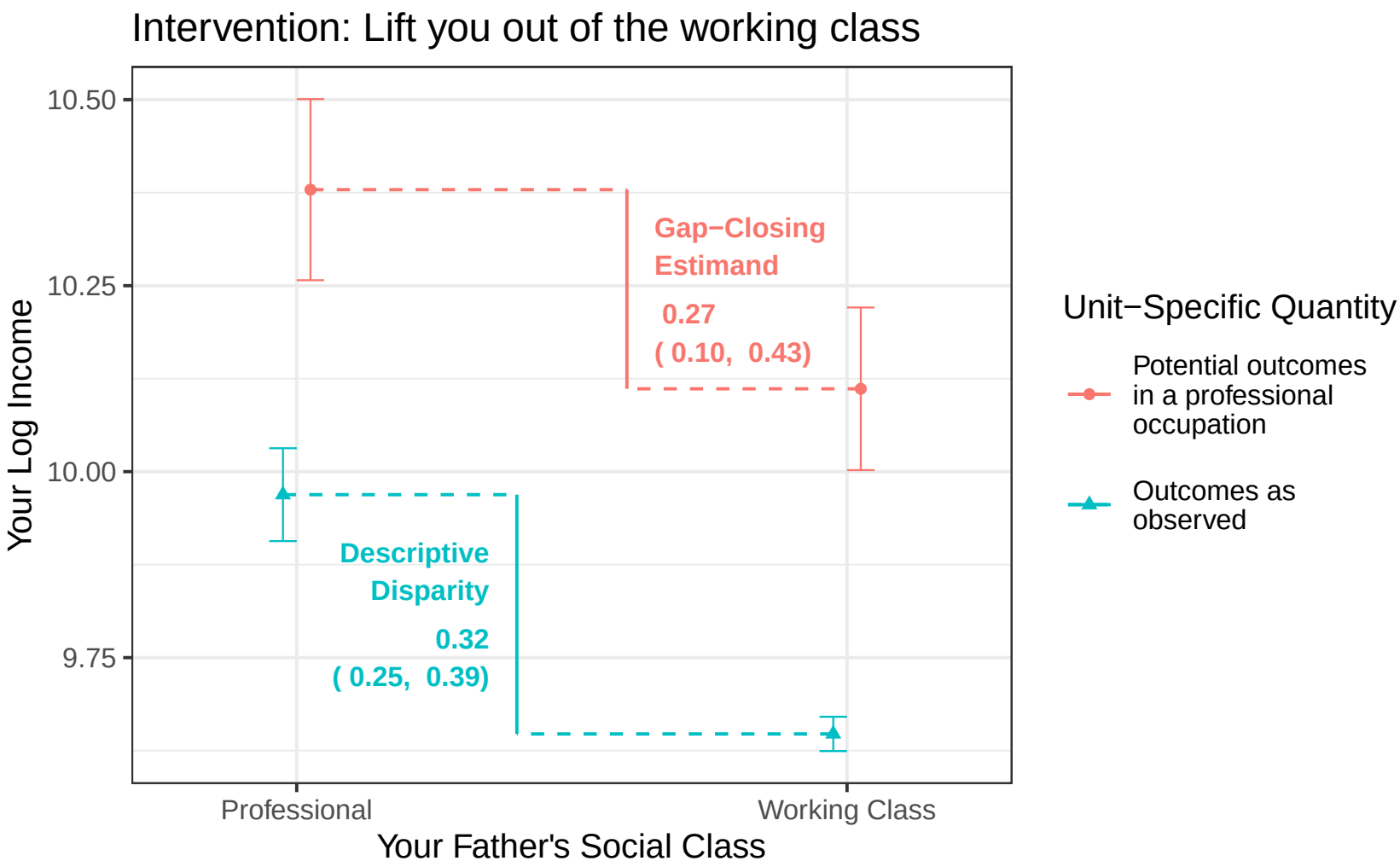
Empirical Example 1: Economic Mobility



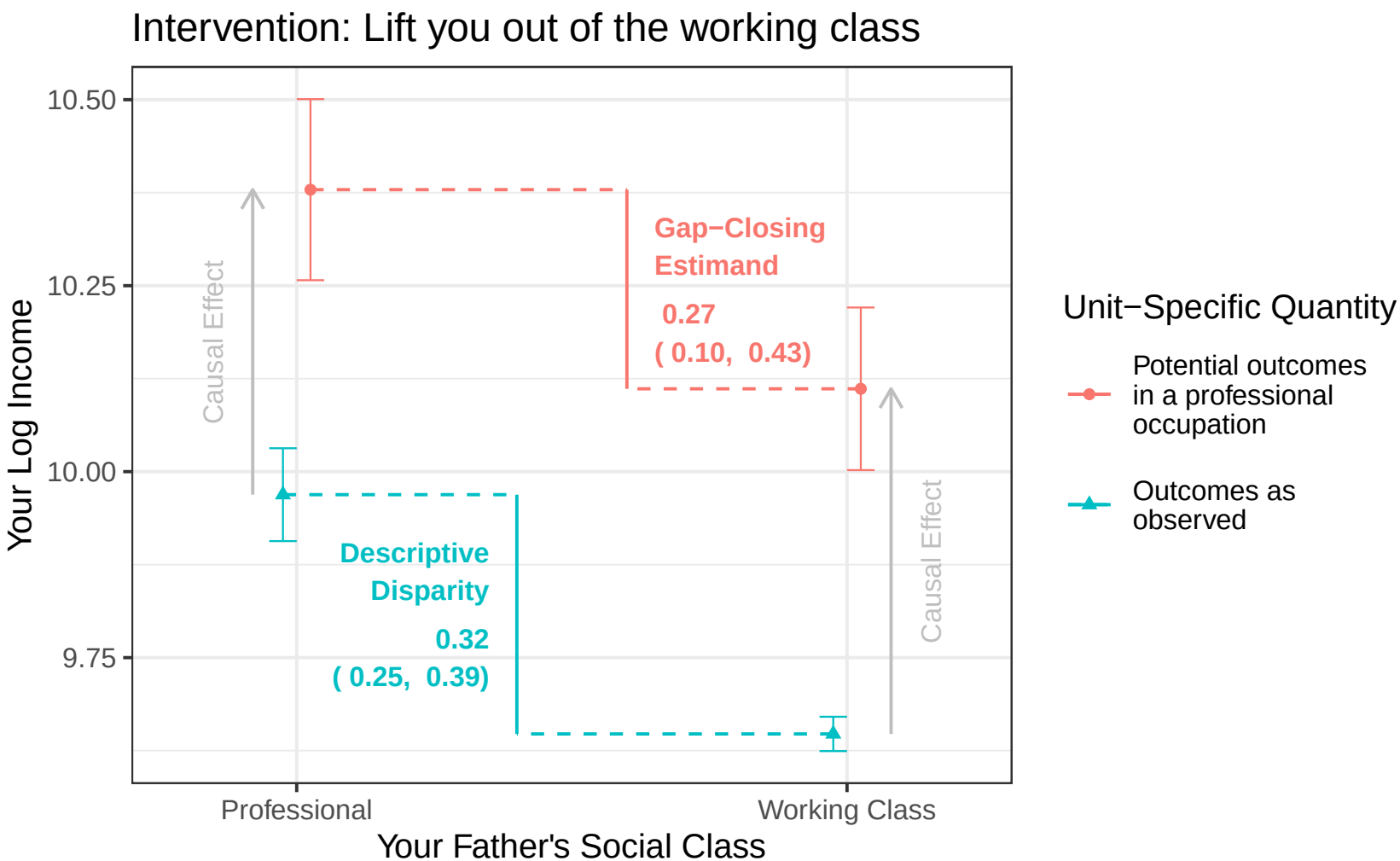
Empirical Example 1: Economic Mobility



Empirical Example 1: Economic Mobility

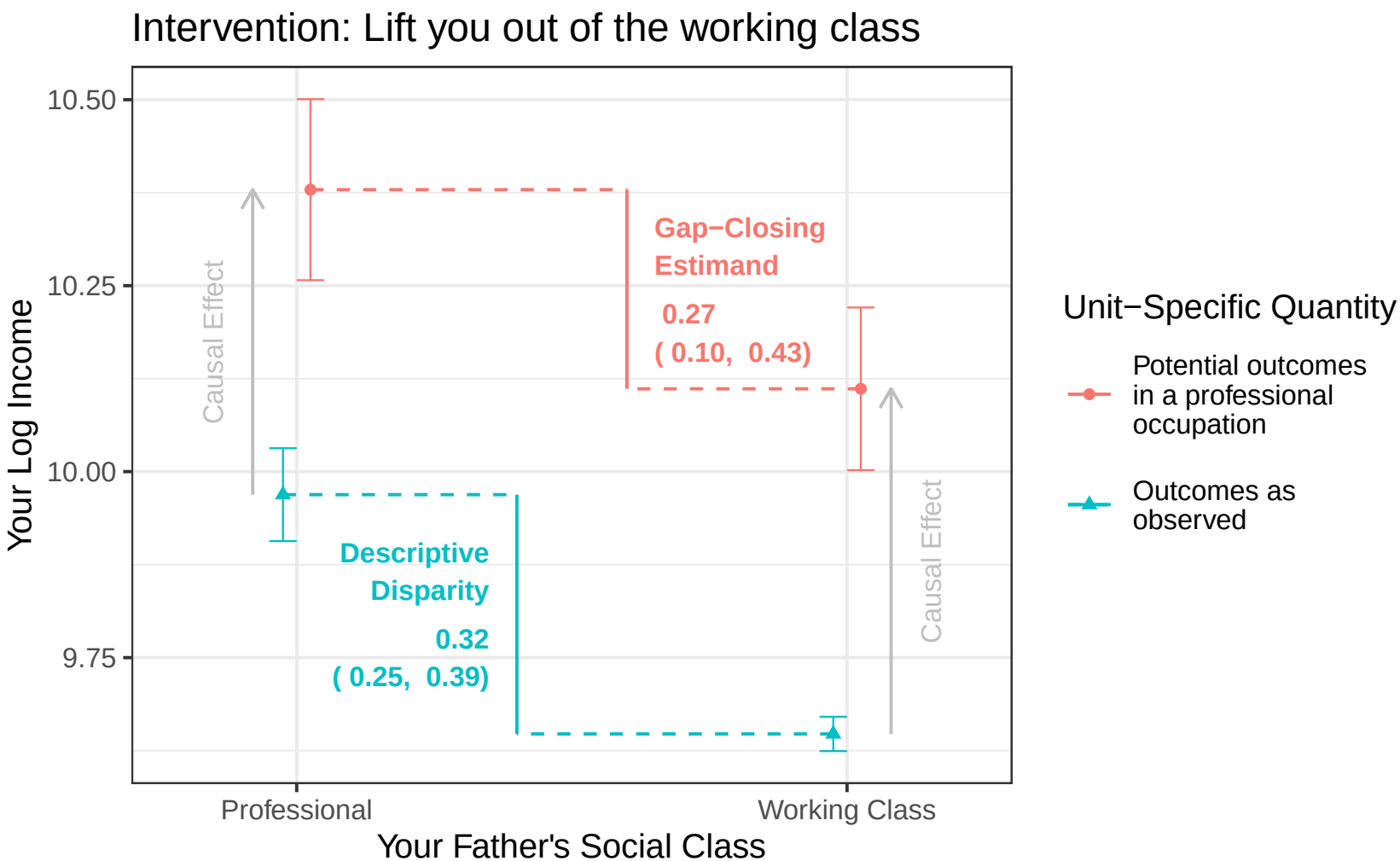


Empirical Example 1: Economic Mobility

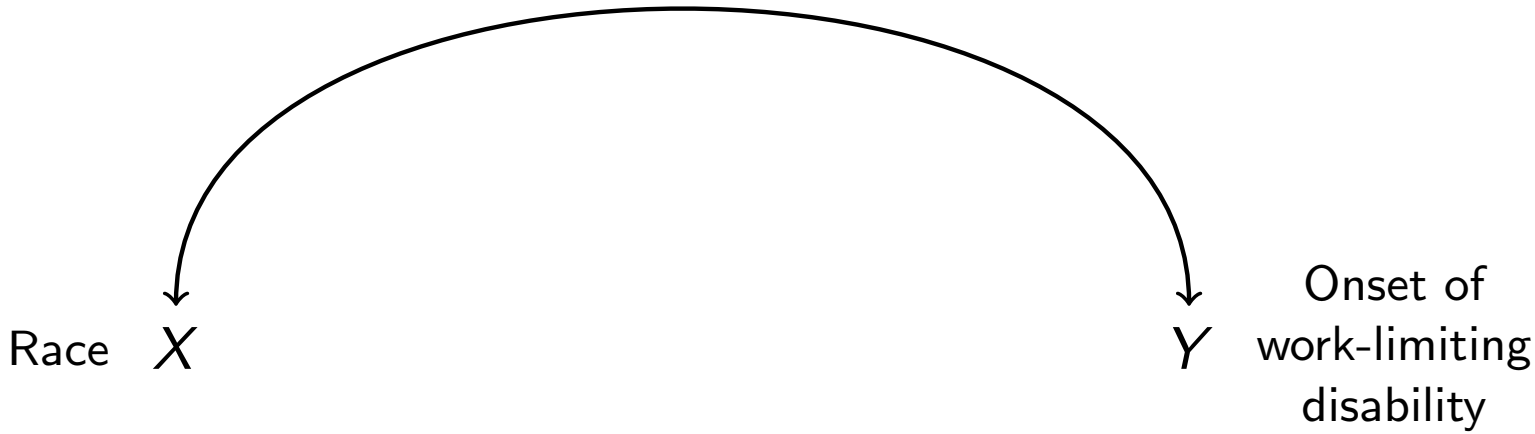


Empirical Example 1: Economic Mobility

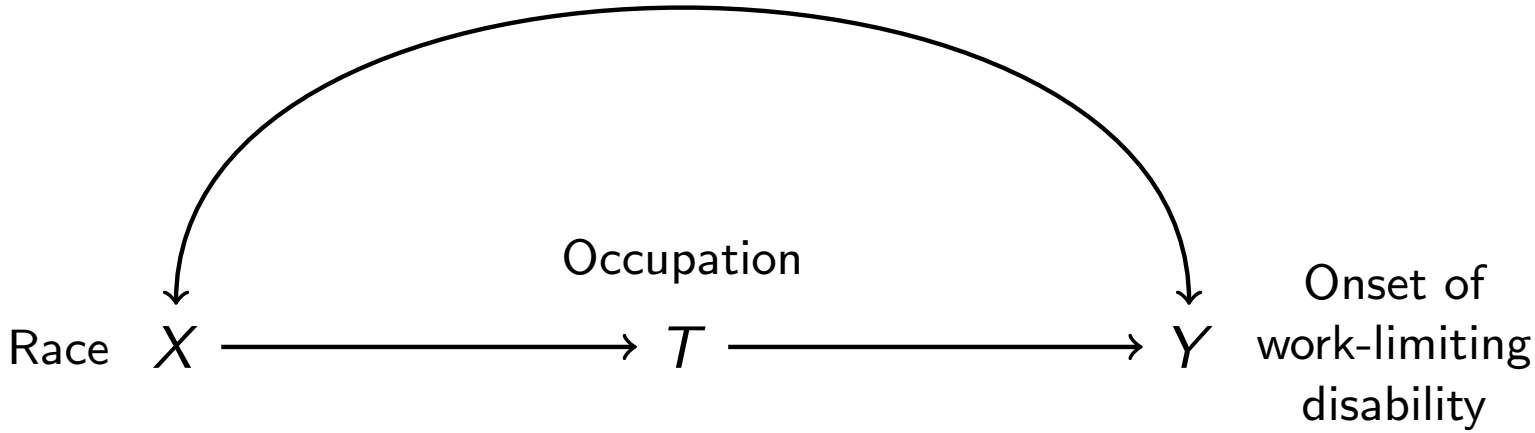
`plot_two_categories()`



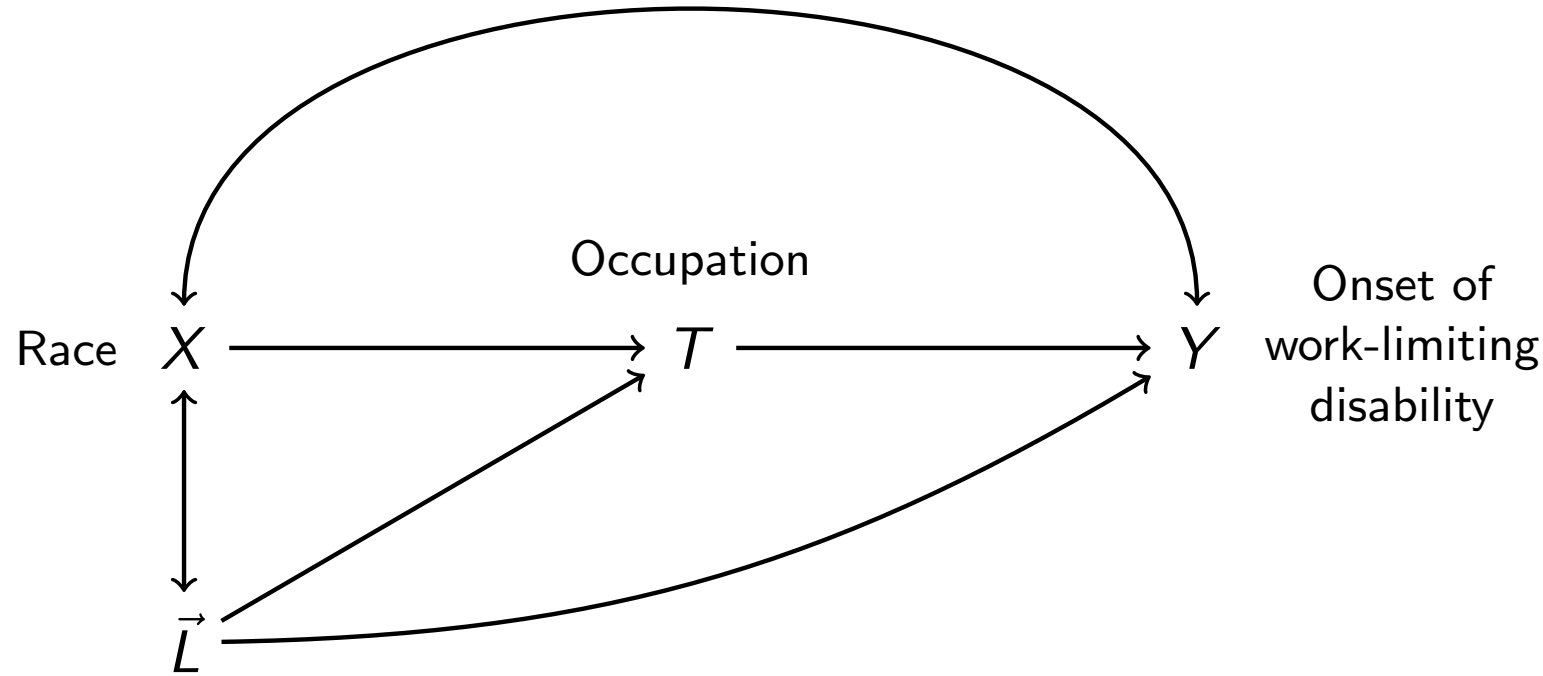
Empirical Example 2: Racial Disparities in Health



Empirical Example 2: Racial Disparities in Health

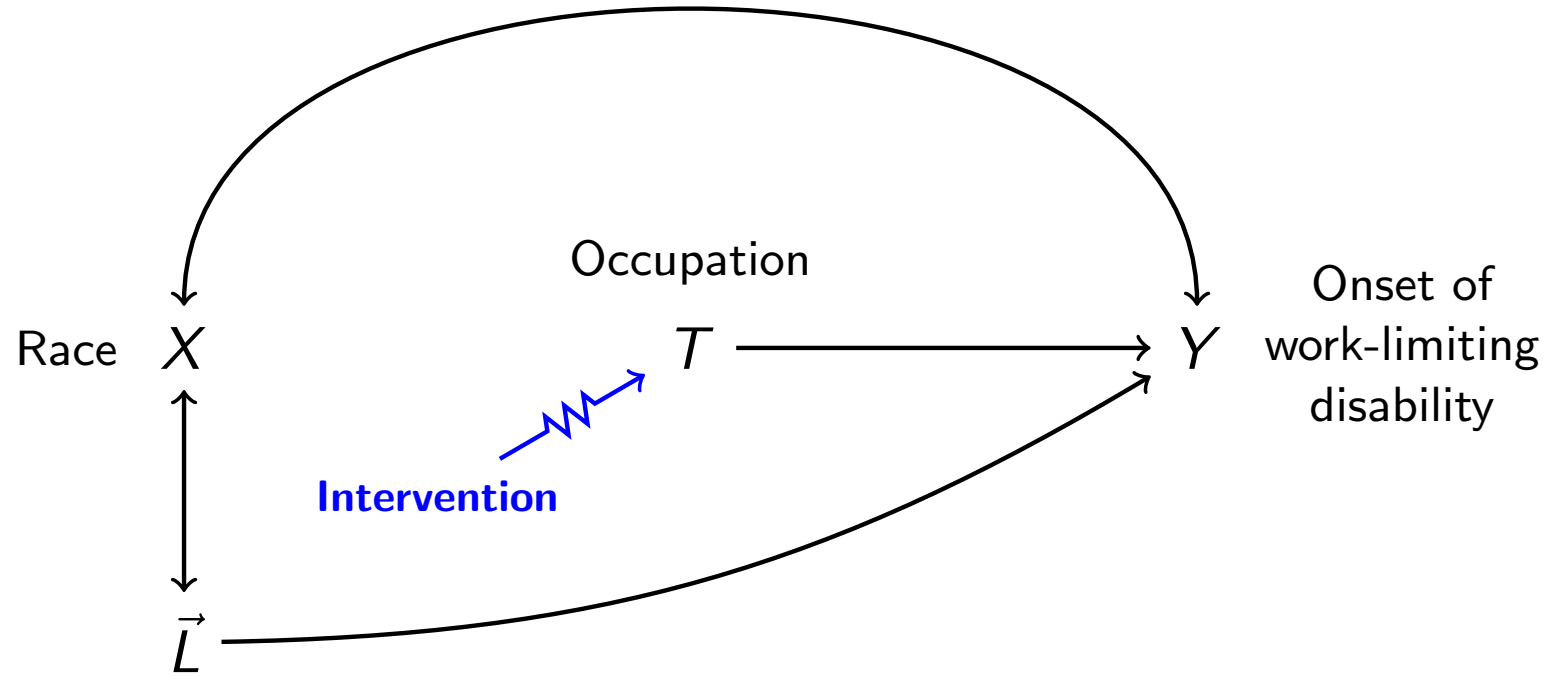


Empirical Example 2: Racial Disparities in Health



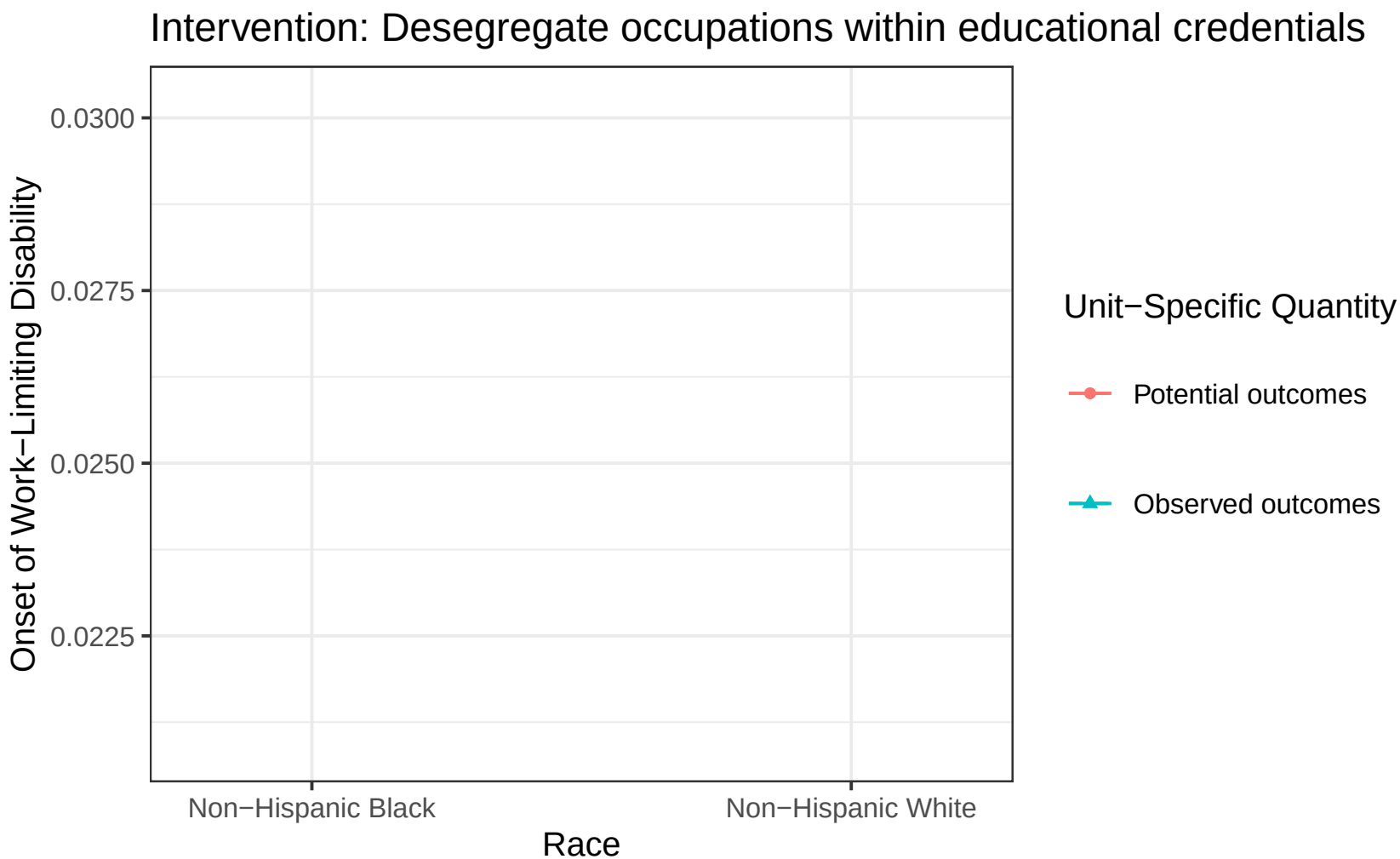
Sex,
Age, Education
Foreign born,
Lagged outcome,
Lagged health

Empirical Example 2: Racial Disparities in Health

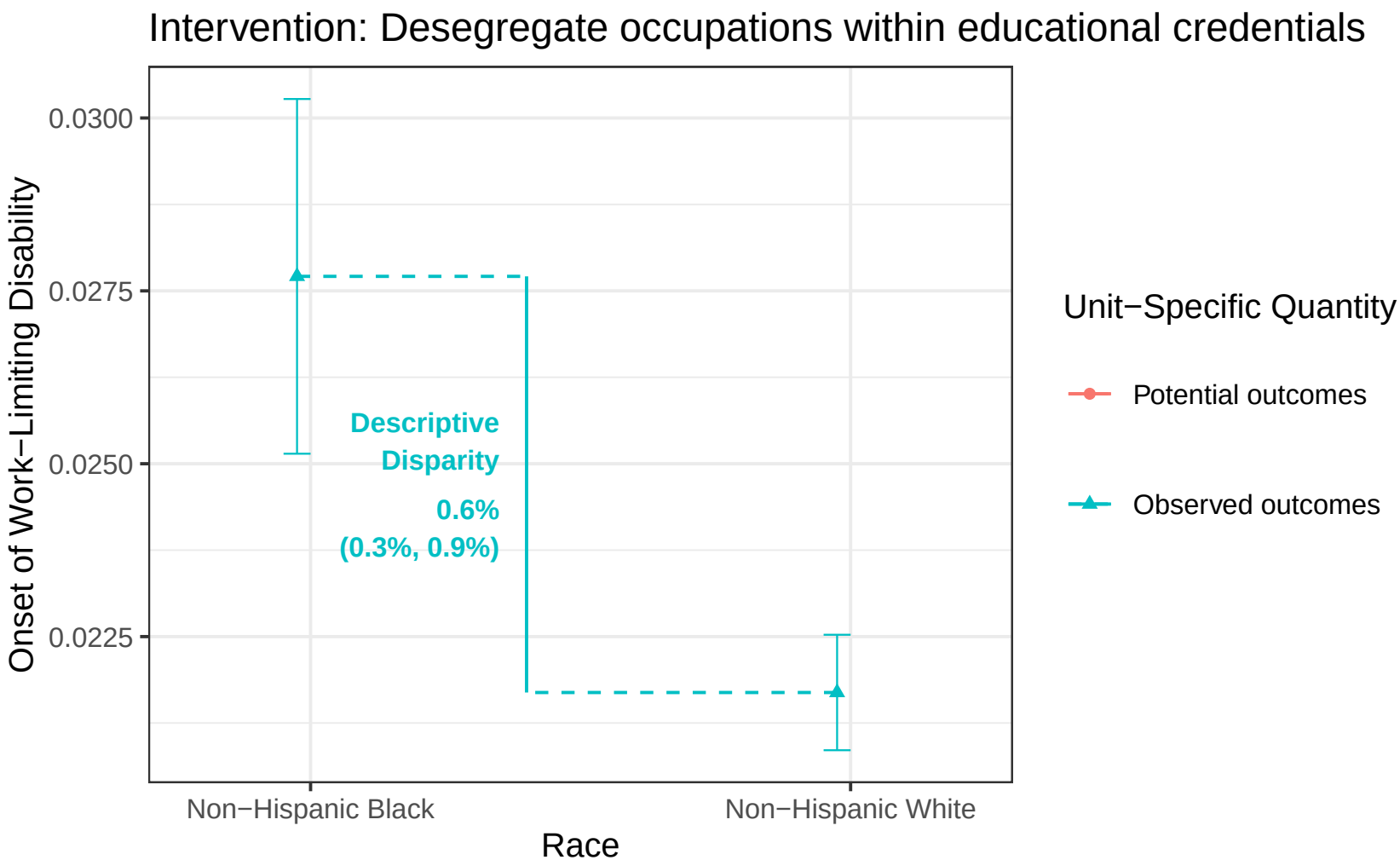


Sex,
Age, Education
Foreign born,
Lagged outcome,
Lagged health

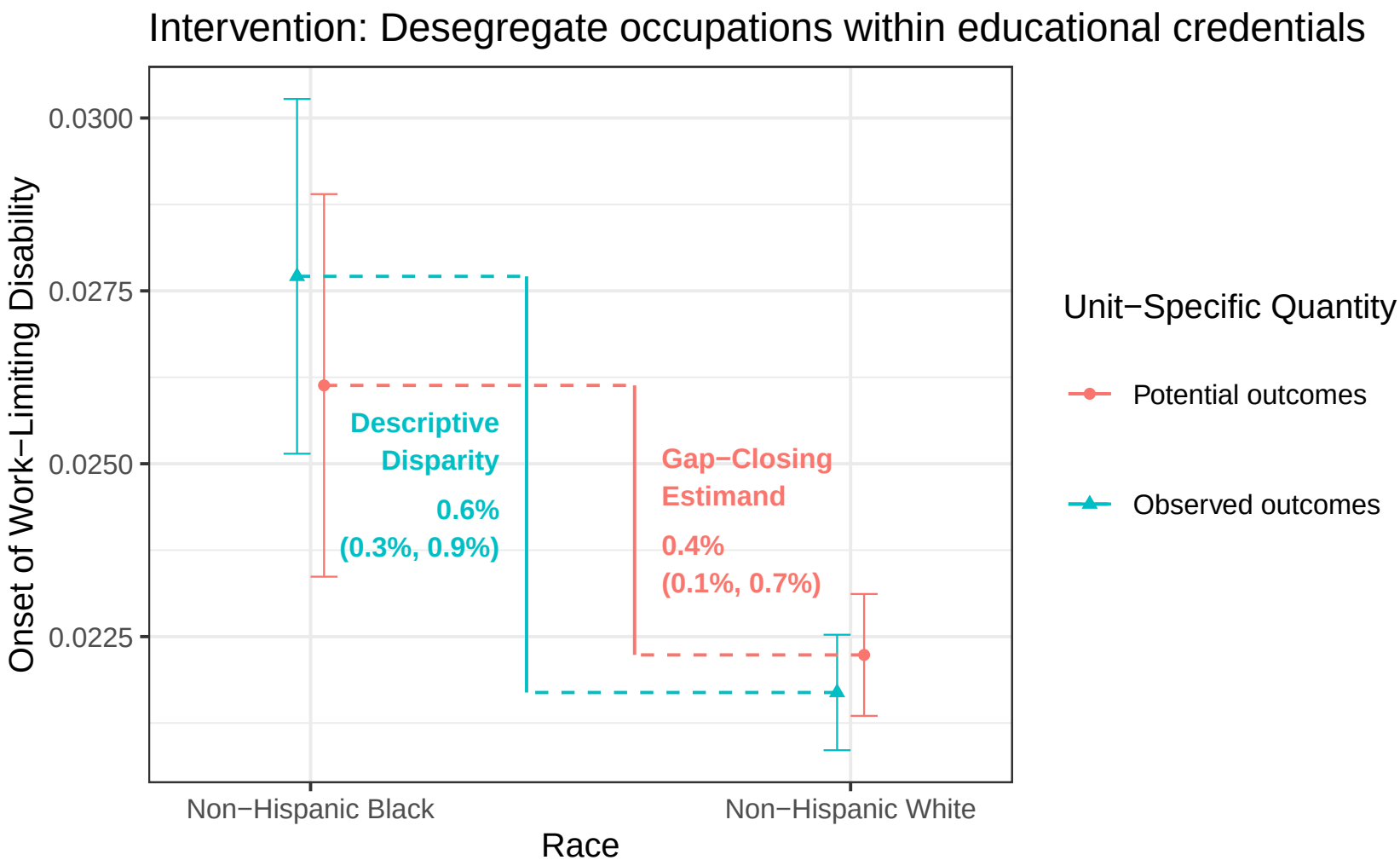
Empirical Example 2: Racial Disparities in Health



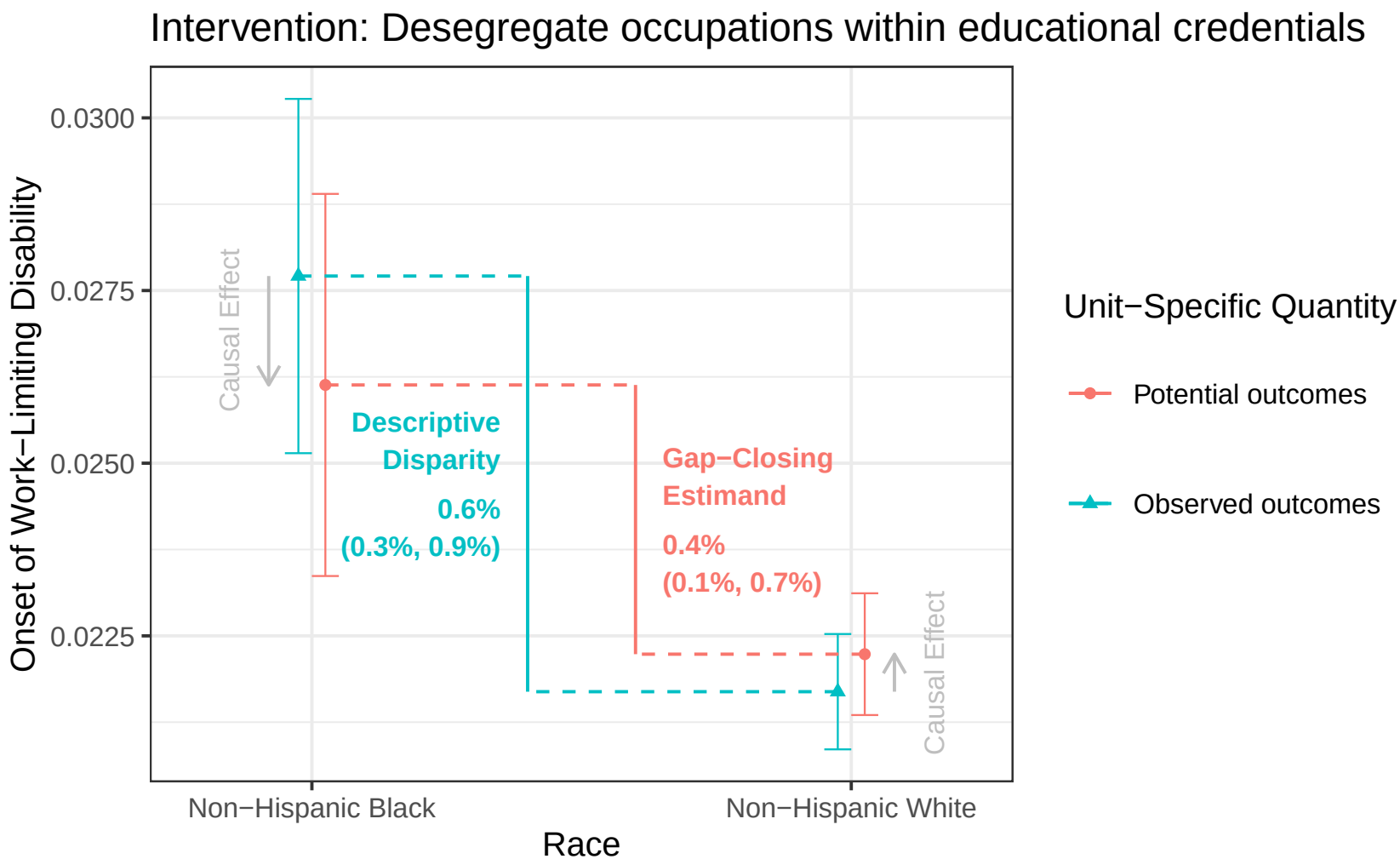
Empirical Example 2: Racial Disparities in Health



Empirical Example 2: Racial Disparities in Health



Empirical Example 2: Racial Disparities in Health



The Gap-Closing Estimand

A causal approach to study interventions that close disparities across social categories

- Past work: Causal decomposition analysis
- Define the estimand
  - what is the treatment?
  - what is the scope of the intervention?
- Identify the estimand
  - choose a sufficient adjustment set
- Estimate
  - by outcome modeling
  - by weighting
  - by doubly robust estimation
- Produce software
- **Impact: Others using these ideas**

Impact: We can now study gaps in a **new way**

Impact: We can now study gaps in a **new way**

Demography (2022) 59(5):1739–1761 Published online: 6 September 2022
DOI 10.1215/00703370-10188919 © 2022 The Authors
This is an open access article distributed under the terms of a Creative Commons license (CC BY-NC-ND 4.0).

Long-Term Exposure to Neighborhood Policing and the Racial/Ethnic Gap in High School Graduation

Joscha Legewie and Nino José Cricco

Impact: We can now study gaps in a **new way**

Demography (2022) 59(5):1739–1761 Published online: 6 September 2022
DOI 10.1215/00703370-10188919 © 2022 The Authors
This is an open access article distributed under the terms of a Creative Commons license (CC BY-NC-ND 4.0).

Long-Term Exposure to Neighborhood Policing and the Racial/Ethnic Gap in High School Graduation

Joscha Legewie and Nino José Cricco

There are **gaps** in high school graduation by race

Impact: We can now study gaps in a **new way**

Demography (2022) 59(5):1739–1761 Published online: 6 September 2022
DOI 10.1215/00703370-10188919 © 2022 The Authors
This is an open access article distributed under the terms of a Creative Commons license (CC BY-NC-ND 4.0).

Long-Term Exposure to Neighborhood Policing and the Racial/Ethnic Gap in High School Graduation

Joscha Legewie and Nino José Cricco

There are **gaps** in high school graduation by race

We could **intervene** to equalize neighborhood policing

Impact: We can now study gaps in a **new way**

Demography (2022) 59(5):1739–1761 Published online: 6 September 2022
DOI 10.1215/00703370-10188919 © 2022 The Authors
This is an open access article distributed under the terms of a Creative Commons license (CC BY-NC-ND 4.0).

Long-Term Exposure to Neighborhood Policing and the Racial/Ethnic Gap in High School Graduation

Joscha Legewie and Nino José Cricco

There are **gaps** in high school graduation by race

We could **intervene** to equalize neighborhood policing

The Black-white gap would close by more than one-fourth

Impact: We can now study gaps in a **new way**

American Sociological Review
2023, Vol. 88(1) 154–188
© American Sociological Association 2022
DOI:10.1177/00031224221141887
journals.sagepub.com/home/asr
SAGE

Higher Education and the Black-White Earnings Gap

Xiang Zhou[a] and Guanghui Pan[b]

Impact: We can now study gaps in a **new way**

American Sociological Review
2023, Vol. 88(1) 154–188
© American Sociological Association 2022
DOI:10.1177/00031224221141887
journals.sagepub.com/home/asr
SAGE

Higher Education and the Black-White Earnings Gap

Xiang Zhou[a] and Guanghui Pan[b]

There are **gaps** in earnings by race

Impact: We can now study gaps in a **new way**

American Sociological Review
2023, Vol. 88(1) 154–188
© American Sociological Association 2022
DOI:10.1177/00031224221141887
journals.sagepub.com/home/asr
SAGE

Higher Education and the Black-White Earnings Gap

Xiang Zhou[a] and Guanghui Pan[b]

There are **gaps** in earnings by race

We could **intervene** to promote college completion

Impact: We can now study gaps in a **new way**

American Sociological Review
2023, Vol. 88(1) 154–188
© American Sociological Association 2022
DOI:10.1177/00031224221141887
journals.sagepub.com/home/asr
SAGE

Higher Education and the Black-White Earnings Gap

Xiang Zhou[a] and Guanghui Pan[b]

There are **gaps** in earnings by race

We could **intervene** to promote college completion

The gap would close by more than half for men, but not for women

Impact: We can now study gaps in a **new way**

CHILD DEVELOPMENT

EMPIRICAL ARTICLE

Environmental inequality and disparities in school readiness: The role of neurotoxic lead

Jared N. Schachner[1] | Geoffrey T. Wodtke[2]

Impact: We can now study gaps in a **new way**

EMPIRICAL ARTICLE

CHILD DEVELOPMENT

Environmental inequality and disparities in school readiness: The role of neurotoxic lead

Jared N. Schachner[1] | Geoffrey T. Wodtke[2]

There are **gaps** in vocabulary skills by race and class

Impact: We can now study gaps in a **new way**



EMPIRICAL ARTICLE

CHILD DEVELOPMENT

Environmental inequality and disparities in school readiness: The role of neurotoxic lead

Jared N. Schachner[1] | **Geoffrey T. Wodtke**[2]

There are **gaps** in vocabulary skills by race and class

We could **intervene** to equalize exposure to lead

Impact: We can now study gaps in a **new way**



EMPIRICAL ARTICLE

CHILD DEVELOPMENT

Environmental inequality and disparities in school readiness: The role of neurotoxic lead

Jared N. Schachner[1] | Geoffrey T. Wodtke[2]

There are **gaps** in vocabulary skills by race and class

We could **intervene** to equalize exposure to lead

Gaps would close by 15–25%

The Gap-Closing Estimand

A causal approach to study interventions that close disparities across social categories

- Past work: Causal decomposition analysis
- Define the estimand
  - what is the treatment?
  - what is the scope of the intervention?
- Identify the estimand
  - choose a sufficient adjustment set
- Estimate
  - by outcome modeling
  - by weighting
  - by doubly robust estimation
- Produce software
- Impact: Others using these ideas

Discussion: Potential outcomes and methodology

Discussion: Potential outcomes and methodology

We often focus on average treatment effects:

$$\mathrm{E}(Y^1 - Y^0)$$

Discussion: Potential outcomes and methodology

We often focus on average treatment effects:

$$\mathrm{E}(Y^1 - Y^0)$$

Gap-closing is a different aggregation of potential outcomes:

$$\mathrm{E}(Y^1 \mid X = 1) - \mathrm{E}(Y^1 \mid X = 0)$$

Discussion: Potential outcomes and methodology

We often focus on average treatment effects:

$$\mathrm{E}(Y^1 - Y^0)$$

Gap-closing is a different aggregation of potential outcomes:

$$\mathrm{E}(Y^1 \mid X = 1) - \mathrm{E}(Y^1 \mid X = 0)$$

Many advances may be possible with other aggregations

Discussion: Potential outcomes and methodology

We often focus on average treatment effects:

$$\mathrm{E}(Y^1 - Y^0)$$

Gap-closing is a different aggregation of potential outcomes:

$$\mathrm{E}(Y^1 \mid X = 1) - \mathrm{E}(Y^1 \mid X = 0)$$

Many advances may be possible with other aggregations

Effect on median $\quad \mathrm{Median}(Y^1) - \mathrm{Median}(Y^0)$

Discussion: Potential outcomes and methodology

We often focus on average treatment effects:

$$\mathrm{E}(Y^1 - Y^0)$$

Gap-closing is a different aggregation of potential outcomes:

$$\mathrm{E}(Y^1 \mid X = 1) - \mathrm{E}(Y^1 \mid X = 0)$$

Many advances may be possible with other aggregations

Effect on median	$\mathrm{Median}(Y^1) - \mathrm{Median}(Y^0)$
Effect on variance	$\mathrm{Var}(Y^1) - \mathrm{Var}(Y^0)$

Discussion: Potential outcomes and methodology

We often focus on average treatment effects:

$$\mathrm{E}(Y^1 - Y^0)$$

Gap-closing is a different aggregation of potential outcomes:

$$\mathrm{E}(Y^1 \mid X = 1) - \mathrm{E}(Y^1 \mid X = 0)$$

Many advances may be possible with other aggregations

Effect on median	$\mathrm{Median}(Y^1) - \mathrm{Median}(Y^0)$
Effect on variance	$\mathrm{Var}(Y^1) - \mathrm{Var}(Y^0)$
Gap-closing for variance	$\mathrm{Var}(Y^1 \mid X = 1) - \mathrm{Var}(Y^1 \mid X = 0)$

Discussion: Potential outcomes and methodology

We often focus on average treatment effects:

$$\mathrm{E}(Y^1 - Y^0)$$

Gap-closing is a different aggregation of potential outcomes:

$$\mathrm{E}(Y^1 \mid X = 1) - \mathrm{E}(Y^1 \mid X = 0)$$

Many advances may be possible with other aggregations

Effect on median	$\mathrm{Median}(Y^1) - \mathrm{Median}(Y^0)$
Effect on variance	$\mathrm{Var}(Y^1) - \mathrm{Var}(Y^0)$
Gap-closing for variance	$\mathrm{Var}(Y^1 \mid X = 1) - \mathrm{Var}(Y^1 \mid X = 0)$
Counterfactual slopes	$\frac{\mathrm{Cov}(Y^1, X)}{\mathrm{Var}(X)}$ for continuous X (Yu & Zhao 2024)

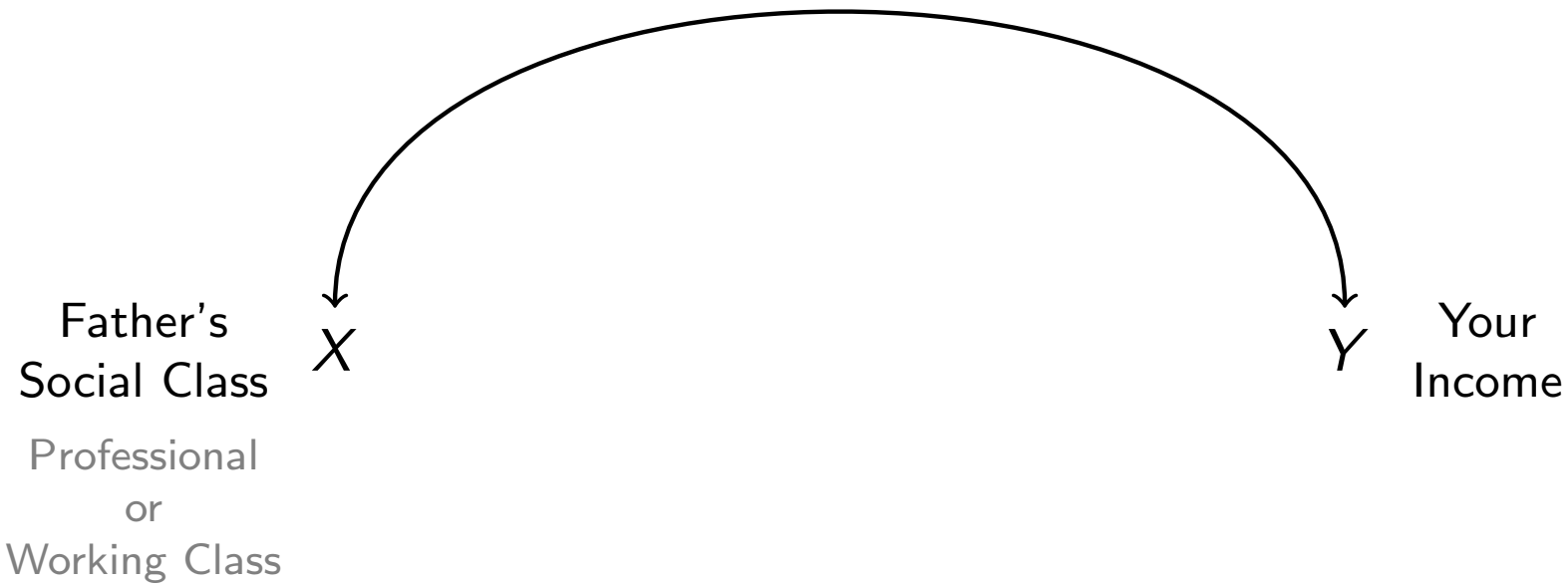
Thanks!

Ian Lundberg
ianlundberg.org
ianlundberg@ucla.edu

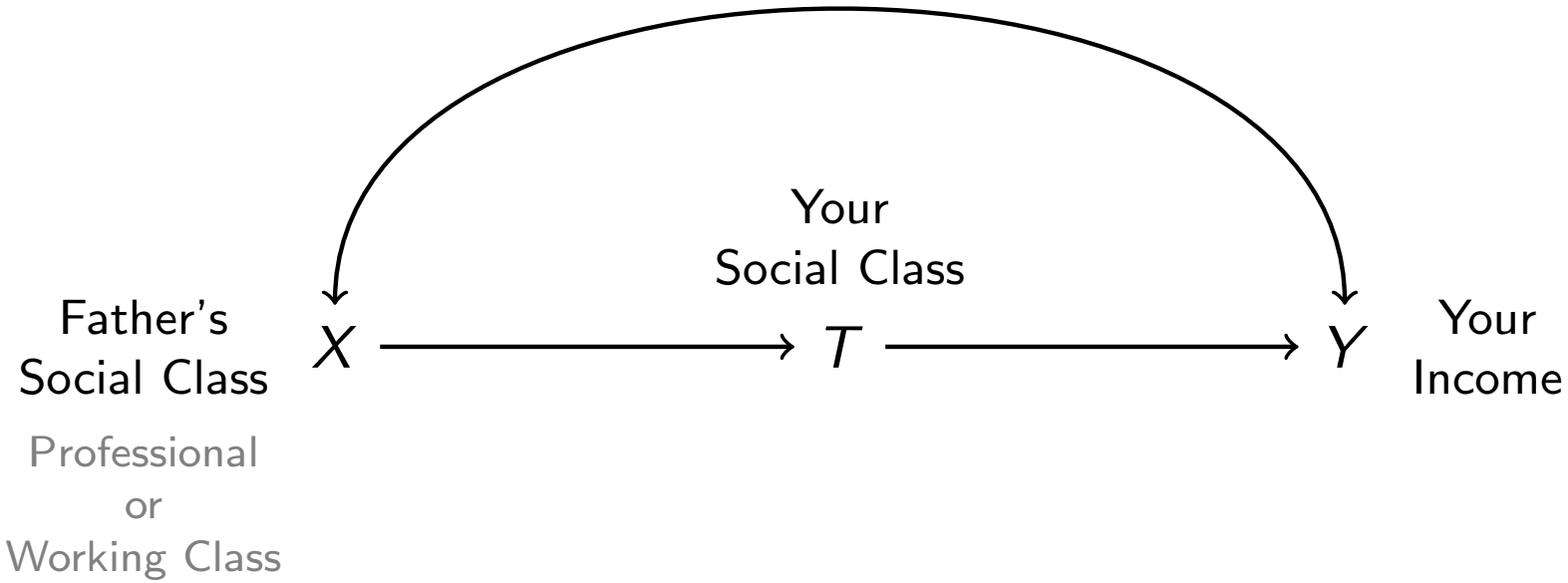
More about this project with a code tutorial:
ilundberg.github.io/gapclosing

Appendix

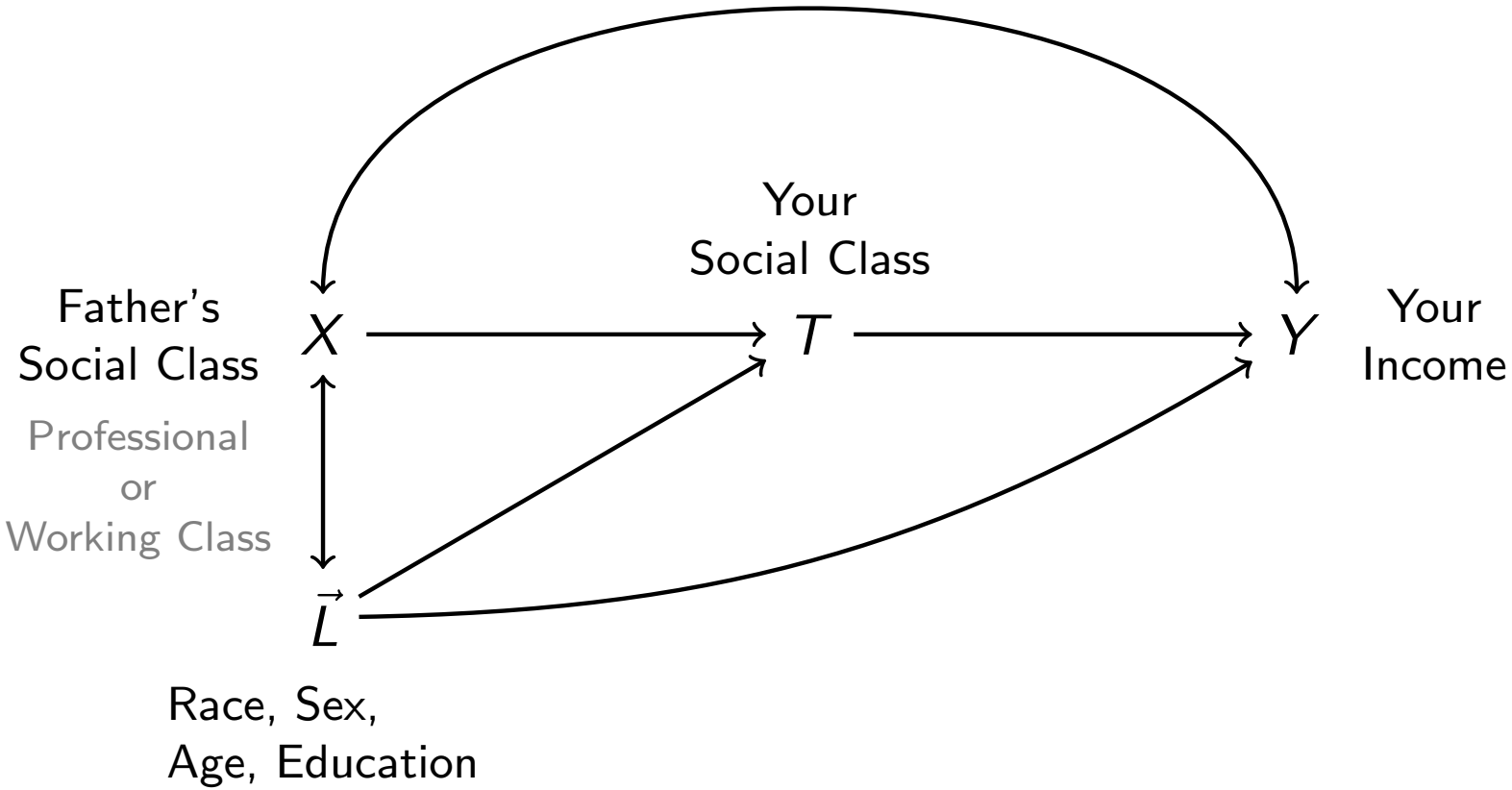
Empirical Example 1: Economic Mobility



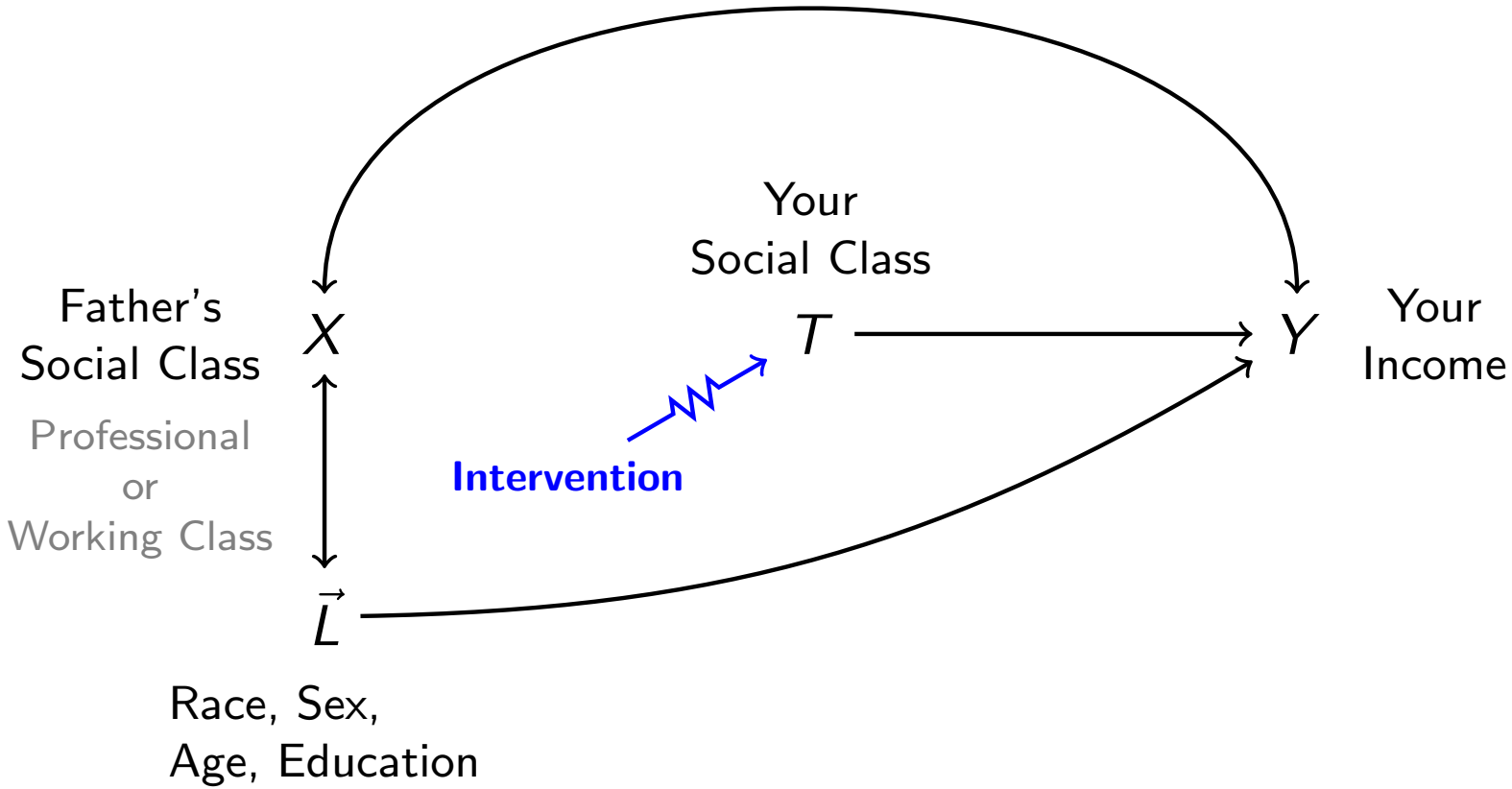
Empirical Example 1: Economic Mobility



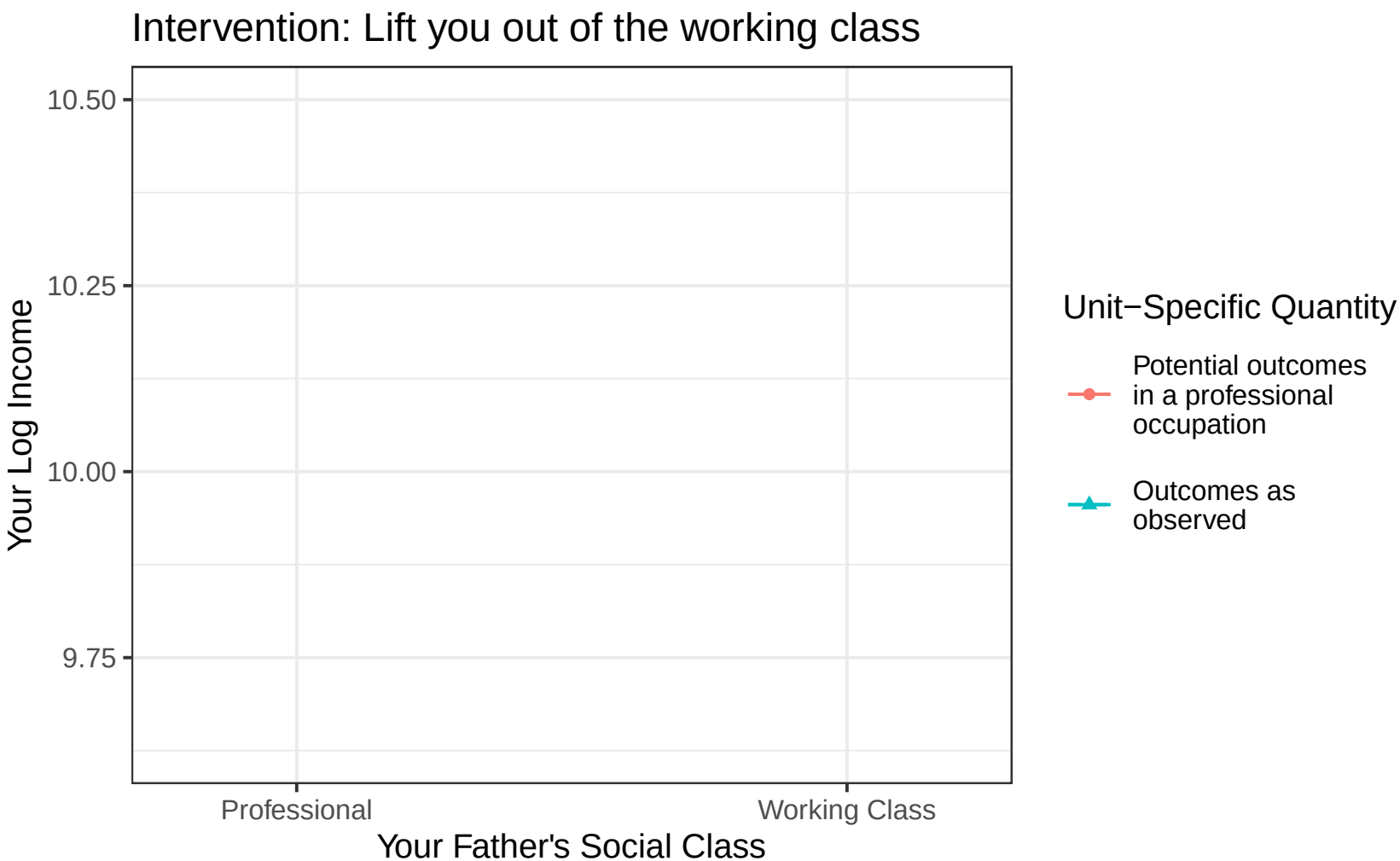
Empirical Example 1: Economic Mobility



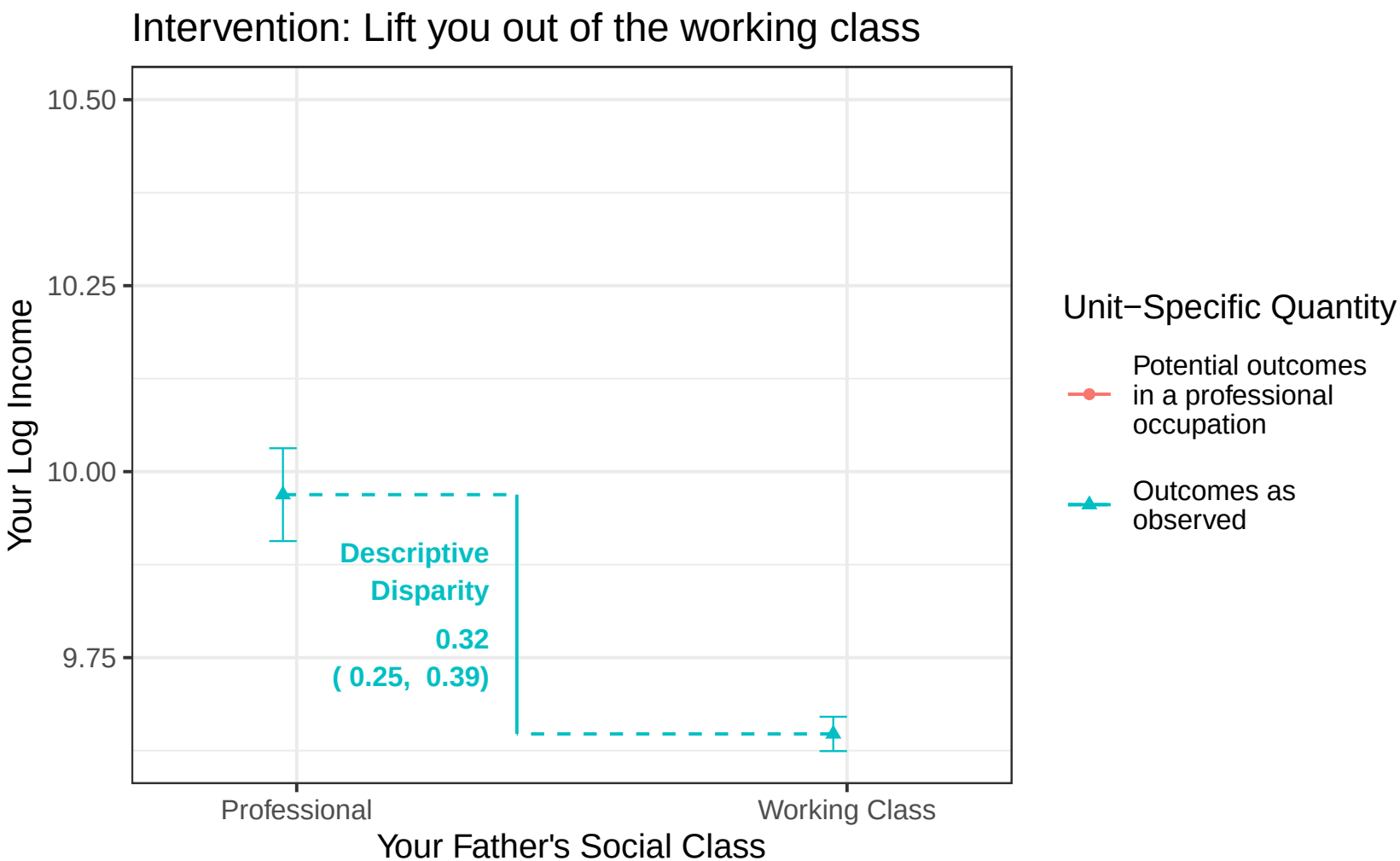
Empirical Example 1: Economic Mobility



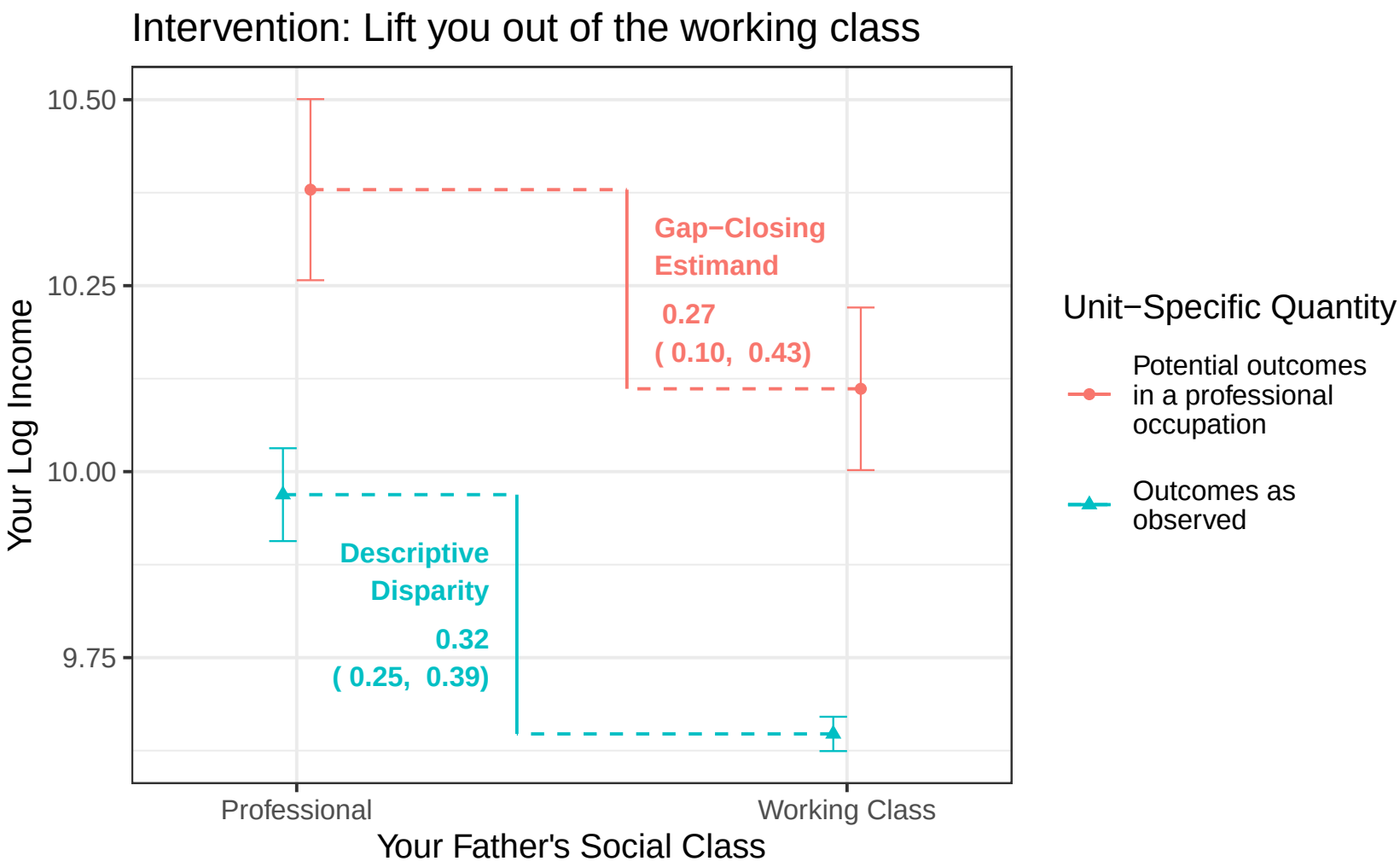
Empirical Example 1: Economic Mobility



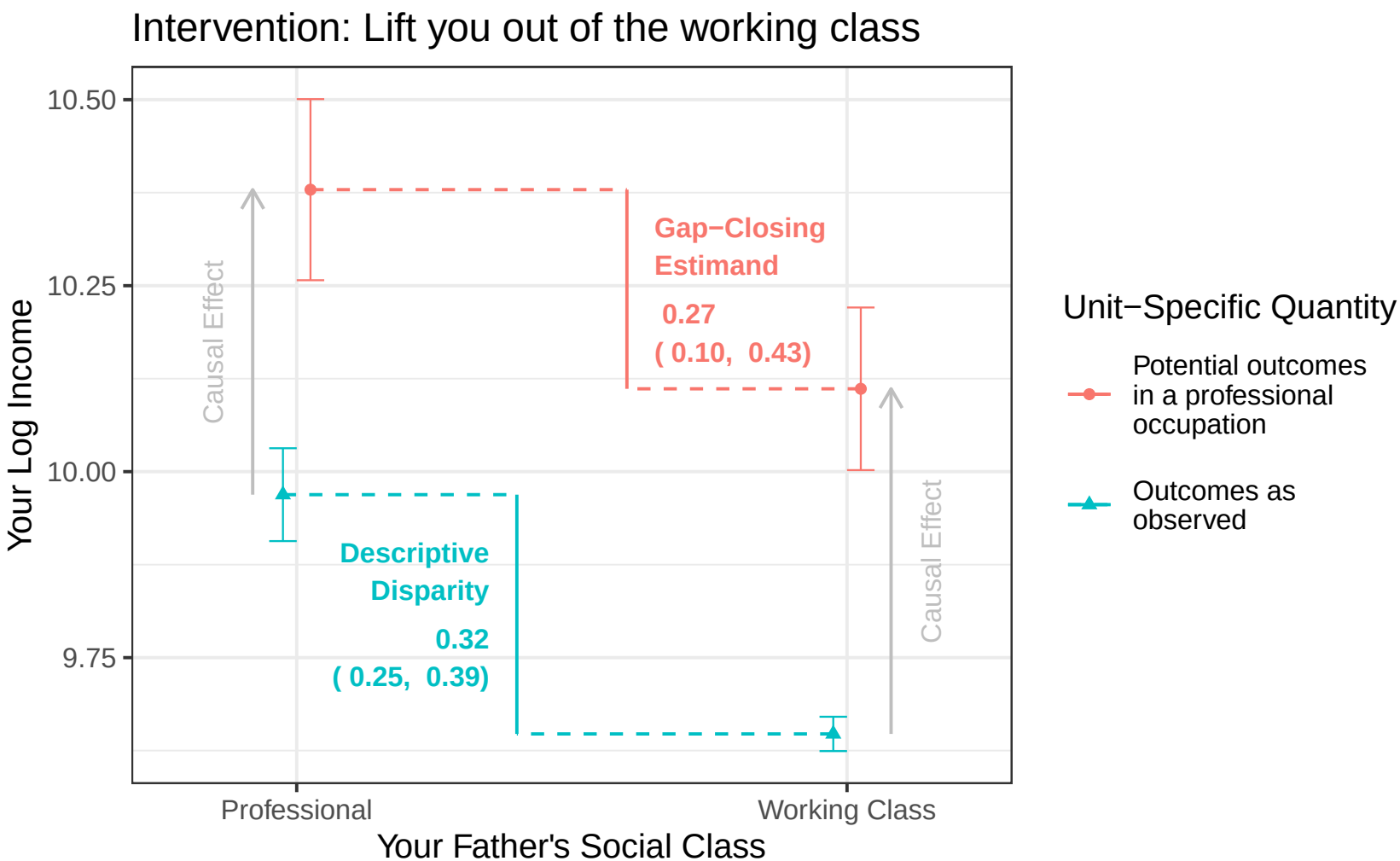
Empirical Example 1: Economic Mobility



Empirical Example 1: Economic Mobility

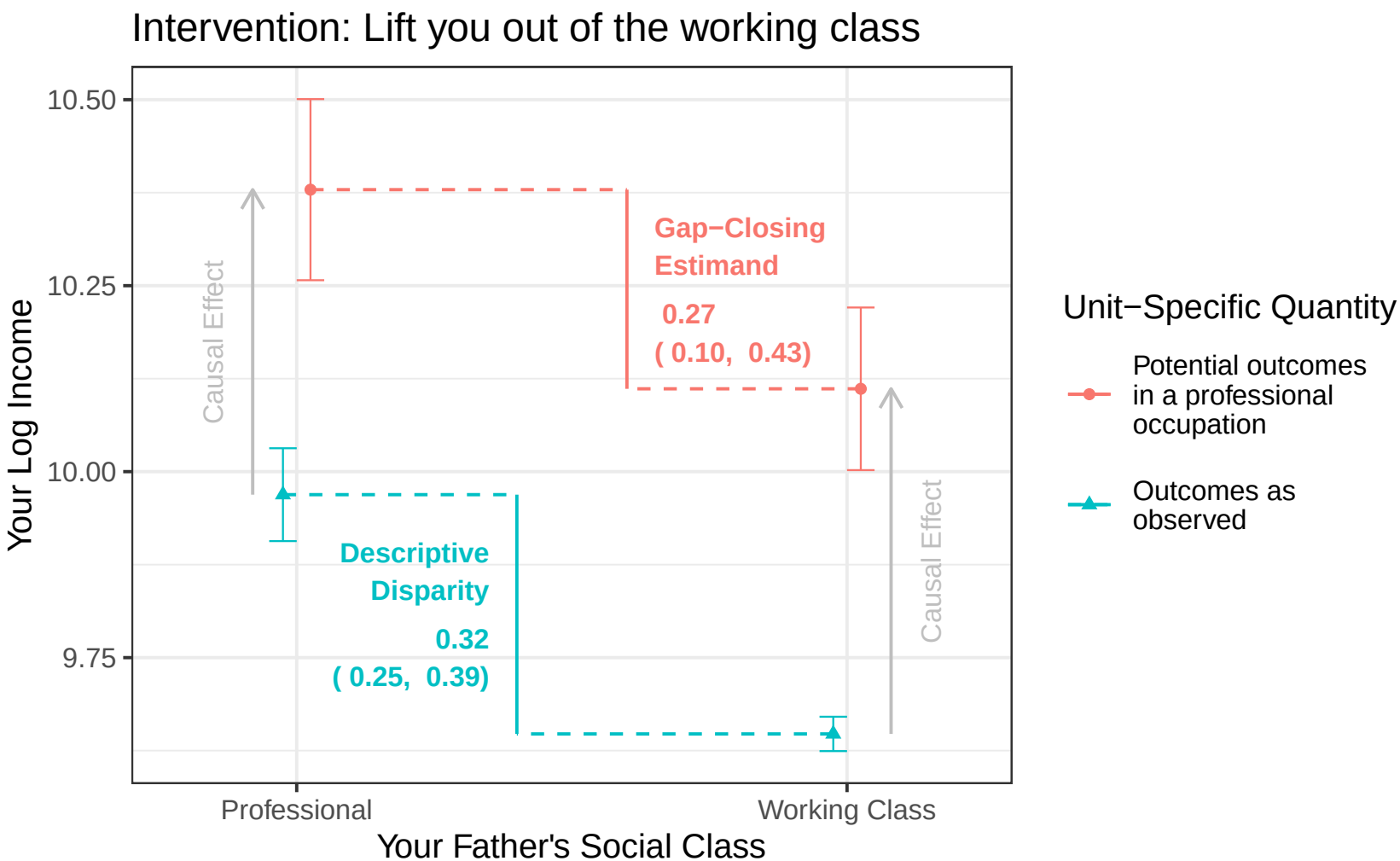


Empirical Example 1: Economic Mobility

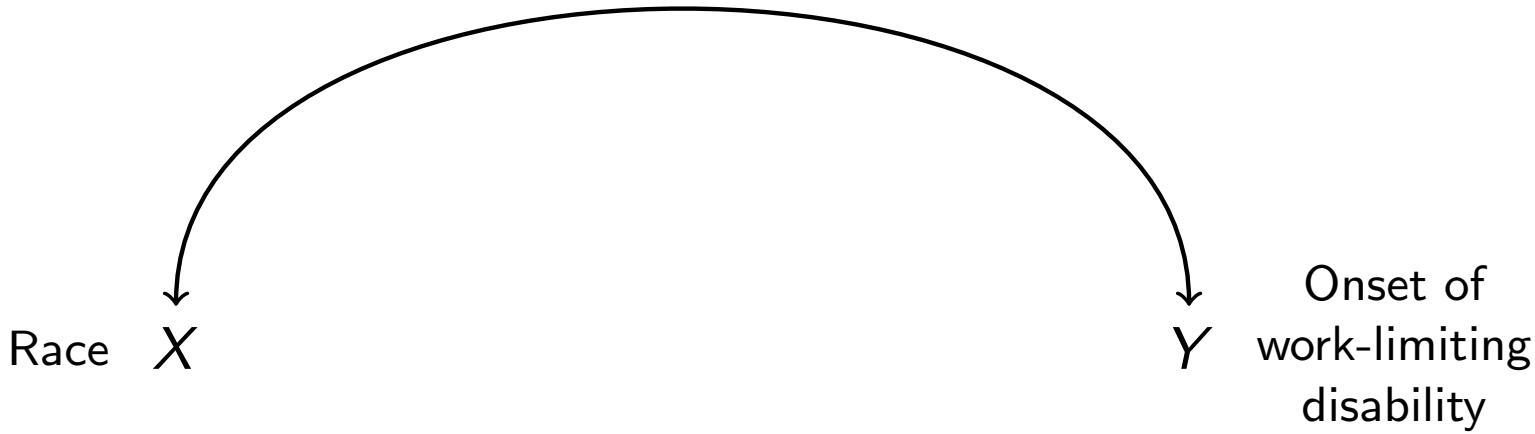


Empirical Example 1: Economic Mobility

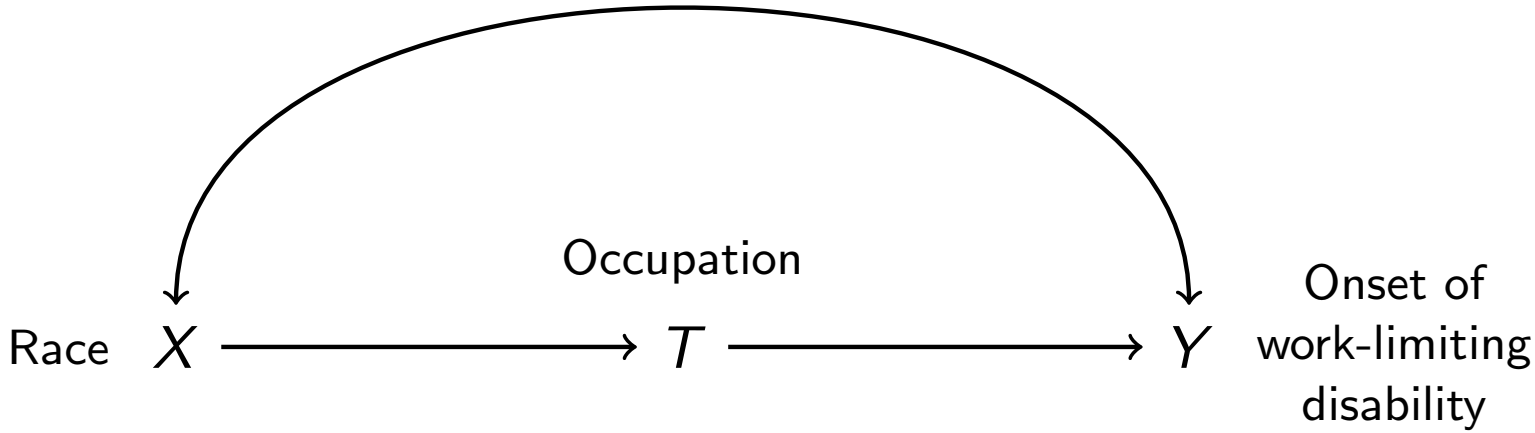
`plot_two_categories()`



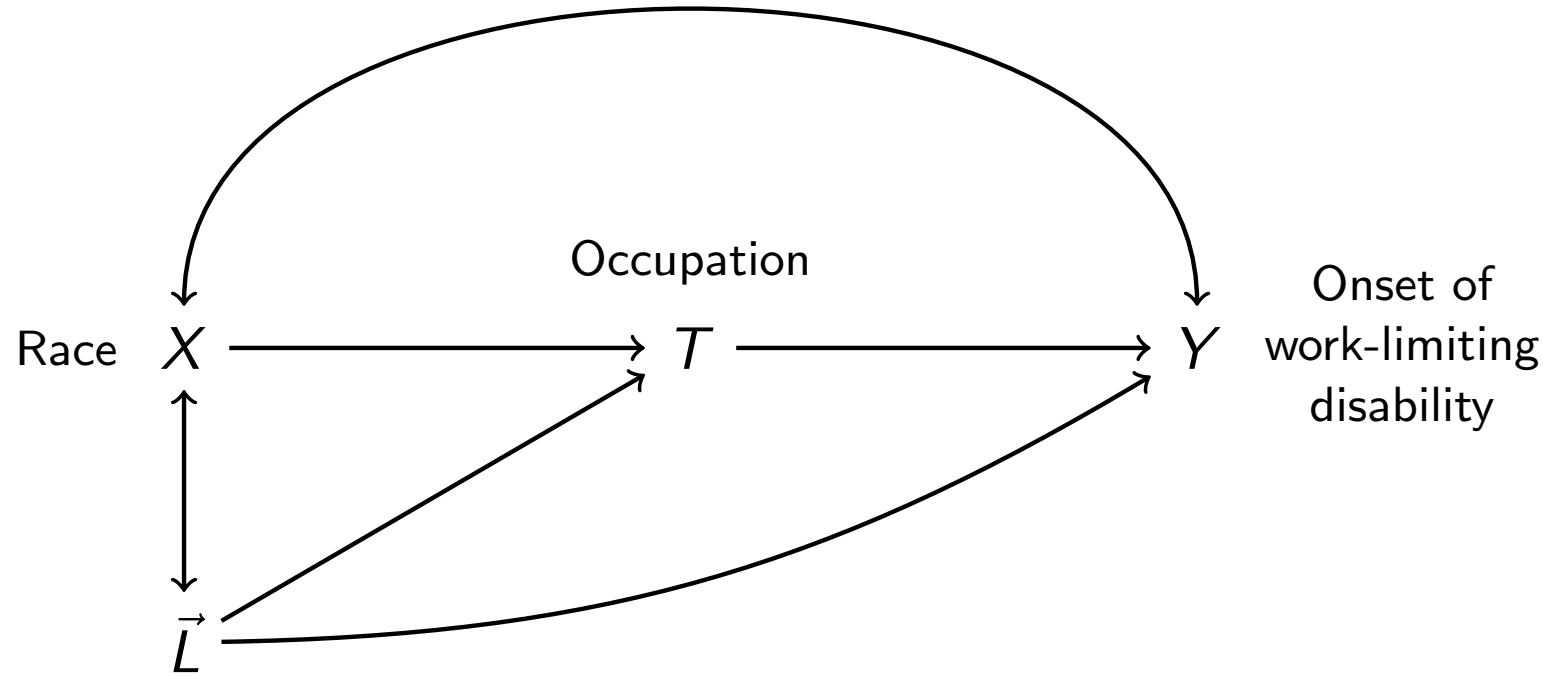
Empirical Example 2: Racial Disparities in Health



Empirical Example 2: Racial Disparities in Health

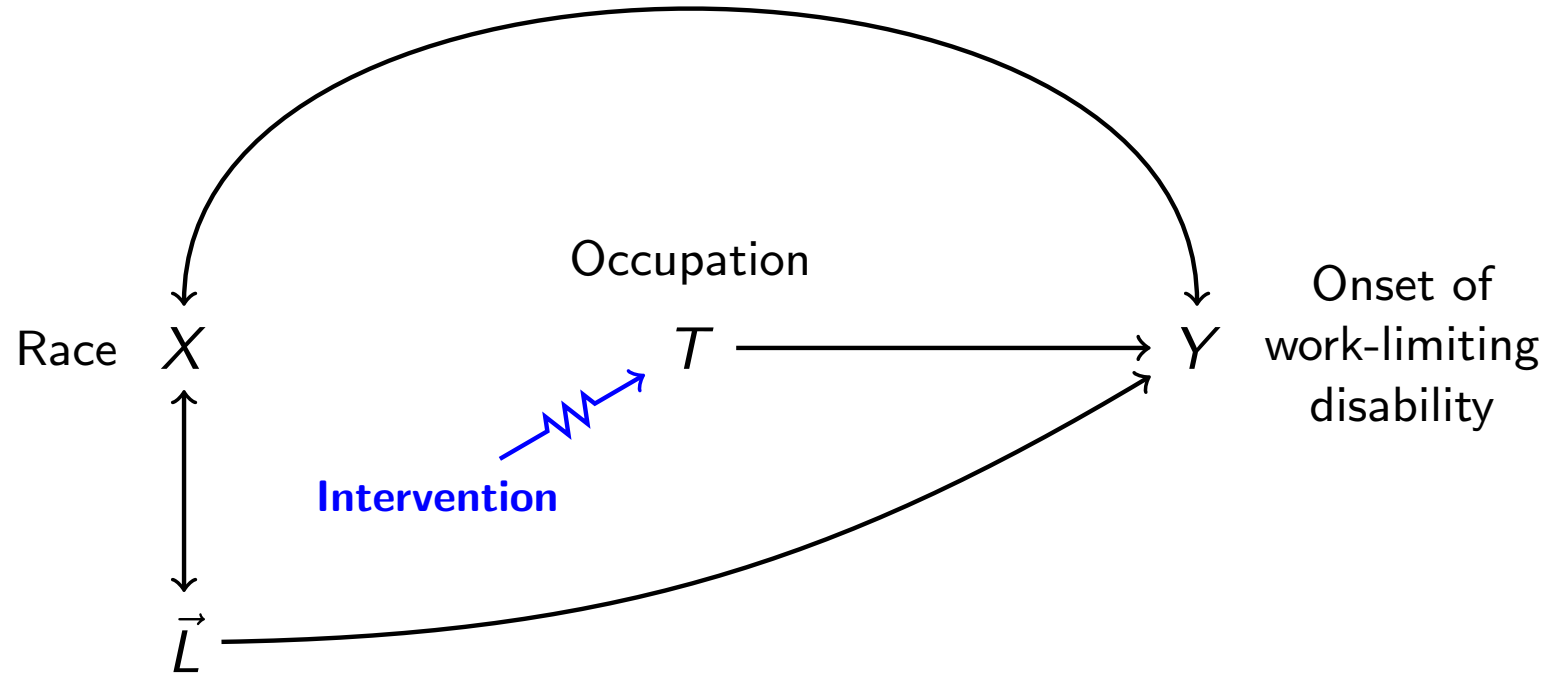


Empirical Example 2: Racial Disparities in Health



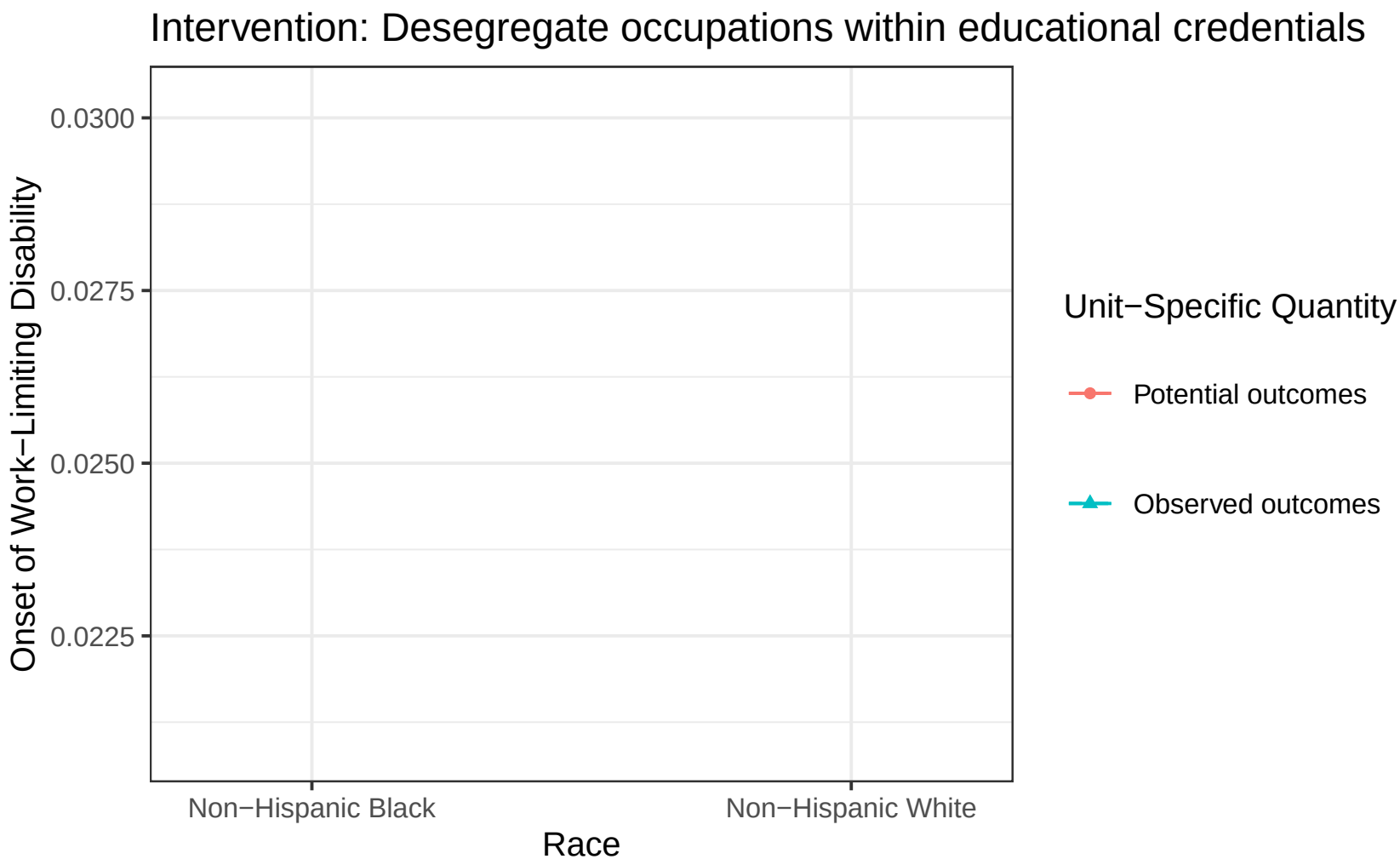
Sex,
Age, Education
Foreign born,
Lagged outcome,
Lagged health

Empirical Example 2: Racial Disparities in Health

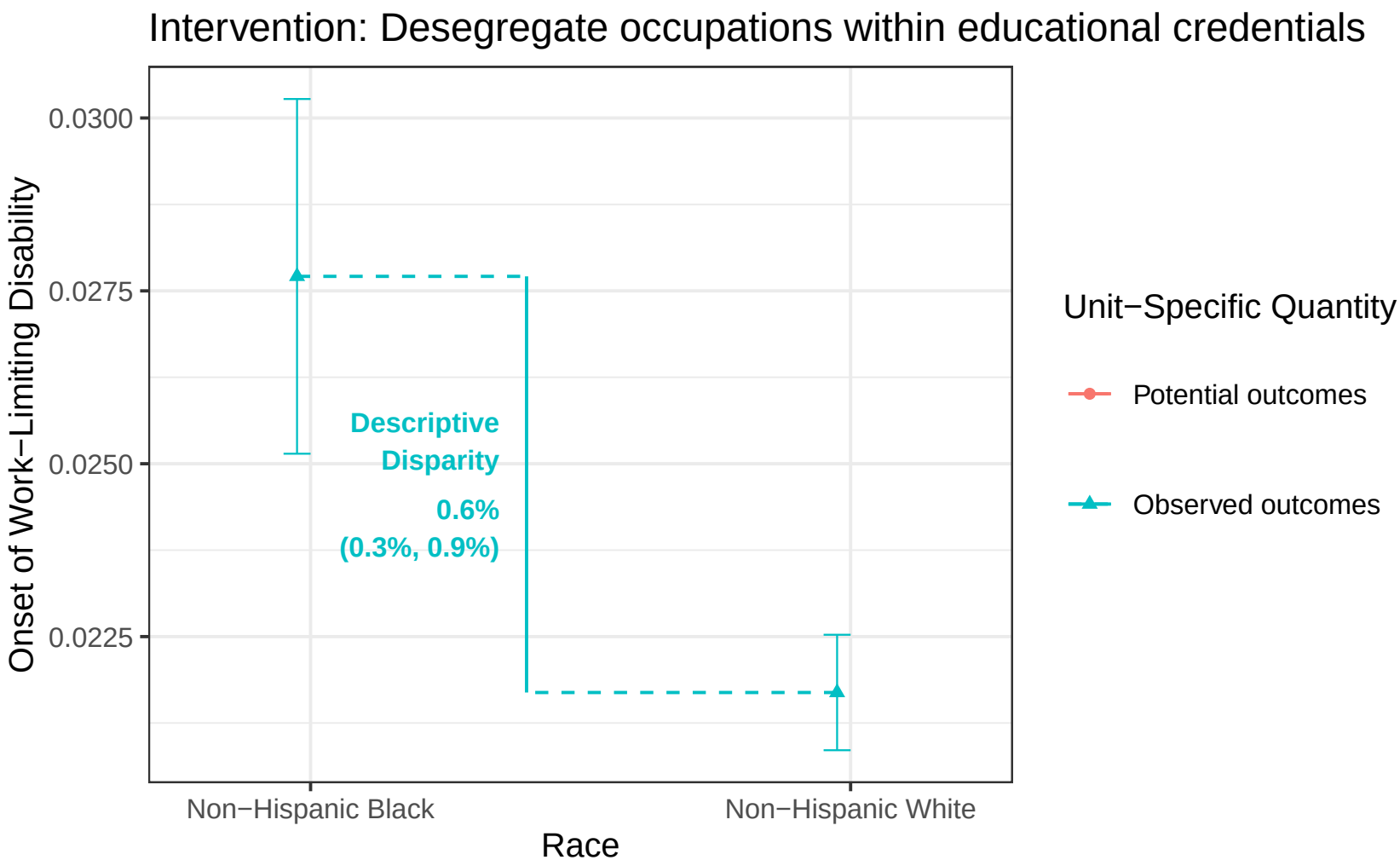


Sex,
Age, Education
Foreign born,
Lagged outcome,
Lagged health

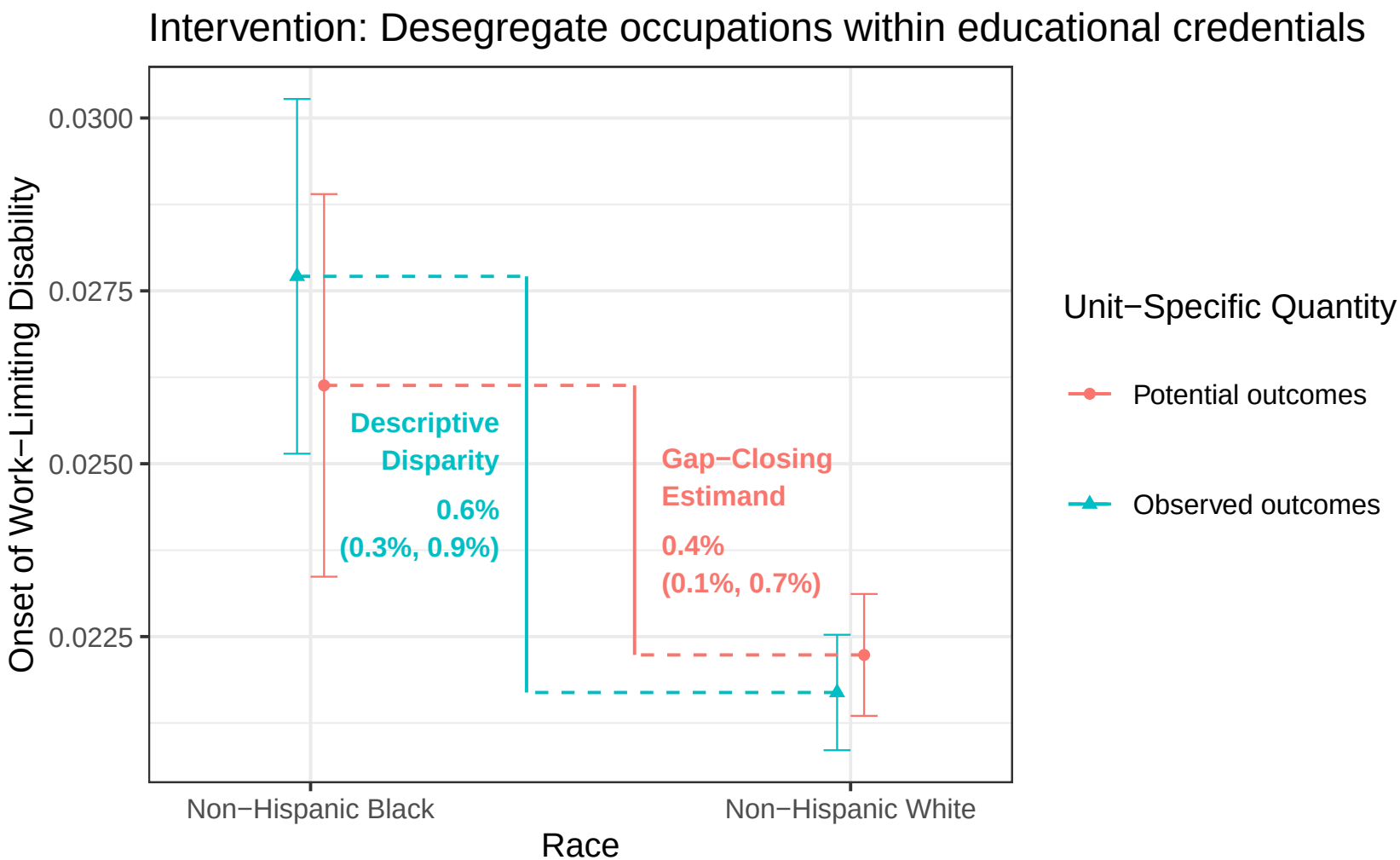
Empirical Example 2: Racial Disparities in Health



Empirical Example 2: Racial Disparities in Health



Empirical Example 2: Racial Disparities in Health



Empirical Example 2: Racial Disparities in Health

